How to create a Platform Application In Azure

Introduction

A Platform is a set of applications working together.

This document provides a complete template describing all steps required to create a Platform Application in Azure.

The goal is to run multiple applications (each encapsulated in its own container) together with back-end resources in parallel. These applications must only exchange information when an internal credential or identity (an Azure User Assigned Managed Identity) is provided.

The entry point of the platform is a public API. External clients — such as Angular applications, mobile applications, or testing tools like Postman — can connect to this API only by presenting a valid token issued by the Platform.

To be precise, the platform contains the following components:

- SampleApi — the main API exposed to external clients.
- SampleDataAccessApi — the internal API responsible for database access
- Azure SQL Database — the persistent data store

The platform has been developed using .NET Core (.NET 8.0) and Entity Framework Core as the ORM (Object-Relational Mapping).

As mentioned earlier, the platform implements two security layers:

- Service-to-Service Authentication — implemented with an Azure User Assigned Managed Identity
- User Authentication — where users must present a valid token to access the application (for example, to add or delete records). Here JWT, Microsoft Entra ID, and Role-Based Authorization play an important role.

This document illustrates how to containerize this platform in Azure and how Azure DevOps can be used to host the code and test the platform using Postman.

Azure Concepts

The first thing we need to do is to understand or to clarify some important concepts in Azure:

- **Resource:** In Azure when create objects/instances it will also create a resource for this instance. A resource stores metadata about the instance: resource ID, Object ID, Resource Group, Region, RBAC rules (who can use based on roles), Tags, Policies (for compliance), Life-cycle (when will be created, deleted or moved), Monitoring which that the resource can

generate Logs. Metrics and alerts. This metadata is necessary to manage the object inside the Azure Universe: where is the object, who has access etc

And also Azure will allow us to group similar resources in groups: resource groups. An analogy would be: the instance=letter, the resource=envelop, resource group=folder.

Some Azure objects/instances can be shared between applications, but each of them always has its own Resource (envelope). Shared objects correspond to infrastructure components such as Logs (Log Analytics Workspace), User Assigned Identity, Key Vault, Storage Account, or Virtual Network. Objects that contain code or data must always have their own dedicated Resource because they need to be isolated: Container App, SQL Database, App Service. The reason why some objects/instances are shared is cost, central management, and security — but the underlying Resource is never shared.

In order to group resources in resource groups the rule would be:

Same Life-cycle: when resources are deployed, deleted, versioned, migrated together.

Same Responsibility: they are developed, deployed by they same team and they use common pipelines.

Same Security Limits: apply to them same RBAC rules, policies and compliance.

In case that all these rules apply then means that we can put all the resources that have all this in common in the same resource group.

For our project, the SampleApiPlatform, where we can have several containers as well as Databases we have the following Resource Groups:

For our global project the SampleApiPlatform where we have databases and different containers running in parallel we will have the following resource groups:

Platform Resource Group Structure (Optimal)

- **sampleApiPlatform_rg** The central resource group (RG) for platform-wide infrastructure. Everything that is shared, long-lived, security-critical, or platform-wide belongs here: **User Assigned Identities, Key Vault, Log Analytics Workspace, Virtual Network, Storage Accounts (if used platform-wide), Private DNS Zones, Shared Policies / RBAC**

  This is the heart of your platform.

  Everything that multiple apps use belongs here.

- **sampleApiApps_rg** The resource group for all container apps, meaning the actual microservices: **SampleDataAccessApi (Container App), SampleMarketingApi, (Container App), SampleSalesApi (Container App), Frontend apps, Sidecar services, Revision management / deployments**

  Apps have a different lifecycle than infrastructure.

  Apps are redeployed frequently; identities are not.

- **sampleApiData_rg** The resource group for data, meaning everything persistent: **Azure SQL Database, Azure SQL Server / Elastic Pool, Private Endpoint for SQL, Auditing / Threat Detection, Backup policies**

  Data has the strictest security boundary.

  Data outlives apps and outlives identities.

- sampleApiNetwork_rg (optional) If you want to keep networking separate (enterprise standard):Vnet, Subnets, Private Endpoints, Routing / Firewall / NSGs

  Optional for small projects, standard for large platforms.

Why this structure is perfect?

- Lifecycle: Platform resources live a long time, Apps are constantly redeployed, Data lives the longest. Conclusion: Three distinct lifecycles mean three resource groups.
- Responsibility: The platform team manages identities, Key Vault, and logs.The app team manages container apps.The data team manages SQL. Conclusion: Three responsibilities mean three resource groups.
- Security Limits: The platform has shared RBAC. Apps have app-specific RBAC. SQL has extremely strict RBAC. Conclusion: Three security boundaries mean three resource groups.

The rule is : So everything that is tighed to an app and not shared with other apps is advisable to put all in the same resource group

- **Container**: packed application that contains everything to run: code, runtime (.net core), libraries, Configuration

  Containers are isolated, portable, reproducible, start start, and they do not have conflicts with other applications.

- **Container Image or Image:** instructions to build a container. They are specified in Docker file , stored in a registry container like ACR, Dockerhub. The Azure Container Apps pulls the image from the registry and started it.
- **Container App:** is a container that is hosted and run in Azure, with the following extras: autoscaling, Ingress (HTTP / HTTPS), Secrets, Revisions, Logs, Managed Identity.

Ingress: when the container app will provide services to other users though the internet then we need to activate Ingress. Ingress provides the app an url, routing by redirecting the app to the right Revision, Load balancing between different container instances. Ingress also manage certificates such as: TLS/HTTPS as well as port mapping.

Revision = Snapshot of the Image and properties of the container app. It is a version of the container app. A new version is created automatically by Azure when the image, the environment variables, the port, a secret, or an ingress property has changed. The revisions are very useful for the following scenarios:

Rollback — You can revert to an older revision at any time.

Traffic Splitting — You can route 10% of traffic to the new revision and 90% to the old one.

Blue/Green Deployments — Test the new version before you push it live.

Stable Releases — The old revision remains active until the new one is running stably.

Zero Downtime — The new revision starts up while the old one continues to run.

Debugging — You can view logs per revision.

A Container App has only one container PER REVISION. If we have 3 revisions we have 3 containers: Revision A (1 container), Revision B(1 Container), Revision C(1 Container). In other environments instead of Container Apps they use Pods where the containers are hosted and each Pod can contain several containers.

Replica: it is an instance of a container and they exist to manage cases of elastic demand. By default we have 1 Revision with one container running, i.e 1 Replica. If suddenly the demand grow, another instance of the container (replica) is created. It also can happen that our replica for any reason falls over. Then another replica will be created automatically for us. Azure will create a maximum of Replicas that we define in the maxReplicas property and azure decides when to create a new Replica dynamically depending on factors such as: cpu > 70%, httpRequests > 100 /s, queueLength > 50. On the otherhand it can reduce the amount of replicas automatically for us when for instance cpu < 20%.

Managed Identity: Azure give to our Container App an account, an Identity for our Container App and there is no need of Password, Secrets, Certification, Key Vault etc. because everything is managed by Azure itself. This is very useful when our application needs to connect with other Azure services such as Azure SQL, Azure Key Vault, Azure Storage, Azure Container Regitry (ACR), Service Bus, Event Hubs, etc.

Then we can assign a role to our identity to limit the scope of what can do or Role Based Identity (RBAC). For instance these roles:

AcrPull: container app can pull images from ACR

Key Vault Reader: the container App can read secrets

Storage Blob Data Container: Container App can write Blobs

We can configure an API at Enterprise Level Security, meaning that it does not need any connection strings, any password, and that Login is managed through Managed Identity.

However Managed Identity cannot be used in case that this application is called from outside Azure: Postman, local application from our laptop, mobile, an AWS application, etc.

The Managed Identity is not automatically activated for Container Apps and has to be explicitly activated. To activate the managed identity from the portal, go to the container > Security > Identity > System Assigned > Set the status = on.

Other authentication methods can be: User defined authentication (go to the container > Security > Identity > User Assigned) or use an identity provider such as Microsft, Apple, Facebook, Google, Twitter, Open ID connect (go to the container > Security >

Authentication). The identity provider allow us to login into anther application using Tokens.

The main difference between User Assigned Identity and Managed Identity:

System Assigned (Managed Identity)

- Azure automatically creates ONE identity for your Container App as soon as you enable it.
- Belongs to this Container App only.
- Is automatically deleted when the app is deleted.
- You can only have one System Assigned Identity per Container App.
- Ideal for ACR Pull, SQL Login, Key Vault, and Storage.
- Simplest approach → covers 95% of use cases

User Defined Identity

- You create the identity yourself as a separate Azure object.
- Can be used by multiple Container Apps
- Persists even when the Container App is deleted
- You can assign multiple User Assigned Identities to a single app
- Ideal for microservices that need to share the same identity
- Ideal for shared access (e.g., multiple apps accessing the same Key Vault)

Just to clarifiy, User Defined or Managed Identities are assigned to applications (conatainer apps) but not to resources (Azure SQL).

Port Mapping: a container App has ONE EXTERNAL port to the outside world, ex 443 (HTTPS), 8080 (HTTP), to where the all containers in each revision listens to (common port), and is used by all revisions. Strictly speaking these external port does not belong to the App Container but to the Ingress Gateway At the same time each container in each revision has and INTERNAL port (ex. 8080) but all share this port number. So all containers in each revision will have the same port number in this case 8080. The port Mapping in the Container App maps the external port to the internal container port: 443 > 8080.

- **Container Apps Environment:** it is the platform where our container app is running. It contains: internal network, ingress controller, secret storage, revision system, log streaming, compute cluster (something similar to Kubernetes), Zone Redundancy. In summary all the infrastructure that makes possible each revision ro run.

- **Availability Zone:** a group of Data centers (minimum 3). An availability zone will have its own electricity supply, cooling system, network, hardware. They are separate from each other but logically connected.

  In West Europe for instance there are 3 Availability zones.

- **Zone Redundancy**: this concept is used at several levels in Azure. Essentially means that when enabled, our App Container will run in different Availability Zones in the same region.

  When disable the App Container run only in one Availability Zone.

  These Availability Zones, are near enough to communicate fast but far enough to avoid the same accident: fire, electricity cut off, hardware-crash.

  When there is an issue in one Availability Zone then the App Container can run from another Availability Zones (back up) and therefore there will not be any down-times. When the zone redundancy is diabled this will not be possible, however, if the issue is local in one data center in the availability zone and other data centers in the same availability zone are not

affected, then the application can continue running in another data center in the same availability zone.

The SampleDataAccessApi Application

Create an image in Dockerhub

The first step before to containerize an application is to have an image repository. For this we have several options: Azure ACR, Github, etc. In this document we have chosen **Dockerhub** mainly because is free and with Githib we have encountered some technical issues.

We will need the first time the image into docker, otherwise App Container is unable to create a container for our App as this data is required at creation time. To avoid creating a local image in an environment that is not the same as in azure, we will create an **azure-pipeline-first-image.yml** that will create an image in dockerhub.

```
trigger:
- main

pool:
  vmImage: 'ubuntu-latest'

steps:
- checkout: self

- script: |
    export DOCKER_BUILDKIT=1
  displayName: Enable BuildKit

- task: DockerInstaller@0
  displayName: Install Docker

- script: |
    docker build -t popdemar/sampledataaccessapi:latest .
  displayName: Build Docker Image

- script: |
    echo $(DOCKERHUB_TOKEN) | docker login -u popdemar --password-stdin
  displayName: Docker Login

- script: |
    docker push popdemar/sampledataaccessapi:latest
  displayName: Push Image to Docker Hub
```

In order to run this script we will need to create an environment variable called **DOCKERHUB_TOKEN** that stores the password to our Docker account (it is not necessary to create a Personal Access Token or PAT for this case). Make sure to make to check the **Keep this Value Secret** checkbox.

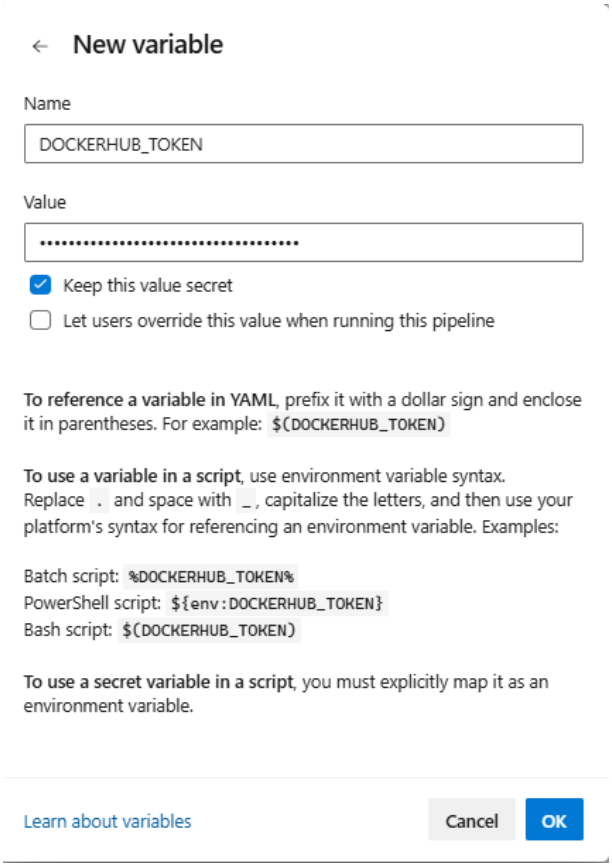


Figure 1: screenshot Screenshot environment variable DOCKERHUB_TOKEN

And this is the **Dockerfile** that we are using (make sure that has been pushed in the repo with the rest of the code):

```
# Build stage
FROM mcr.microsoft.com/dotnet/sdk:8.0 AS build
ARG BUILD_CONFIGURATION=Release

WORKDIR /src

# Copy csproj
COPY SampleDataAccessApi/SampleDataAccessApi.csproj SampleDataAccessApi/

# Restore
RUN dotnet restore "SampleDataAccessApi/SampleDataAccessApi.csproj"

# Copy the rest of the source code
COPY SampleDataAccessApi/ SampleDataAccessApi/
```

```
# Build
RUN dotnet build "SampleDataAccessApi/SampleDataAccessApi.csproj" -c
$BUILD_CONFIGURATION -o /app/build

# Publish
RUN dotnet publish "SampleDataAccessApi/SampleDataAccessApi.csproj" -c
$BUILD_CONFIGURATION -o /app/publish /p:UseAppHost=false

# Runtime stage
FROM mcr.microsoft.com/dotnet/aspnet:8.0 AS final
WORKDIR /app

COPY --from=build /app/publish .

ENV ASPNETCORE_URLS=http://+:8080
EXPOSE 8080

ENTRYPOINT ["dotnet", "SampleDataAccessApi.dll"]
```

The **image and tag** for simplicity we chose the **latest**. It is possible to indicate a specific build.

Once we complete this stage we will have our image, sampleaccessapi, uploaded in the Dockerhub and now we will be able to create a container in Azure.

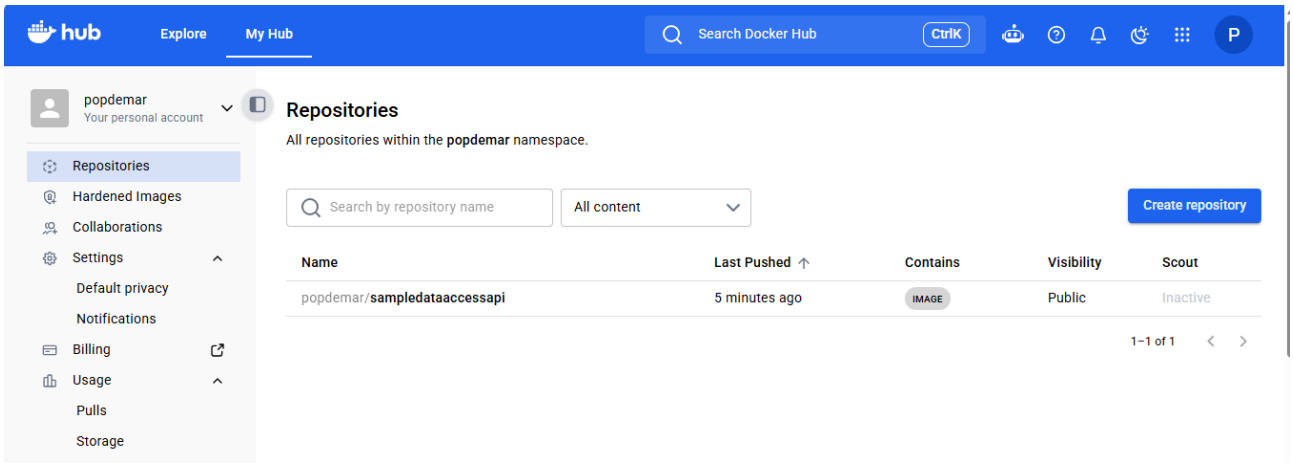
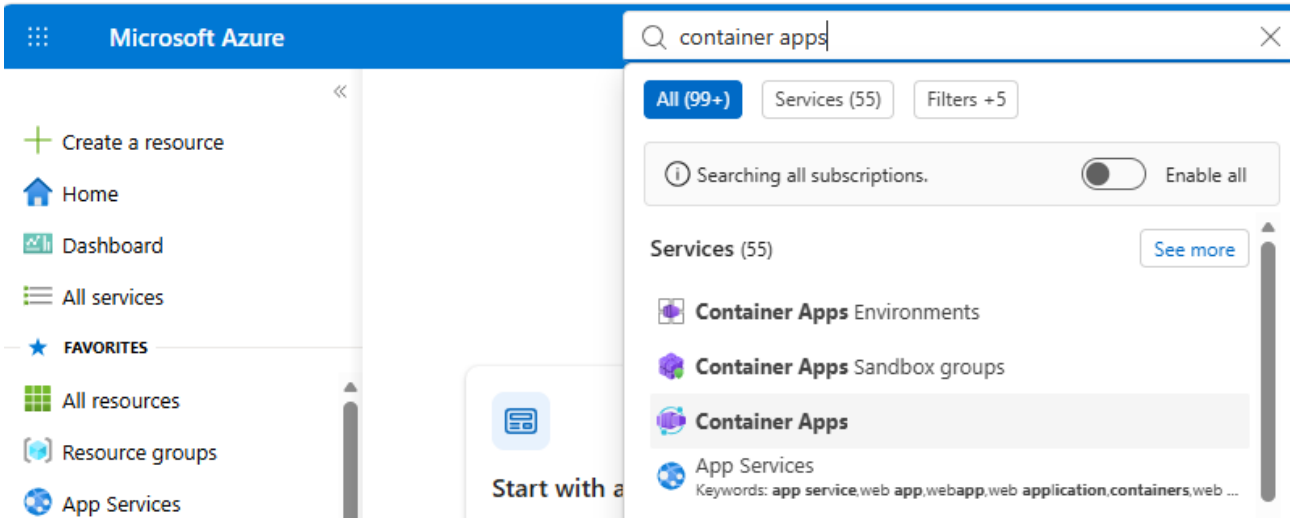


Figure 2: screenshot DockerHub image created

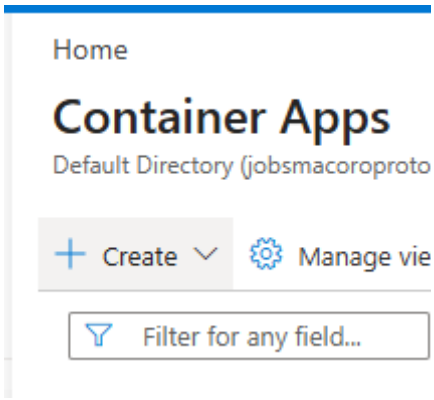
Create Container App in Azure

Before deploying and running our pipeline we need to create in Azure a container in the Container App section as follows:

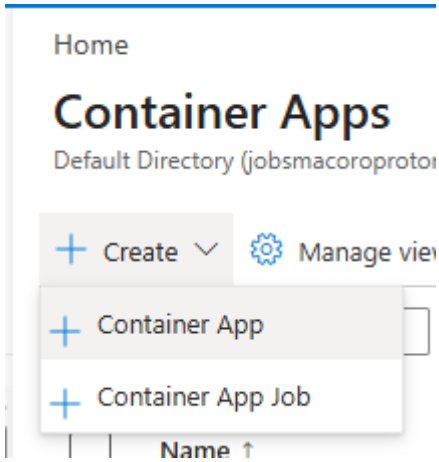
Navigate to Container Apps:



Click on Create:



Select the Container App option:



Fill up the form with the following values:

Azure Subscription = your subscription

Resource Group = SampleDataAccessApi_rg (create a new resource group)

Container app name = sampledataaccessapi

Optimize For Azure Functions = False (Unchecked)

Deployment Source = Container Image

Container Apps Environment:

- Show Environment In All Regions = False (Uncheked)
- Region= France Central (Your Region)
- Container Apps Environment= sampledataaccessapi-env (create a new one)

Create Container App:

- Container Details
  - Name= sampledataaccessapi
  - Image Source = Dockerhub or other registries
  - Image type = public
  - Registry login server = docker.io
  - Image and tag = popdemar/sampledataaccessapi:latest
  - Command override = empty
  - Arguments override = empty

Development Stack Specific Features:

- Development stack = Unspecified

Container Resource Allocation:

- Workload Profile=Consumption-Up to 4 vCPU 8GB memory
- GPU = False (Unchecked)
- CPU and Memory = 0.5 CPU Cores , 1 Gi memory

Environment Variables:

- ConnectionStrings__MoviesDb=secretRefMoviesDb

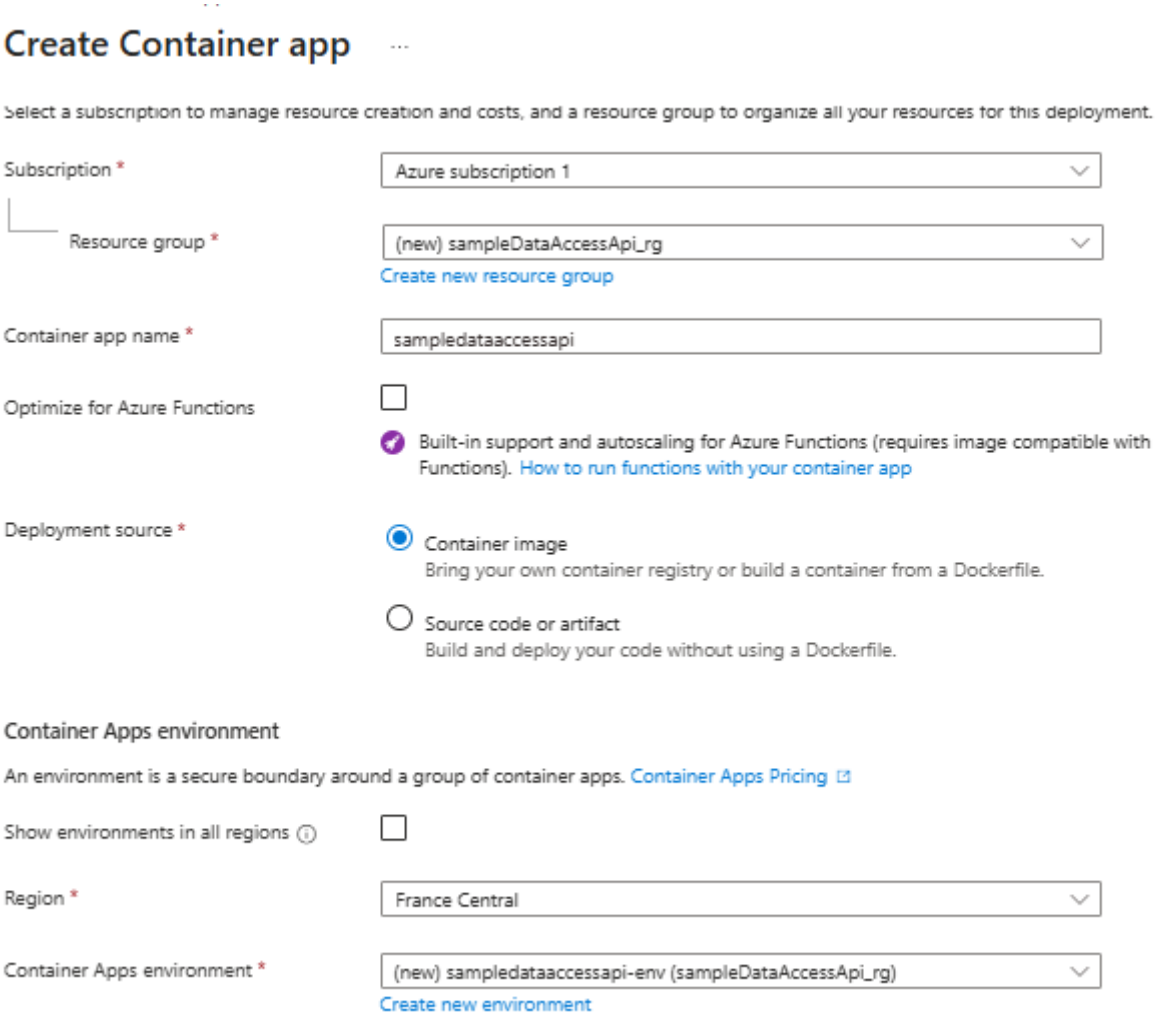


Figure 3: screenshot Create Container App

Notice that we have created:

- new resource = SampleDataAccessApi_rg
- new environment = sampledataaccessapi-env

  For this purpose fill out the form with the following values:

  Environment Name= sampledataaccessapi-env

  Zone Redundancy=disabled

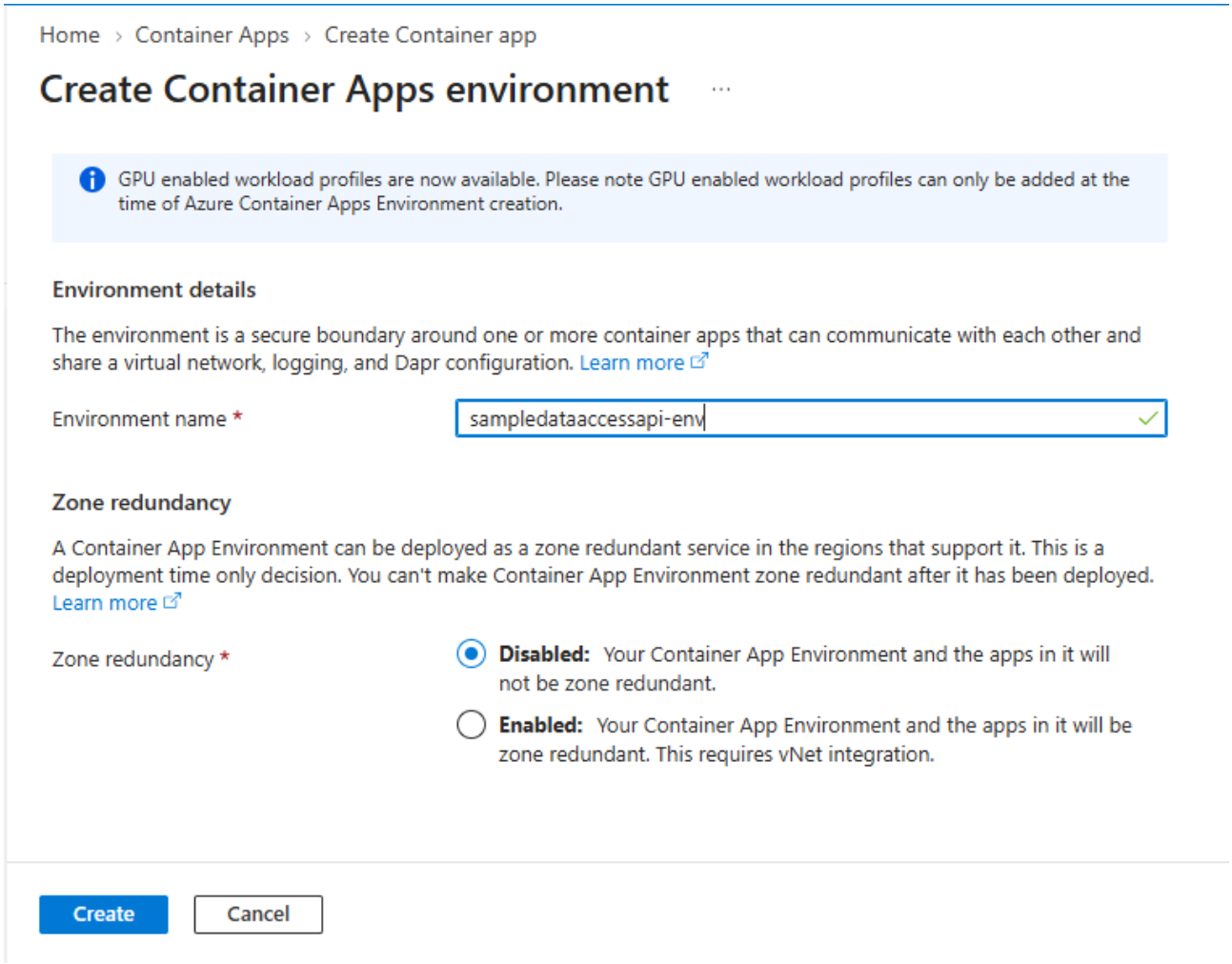


Figure 4: screenshot Createtion of the Container Apps Environment

Next we have to specify from which image and from which repository this container will be created.

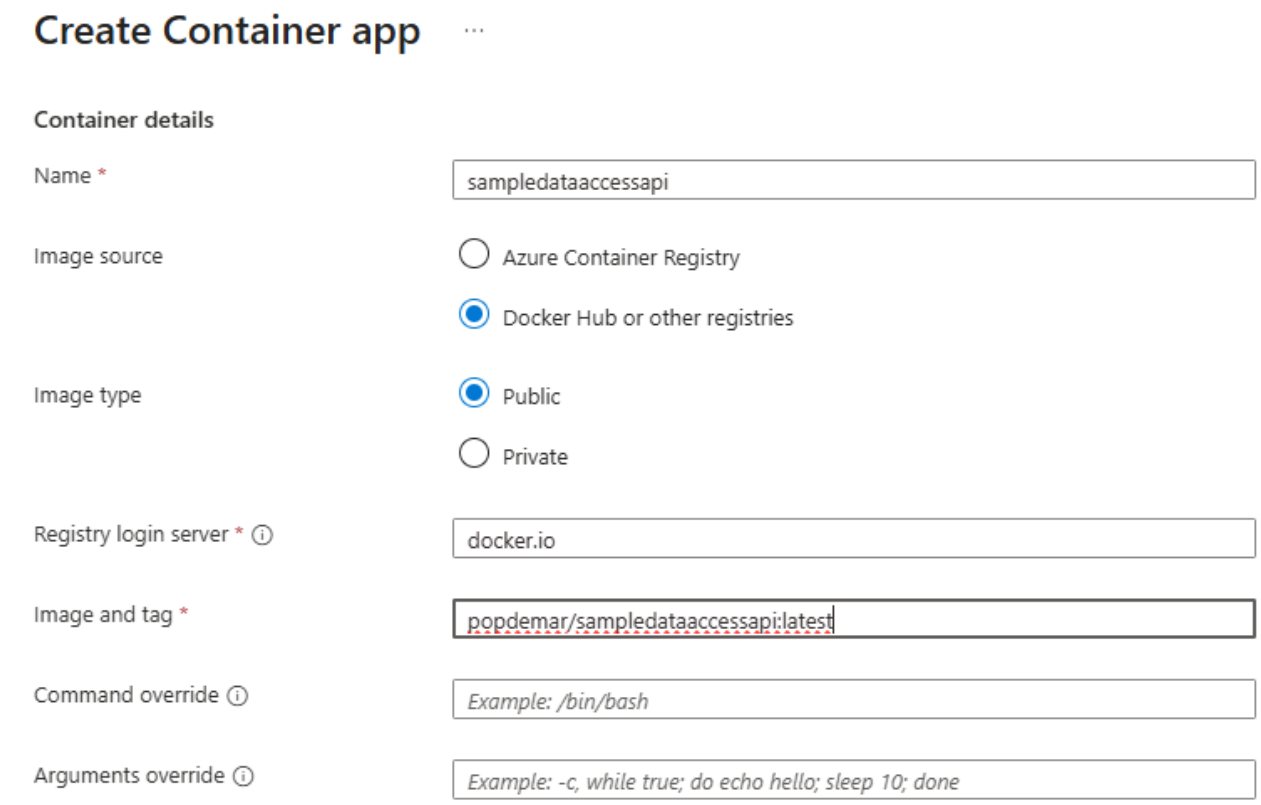


Figure 5: screenshot Create Container App

In our case we are storing our image in Dockerhub and our repositiory is marked in Docker as public.

In a free account Dockerhub provides 1 only repository that can be Private.

As we probably will need more cases like this for the Sample Api Platform, one repository will not be enough, and therefore we opt for a Public.

The Privacy properties in our Docker for our account can be set on the following link:

https://hub.docker.com/repository-settings/default-privacy

And then go to : Settings > Default Privacy..

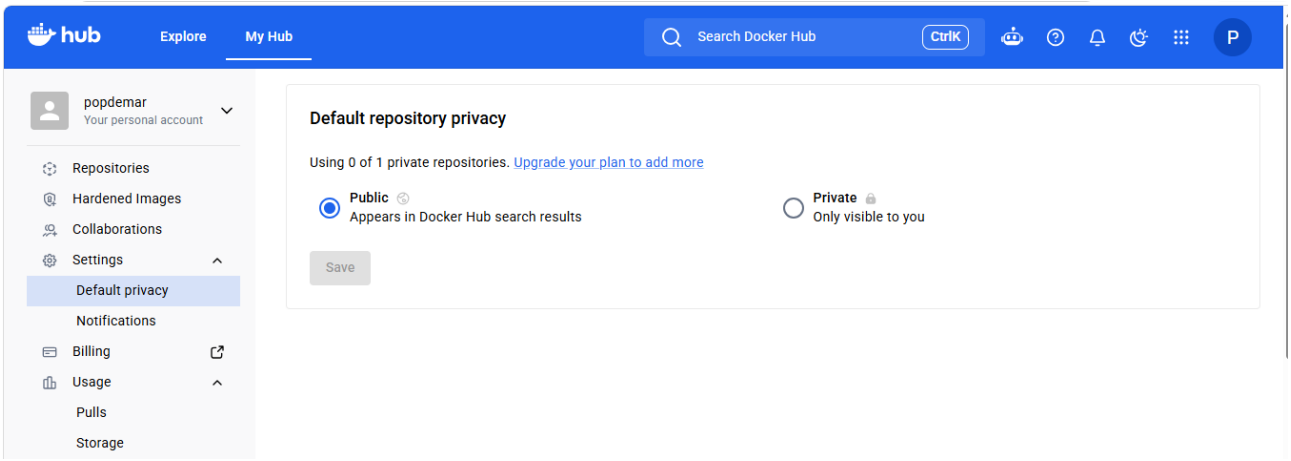


Figure 6: screenshot Dockerhub Repository Privacy = Public

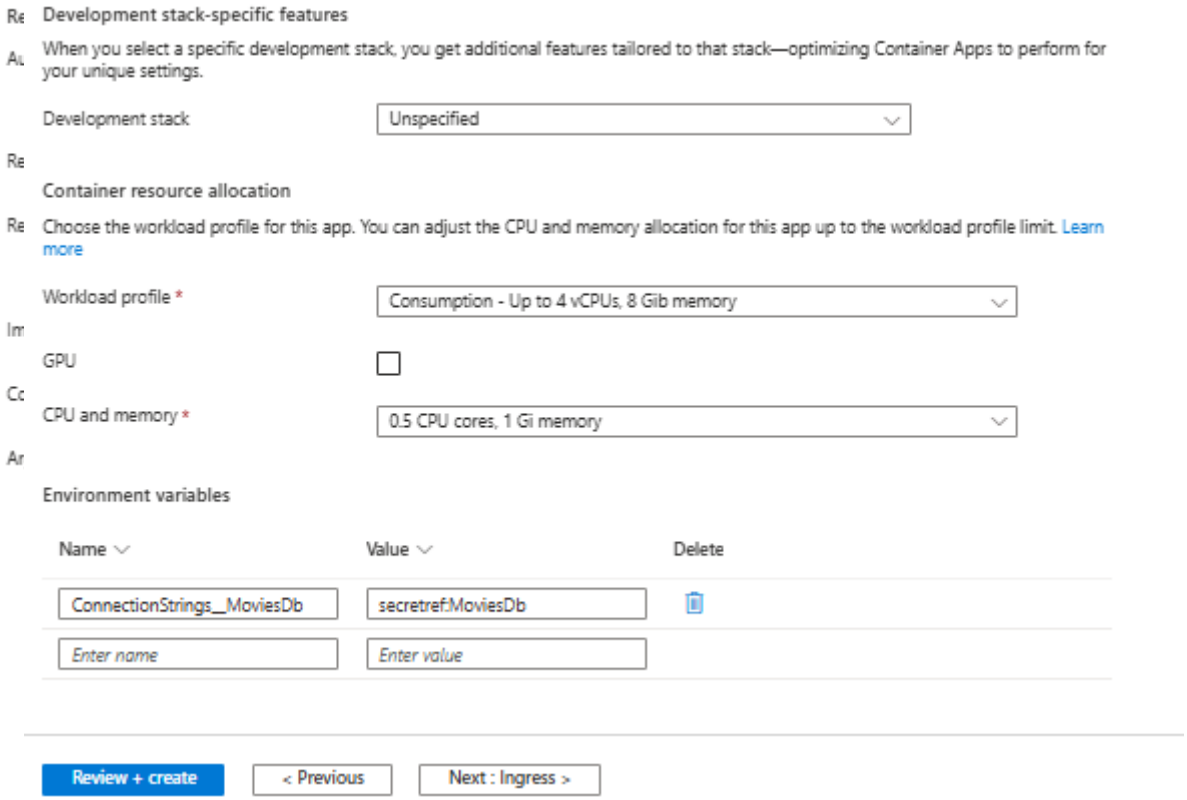


Figure 7: screenshot Container Resource Allocation Properties

The container resource allocation can be left as the default values provided.

Notice a that we provide the name of the connection string to the database as environment variable. For this we have 2 options:

1. Paste the connection string with the password: “ Server=tcp:sampleapiserver.database.windows.net,1433;Initial Catalog=free-sql-

db-5542591;Persist Security Info=False;User ID=sampleapiserveradmin;Password=**{your_password}**;MultipleActiveResultSets=False;Encrypt=True;TrustServerCertificate=False;Connection Timeout=30;".

This is insecure because the password is visible to all who can access this form.

2. To keep it as a secret: for this we need to coordinate this secret with azure devops. So here we only set a reference to the secret:

   ConnectionStrings__MoviesDb = secretref:MoviesDb

   Then in the Azure DevOps go to: Pipeline > Libraries > Variable Group > We create a new variable Group and we will add a new variable called sampleapiDBConnectionstring and the value will be our connectionstring (Server=tcp:sampleapiserver.database.windows.net,1433;Initial Catalog=free-sql-db-5542591;Persist Security Info=False;User ID=sampleapiserveradmin;Password=**{your_password}**;MultipleActiveResultSets=False;Encrypt=True;TrustServerCertificate=False;Connection Timeout=30;) and we will mark it as secret.

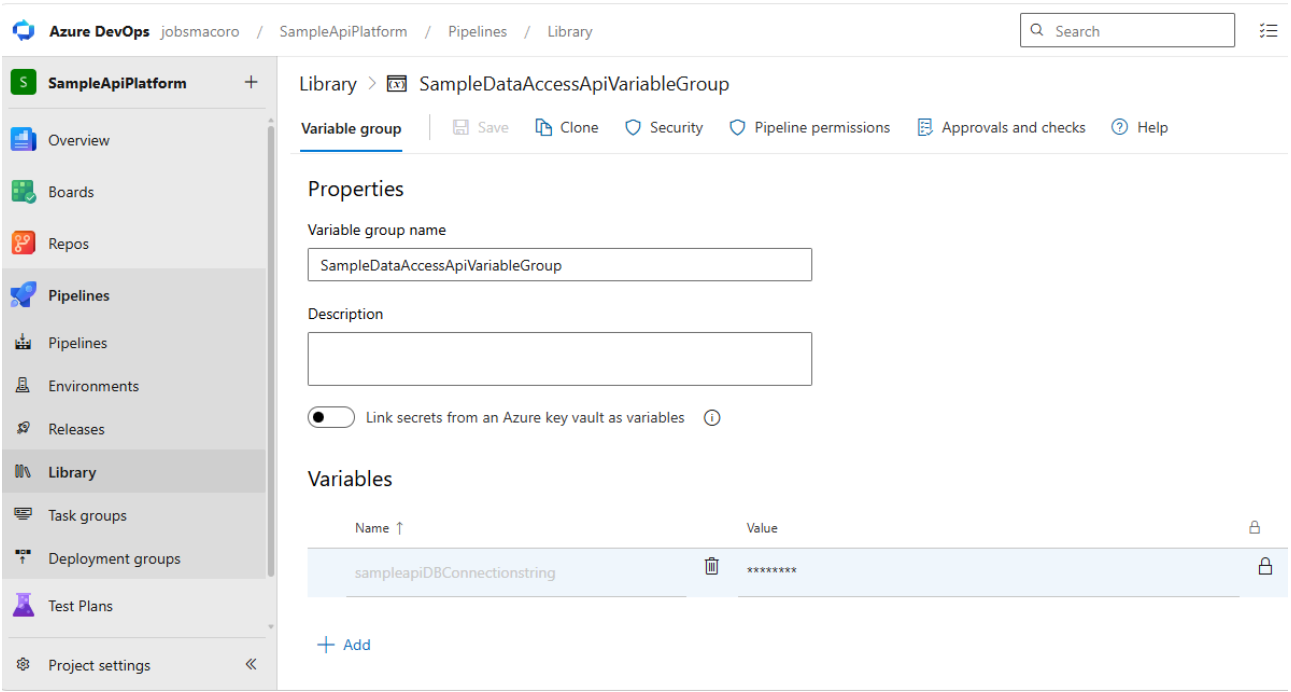


Figure 8: screenshot Azure Devops Environment Variable

And then in the pipeline we inject this secret. We see in the pipeline a reference to this sampleapiDBConnectionstring:

Make sure also that in Program.cs we reference the right connection string:

```
builder.Services.AddDbContext<MoviesDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("MoviesDb"),
        sqlOptions => sqlOptions.EnableRetryOnFailure(
            maxRetryCount: 5,
            maxRetryDelay: TimeSpan.FromSeconds(10),
            errorNumbersToAdd: null
        )
    )
);
```

Figure 9: screenshot Program.cs connection string

And in appsettings.json this value must be empty because they are setup in Azure.

```
// staging: Azure connecti
"ConnectionStrings": {
  "MoviesDb": "",
  //"BillingDb": "Server=t
  //"ProductsDb": "Server=
  //"OrdersDb": "Server=tc
  //"CustomerDb": "Server=

},
```

Figure 10: screenshot appsettings.json connection string has to be empty

After that click on Review + Create > Create

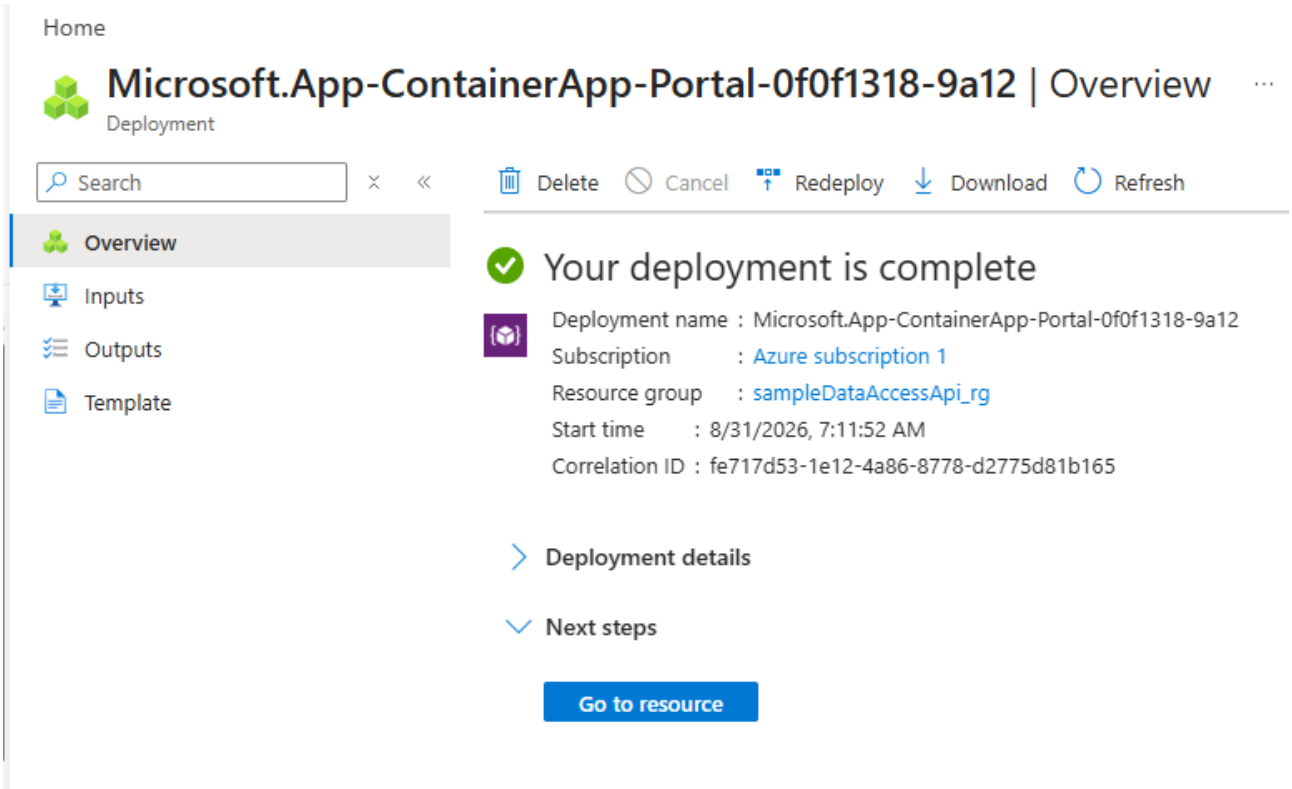


Figure 11: screenshot Container App Created in Azure

Once we receive the confirmation that Azure has created our container, we need to check if the container is running, and if there is a revision active, and if the app is reachable etc.

For this purpose in the Azure Portal go to: **Azure Portal → Container Apps → Deine App → Application → Revisions**

It should show something like this:

- **Active Revision: Yes**
- **Healthy: Yes**
- **Container: Running**
- **Image: popdemar/<repo>:latest**

When all is green then this means that the deployment was Successful.

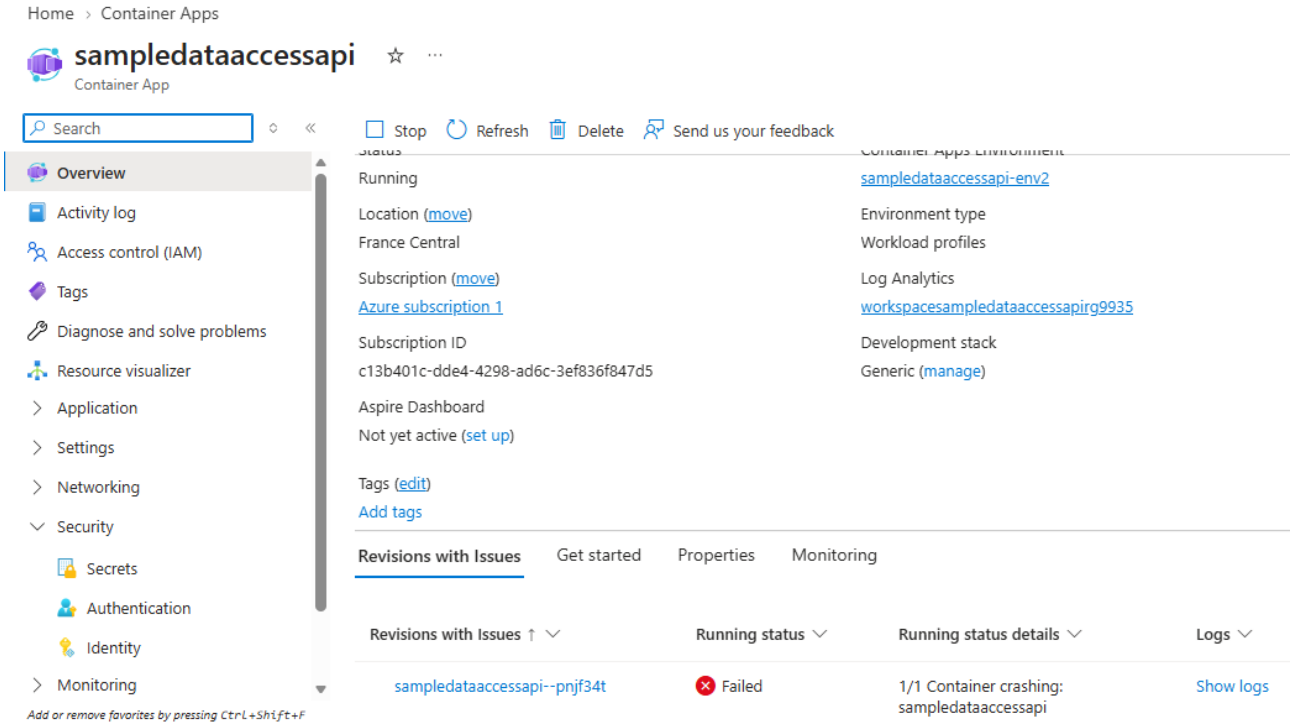


Figure 12: screenshot Container App Created but Deployment Failed

We see that our container is not running.

Troubleshooting The Container App

The quickest way to get the logs is not through the portal but through the Cloud Shell:

```
az containerapp logs show \
  --name sampledataaccessapi \
  --resource-group SampleDataAccessApi_rg \
--follow
```

In this case the error was that program still was reading some empty settings. We commented the code. We pushed everything to the Repo. The pipeline was triggered: a new image was deployed into the Dockerhub. However the container was still failing for the same reason because it is still using the cached image: ACA cached Images pro Revision. When a new Revision id generated the ACA always start the old Revision which still contain the old code and then it throws away the old exception.

The solution is issuing this command to force to get the last image ceated and to start a new revision.

```
az containerapp update \
  --name sampledataaccessapi \
  --resource-group SampleDataAccessApi_rg \
```

--image popdemar/sampledataaccessapi:latest

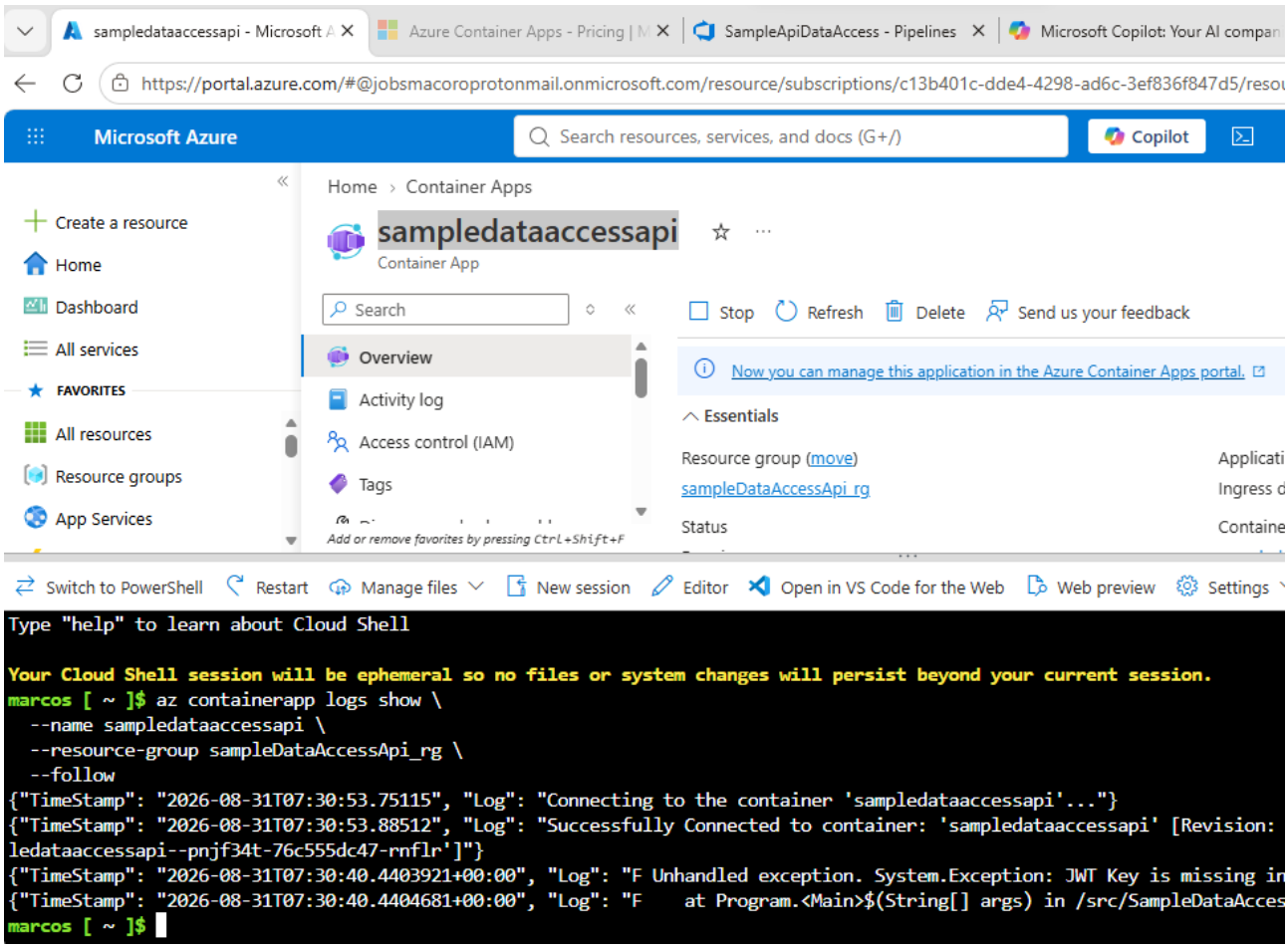


Figure 13: screenshot Azure Cloud Shell

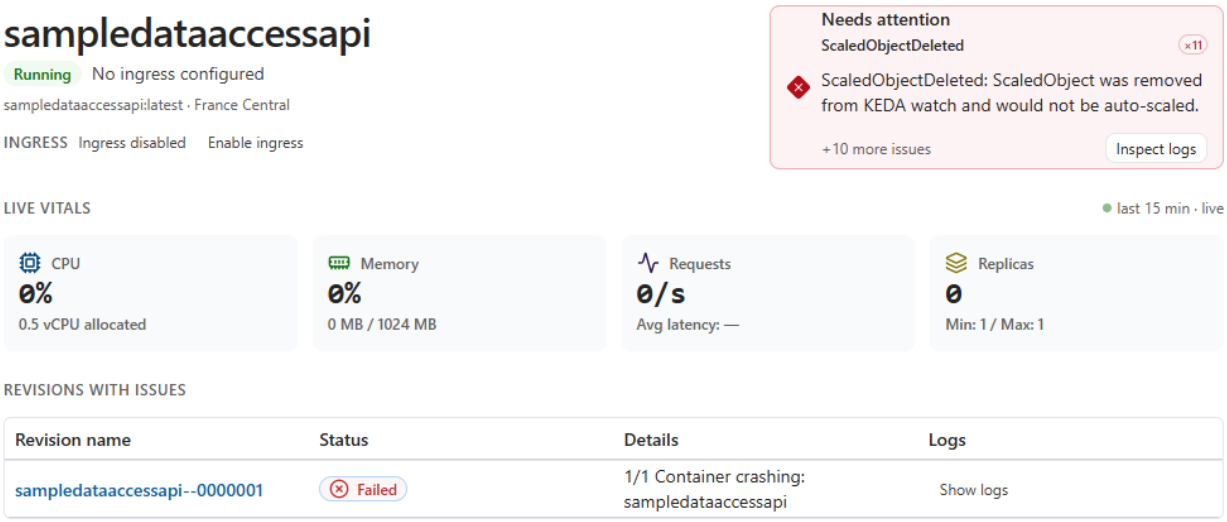


Figure 14: screenshot New Screen To Administer Container Apps In Azure

In this case this command did not fic the issue: we have ran the pipelines but the revision remains the same and therefor we need to kill the revision and to create a new one:

az containerapp revision deactivate \

--name sampledataaccessapi \

--resource-group SampleDataAccessApi_rg \

--revision sampledataaccessapi—0000001

Once the we see the "Deactivate succeeded" then issue the following command:

az containerapp update \

--name sampledataaccessapi \

--resource-group SampleDataAccessApi_rg \

--image popdemar/sampledataaccessapi:latest

At the end we see that the application has been successfully deployed.

What we need to add is the database connection as a secret in this container.

- Server=tcp:sampleapiserver.database.windows.net,1433;Database=free-sql-db-5542591;Authentication=Active Directory Default;Encrypt=True;
- Create a secret in the containeriz
- Reference the secret in the environment variables of the container.

Azure Devops Pipeline Concepts

1) **Task**: a series commands that are encapsulated under the same hood. A pipeline is a list of tasks written in a yaml file. A complete list of the available tasks are available in the azure devops when we edit/create a pipeline:

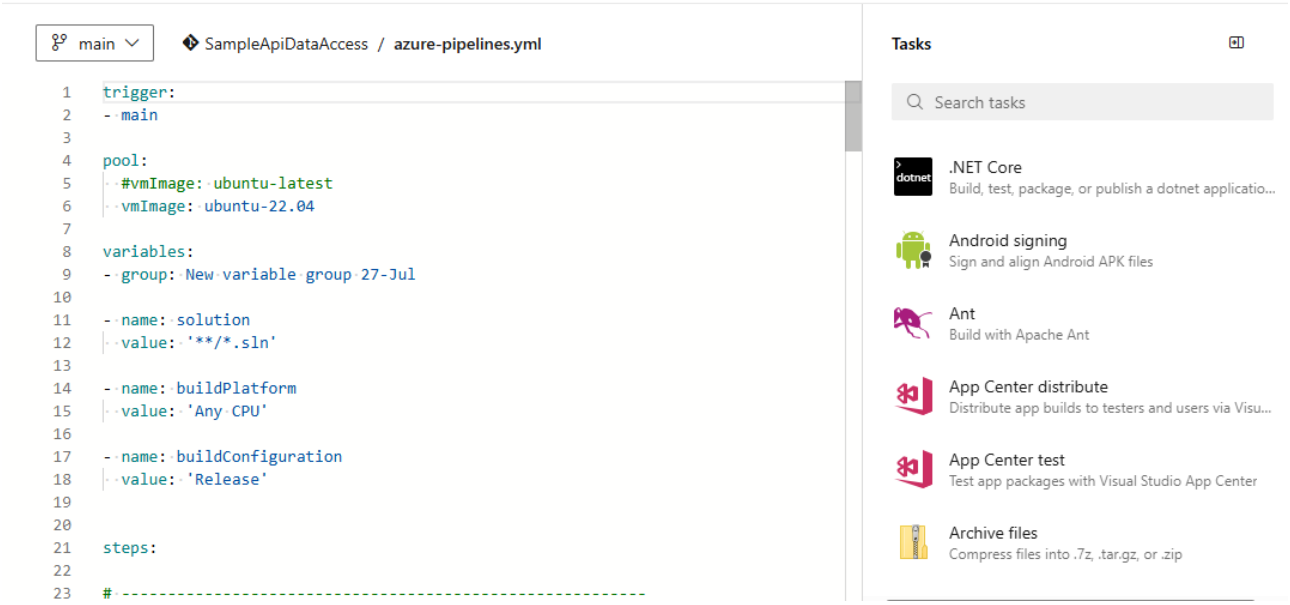


Figure 15: screenshot We can see the tasks on the right hand side

2) **Agent**: it is a machine that executes the tasks described in the pipeline on our behalf: they perform restore, builds, tests, etc on our behalf. Some agents might not have the necessary rights to execute a task (ex: to have root access to a NuGet folder). In these cases we will need to find alternative ways to execute the task.

It is also possible to give directly through commands (**script**) to the agent bypassing the tasks.

3) **Variables**: in a pipeline we can have 2 variable types: **environment variables** or **pipeline variables**. The difference is very subtle. This is important to differentiate for instance if we ask an AI to write a pipeline for us, because if we define the variables in the wrong place the pipeline will fall over.

Pipeline-Variable	`$(VAR)`	`$(DOCKERHUB_TOKEN)`
Bash-Environmentvariable	`$VAR`	`$DOCKERHUB_TOKEN`

Pipeline variables are stored in the variables section when we edit the pipelines.

The environment variables are in Pipelines > Library > Variable group.

4) **Service Connections:** it is necessary to create a service connection whenever a pipeline task or script needs to authenticate with azure or also when we are using the azure CLI. Then in the task or script we specify which Service Connection to use in order to login. For instance, and sometimes it is not obvious:

- **task**: AzureCLI@2
  displayName: Update ACA image
  inputs:

```
    azureSubscription: 'SampleDataAccessApi_ServiceConnection'
    scriptType: bash
    scriptLocation: inlineScript
    inlineScript: |
      az containerapp update \
        --name sampledataaccessapi \
        --resource-group SampleDataAccessApi_rg \
        --image docker.io/$(DOCKERHUB_USERNAME)/sampledataaccessapi:latest
```

In other occasions the Service Connection is not explicit declared in the pipepline but is implicit defined in the task and therefore we need to create a service connection for this pipleine when we use such a task. For instance to load an image from Dockerhub, we need to service connection to connect from docker to azure:

```
- task: Docker@2
  displayName: Build Docker image
  inputs:
    command: build
    Dockerfile: Dockerfile
    repository: $(DOCKERHUB_USERNAME)/sampledataaccessapi
    tags: |
      latest
```

Create a pipeline that deploys the code into the Container Application.

At this stage we have our Image in Dockerhub and an Container App to pull the image from this Image. However there are a lot of steps missing: Restore the Code, Build, Running Tests, Entity Framework Migrations. So we need to update or even to make a new pipeline that covers all these functions.

```
trigger:
- main

pool:
  vmImage: ubuntu-latest

variables:

- name: solution
  value: '**/*.sln'

- name: buildPlatform
  value: 'Any CPU'

- name: buildConfiguration
  value: 'Release'

#this variable is necessary to allow the agent to give access to the agent to the NuGet packages.
- name: NUGET_PACKAGES
  value: $(Agent.TempDirectory)/.nuget/packages
```

```
steps:

# ----------------------------------------------------------
# Restore
# ----------------------------------------------------------
- task: DotNetCoreCLI@2
  displayName: Restore packages
  inputs:
    command: restore
    projects: '$(solution)'

# ----------------------------------------------------------
# Build
# ----------------------------------------------------------
- task: DotNetCoreCLI@2
  displayName: Build solution
  inputs:
    command: build
    projects: '$(solution)'
    arguments: '--configuration $(buildConfiguration)'

# ----------------------------------------------------------
# Install EF Toolsgit add
# ----------------------------------------------------------
- script: |
    dotnet tool install --global dotnet-ef
  displayName: Install EF Tools

# ----------------------------------------------------------
# Build EF migration bundle
# ----------------------------------------------------------
- script: |
    cd SampleDataAccessApi
    dotnet ef migrations bundle --output ../efbundle
  displayName: Build EF migration bundle

# ----------------------------------------------------------
# Make bundle executable
# ----------------------------------------------------------
- script: |
    chmod +x efbundle
  displayName: Make EF bundle executable

# ----------------------------------------------------------
# Run EF migration bundle
# ----------------------------------------------------------
- script: |
    ./efbundle --connection "$(sampleapiDBConnectionstring)"
  displayName: Run EF migration bundle

# ----------------------------------------------------------
```

```
# Test + Coverage
# ----------------------------------------------------------
- script: |
    dotnet test SampleDataAccessApi.sln --configuration Release --collect:"XPlat Code Coverage"
  displayName: Run tests with coverage

# ----------------------------------------------------------
# Publish coverage
# ----------------------------------------------------------
- task: PublishCodeCoverageResults@1
  displayName: Publish coverage
  inputs:
    codeCoverageTool: Cobertura
    summaryFileLocation: '$(Agent.TempDirectory)/**/coverage.cobertura.xml'
    reportDirectory: '$(Agent.TempDirectory)'

# ----------------------------------------------------------
# Build Docker image
# ----------------------------------------------------------
- task: Docker@2
  displayName: Build Docker image
  inputs:
    command: build
    Dockerfile: Dockerfile
    repository: $(DOCKERHUB_USERNAME)/sampledataaccessapi
    tags: |
      latest

- script: |
    echo "$(DOCKERHUB_TOKEN)" | docker login -u "$(DOCKERHUB_USERNAME)" --
password-stdin
  displayName: Docker Hub Login

# ----------------------------------------------------------
# Push Docker image
# ----------------------------------------------------------
- script: |
    docker push $(DOCKERHUB_USERNAME)/sampledataaccessapi:latest
  displayName: Push Docker image

# ----------------------------------------------------------
# Update ACA image
# ----------------------------------------------------------
- task: AzureCLI@2
  displayName: Update ACA image
  inputs:
    azureSubscription: 'SampleDataAccessApi_ServiceConnection'
    scriptType: bash
    scriptLocation: inlineScript
    inlineScript: |
      az containerapp update \
        --name sampledataaccessapi \
```

--resource-group SampleDataAccessApi_rg \
--image docker.io/$(DOCKERHUB_USERNAME)/sampledataaccessapi:latest

With the following pipeline variables: DOCKERHUB_TOKEN, DOCKERHUB_USERNAME, sampleapiDBConnectionstring.

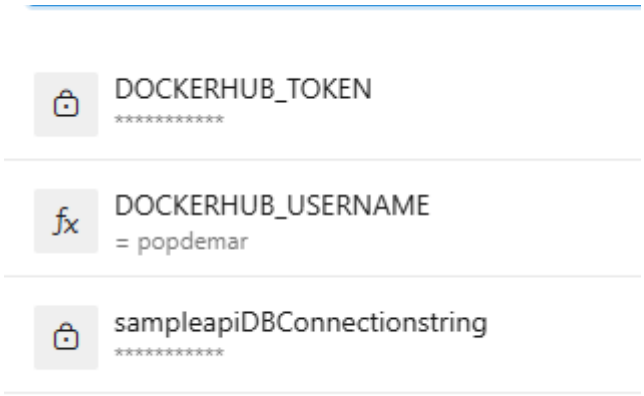


Figure 16: screenshot pipeline variables

Service Connections

In this pipeline we are using Azure-CLI and therfore we need ARM-Service-Connection.

AzureDevOps > Project Settings > Pipelines > Service Connections > New > ARM (Azure Resource Manager) :

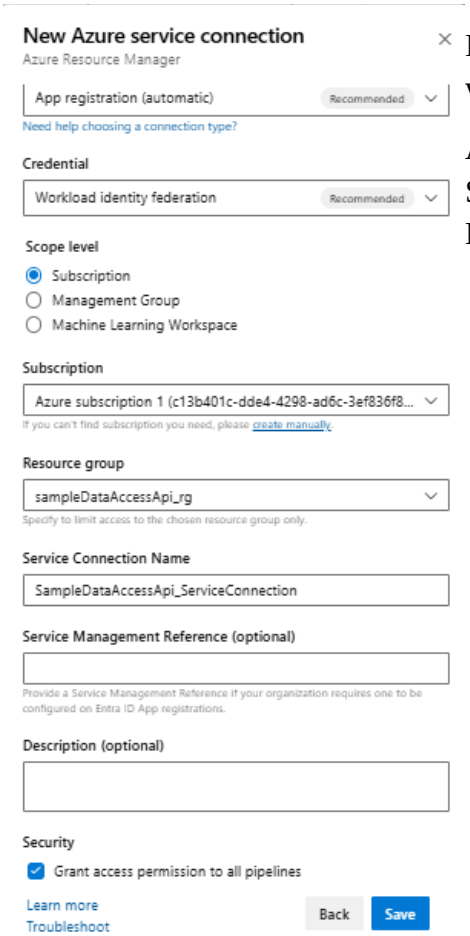


Figure 17: screenshot ARM Service Connection

We need also for this specific pipeline also a docker service connection , type: Docker registry:



Figure 18: screenshot Docker Service Connection Type

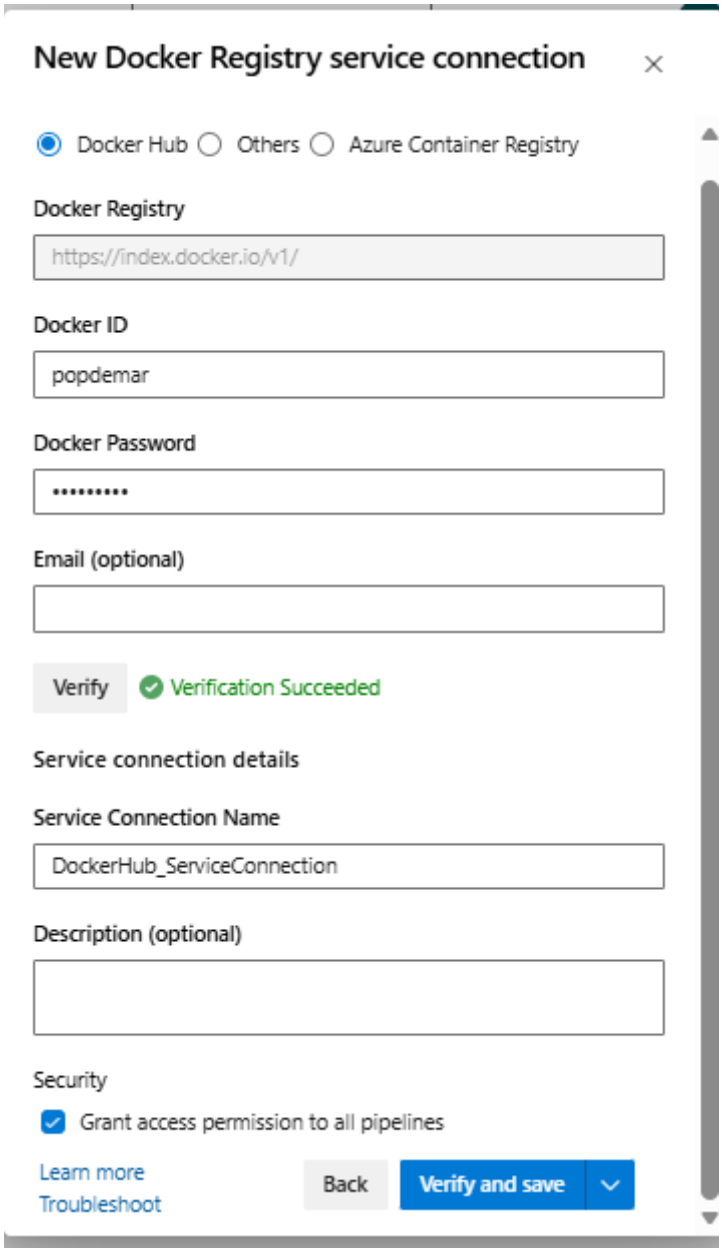
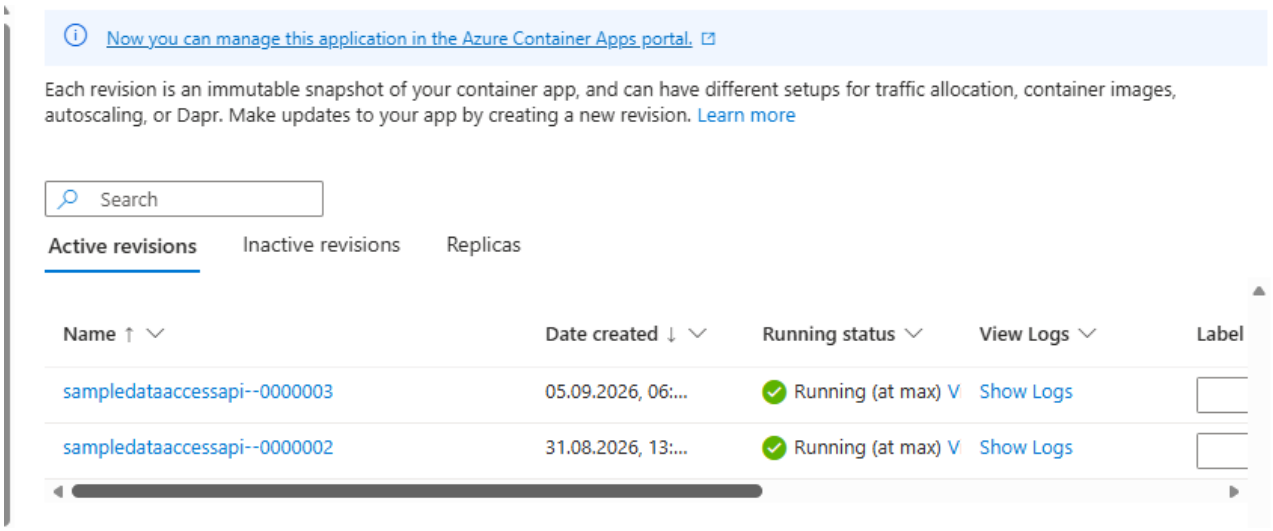


Figure 19: screenshot Inputs For A Docker Service Connection

After all this setup when the pipeline runs a new revsion will be created for our pipeline.



The revision 03 was created.

Creating A Managed Identity

Among the Identity managed by Azure, Managed Identity and User Identity, this is the one that gives more flexibility in terms that is not a 1:1 relation between containers but a 1:n, so a we can have several containers that use the same identity to access a container.

An Identity is an object in itself in Azure. We create a User Identity:

Type in the search box of the Azure Portal: Managed Identies:

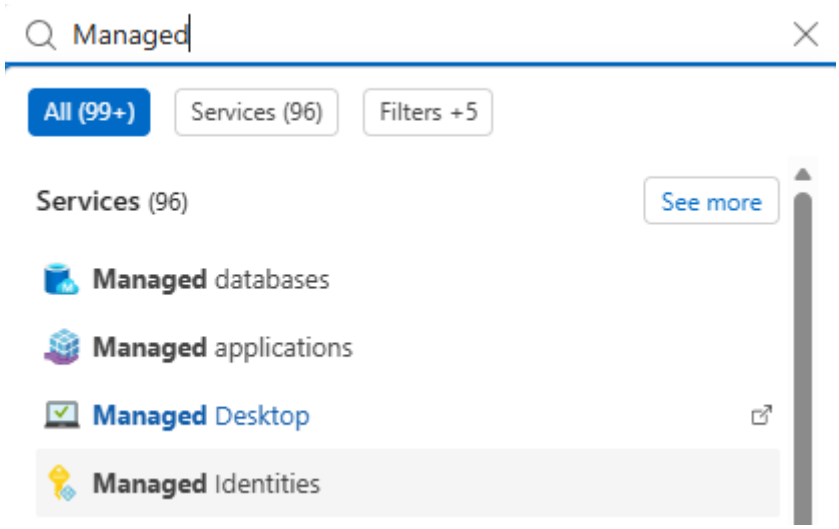


Figure 20: screenshot Managed Identities Search

An screen with the current existing identites will be presented:

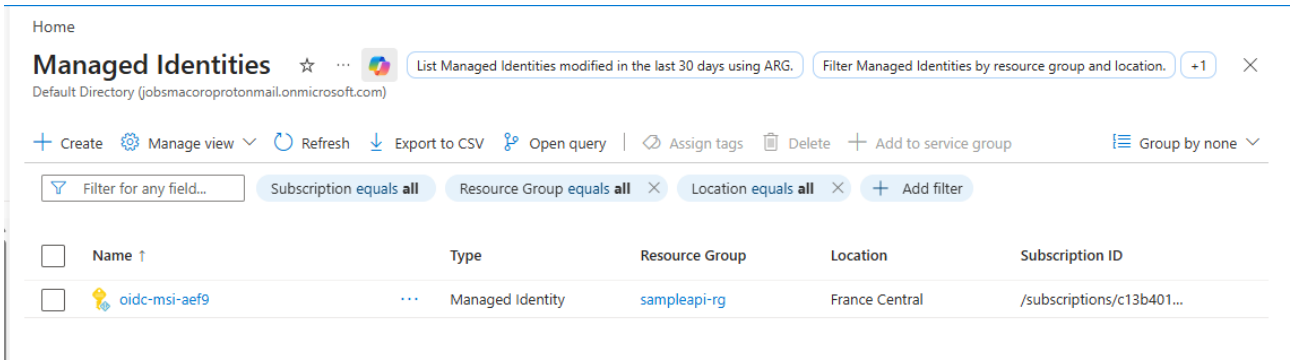


Figure 21: screenshot Current Identities List

Click on the Create button. And the following screen will be presented:

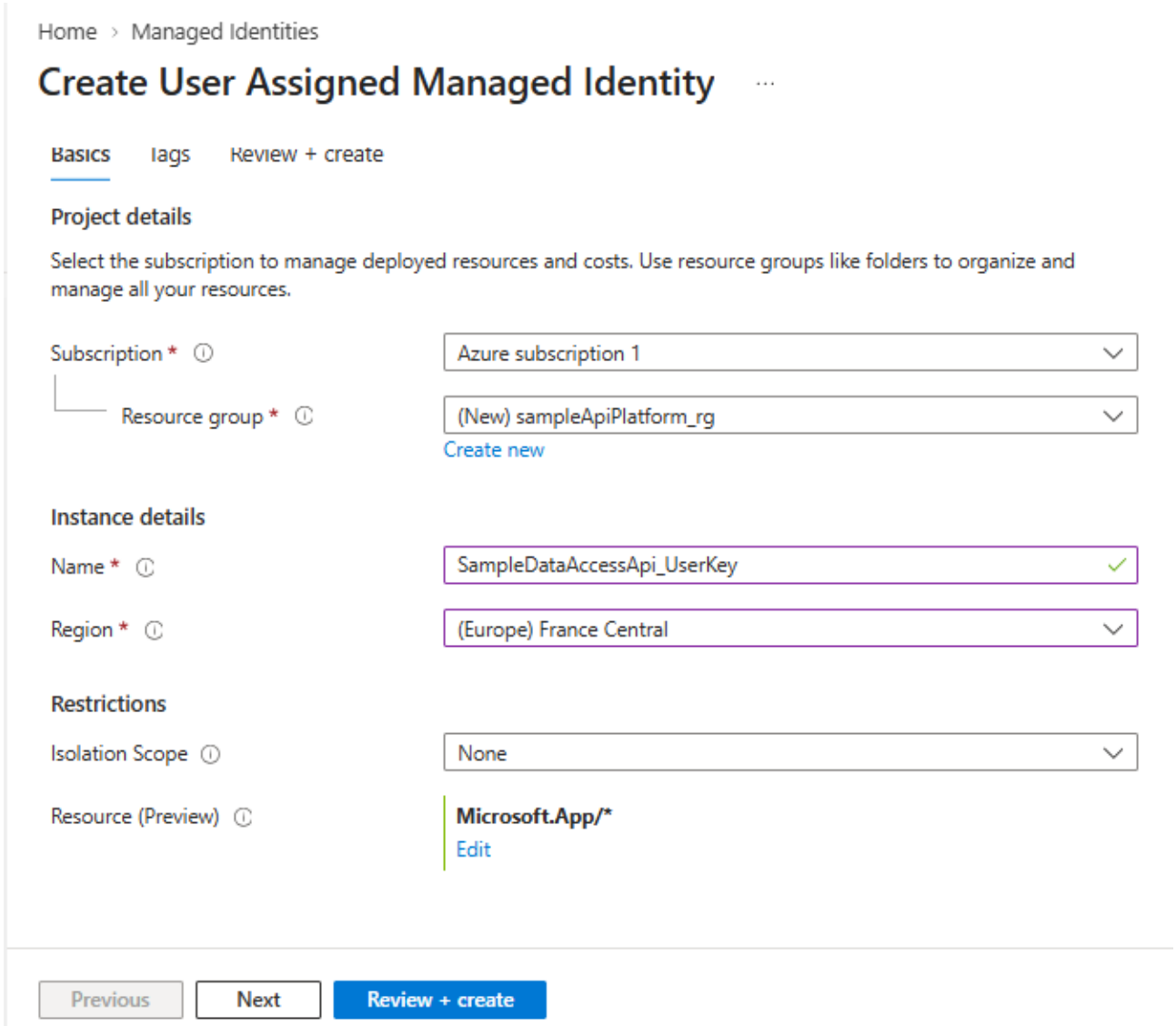


Figure 22: Create A User Defined Identity

Resource group = sampleApiPlatform_rg

Name = SampleDataAccessApi_UserKey

Region = France

Restrictions

> Isolation Scope= **None** (it can be used in any region of the subscription) / Regional (can only be used by resources in the same region in the subscription). None is more flexible. If tomorrow I need to define another resource but I cannot do it in the same region then this can give some issues.
>
> Resource (Preview) = Which resource types the identity is allowed to manage = **Microsoft.App** (we want this User Defined Identity to control the access to the container, so only containers can use this identity)

Now we go to the Container App and we will see this User Defined Identity in the list of available identities that we can use:

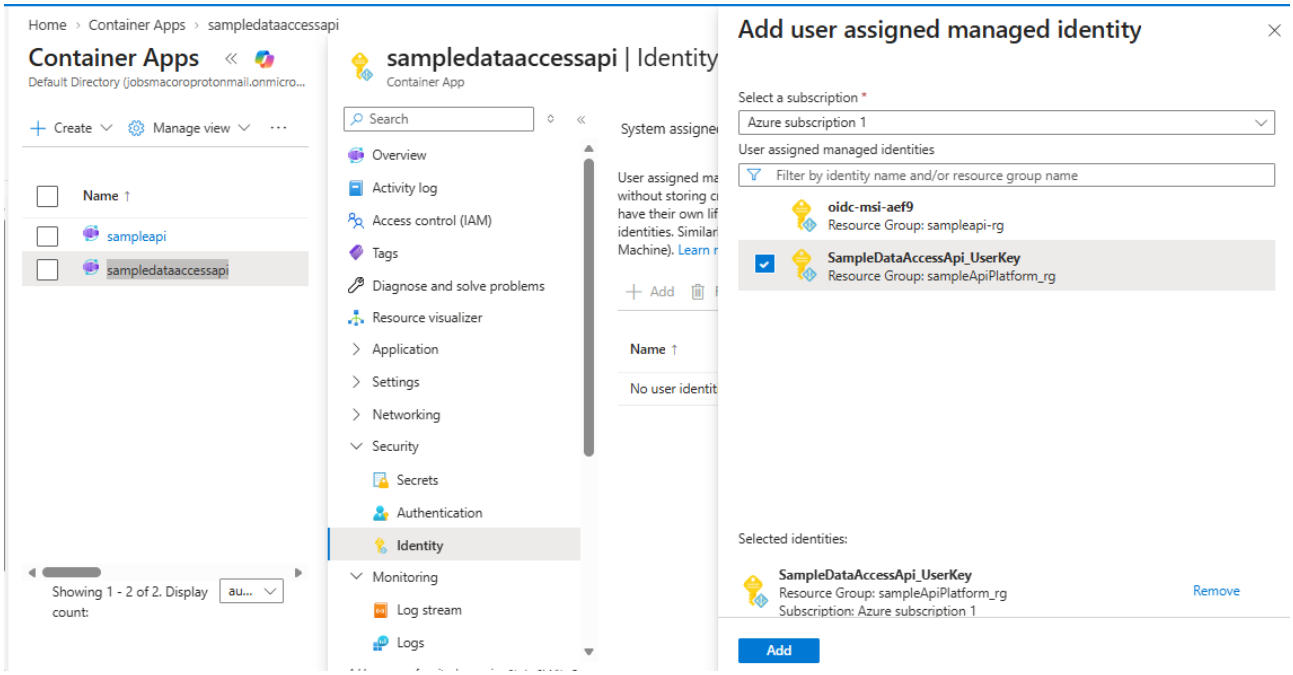


Figure 23: Adding The User Defined Identity To The Container App

After clicking on the Add button:

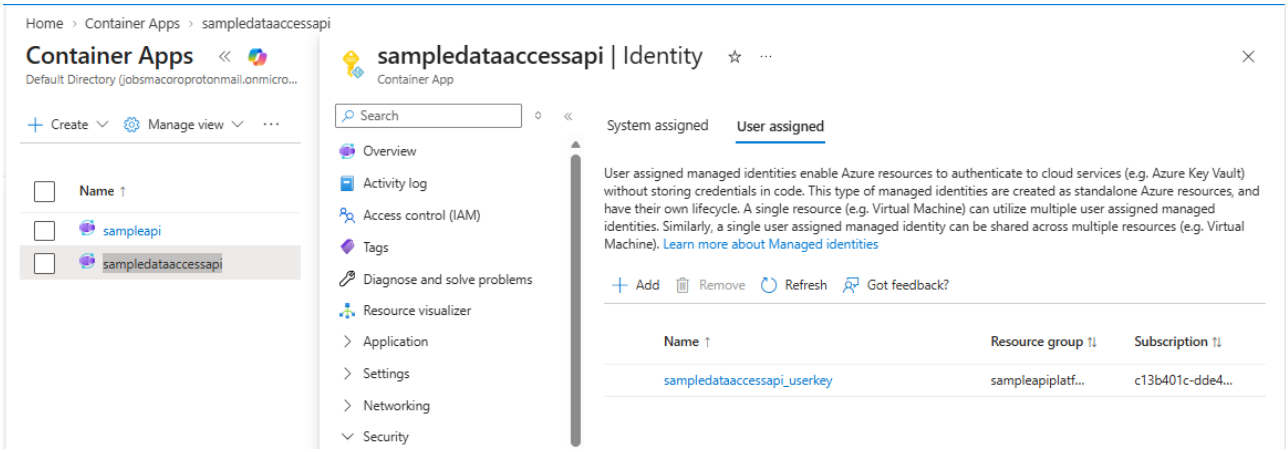
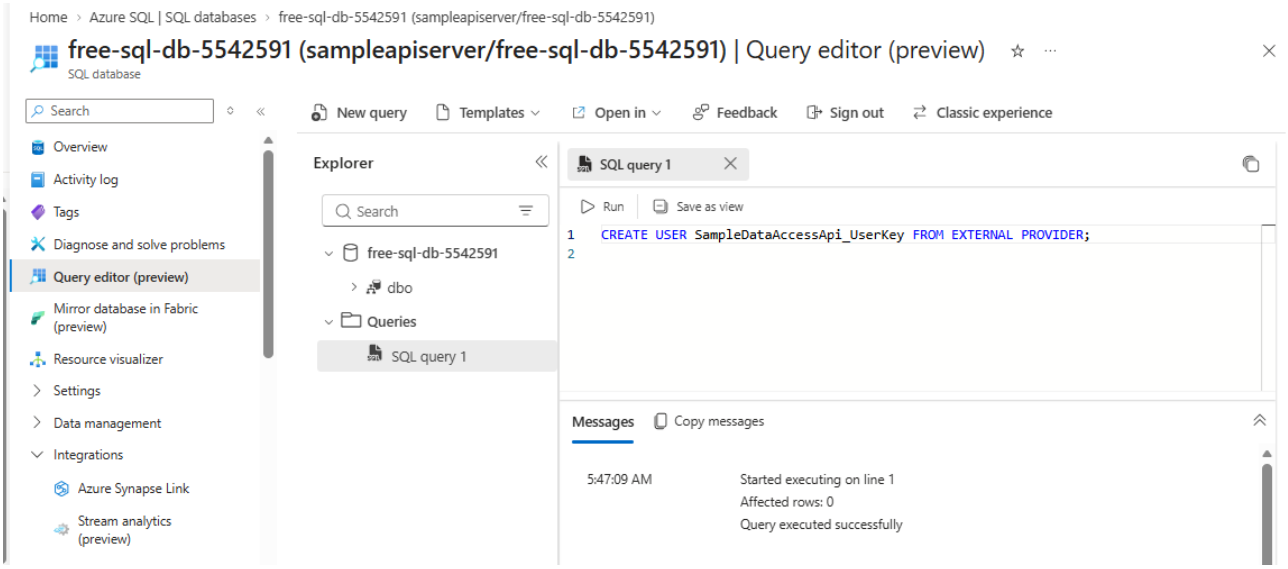


Figure 24: User Defined Identity Added To The Container App

Now we need to add permissions to this User Identity: for instance to give access to our Azure SQL Database.

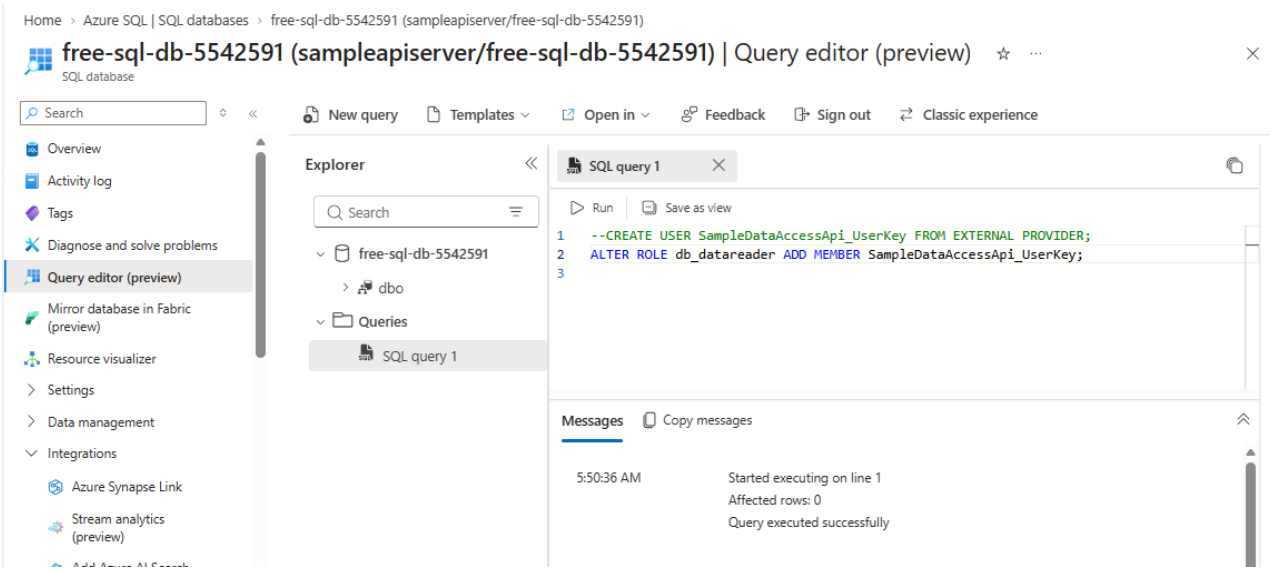
We need to login into our Azure Database with an Entra ID admin credential (in our case as jobs.macoro), not as a database admin (sampleapiserveradmin), and add our User Defined Identity , SampleDataAccessApi_UserKey, so that the database recognizes as a user and can login without a password (behind the scenes Azure will exchange a token between the database and the container).

CREATE USER `SampleDataAccessApi_UserKey` FROM EXTERNAL PROVIDER;

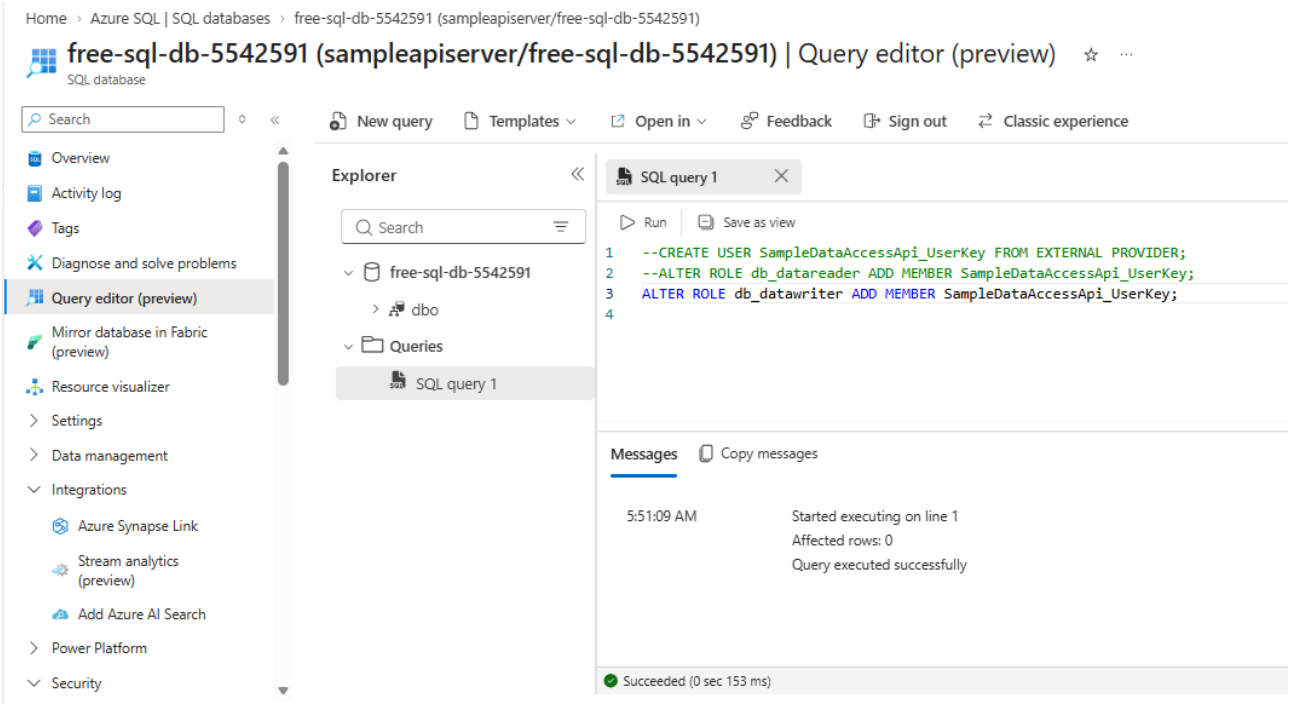


and that allows to login without password and to performed the operations.

ALTER ROLE db_datareader ADD MEMBER `SampleDataAccessApi_UserKey`;



ALTER ROLE db_datawriter ADD MEMBER `SampleDataAccessApi_UserKey`;



The SampleApi Application

Create an image in Dockerhub

azure-pipeline.yml will create an image in dockerhub.

trigger:
- main

pool:

```
  vmImage: 'ubuntu-latest'

steps:
- checkout: self

- script: |
    export DOCKER_BUILDKIT=1
  displayName: Enable BuildKit

- task: DockerInstaller@0
  displayName: Install Docker

- script: |
    docker build -t popdemar/sampledataaccessapi:latest .
  displayName: Build Docker Image

- script: |
    echo $(DOCKERHUB_TOKEN) | docker login -u popdemar --password-stdin
  displayName: Docker Login

- script: |
    docker push popdemar/sampledataaccessapi:latest
  displayName: Push Image to Docker Hub
```

And this is the **Dockerfile** that we are using (make sure that has been pushed in the repo with the rest of the code):

```
# Build stage
FROM mcr.microsoft.com/dotnet/sdk:8.0 AS build
ARG BUILD_CONFIGURATION=Release

WORKDIR /src

# Copy csproj
COPY SampleDataAccessApi/SampleDataAccessApi.csproj SampleDataAccessApi/

# Restore
RUN dotnet restore "SampleDataAccessApi/SampleDataAccessApi.csproj"

# Copy the rest of the source code
COPY SampleDataAccessApi/ SampleDataAccessApi/

# Build
RUN dotnet build "SampleDataAccessApi/SampleDataAccessApi.csproj" -c
$BUILD_CONFIGURATION -o /app/build
```

```
# Publish
RUN dotnet publish "SampleDataAccessApi/SampleDataAccessApi.csproj" -c
$BUILD_CONFIGURATION -o /app/publish /p:UseAppHost=false

# Runtime stage
FROM mcr.microsoft.com/dotnet/aspnet:8.0 AS final
WORKDIR /app

COPY --from=build /app/publish .

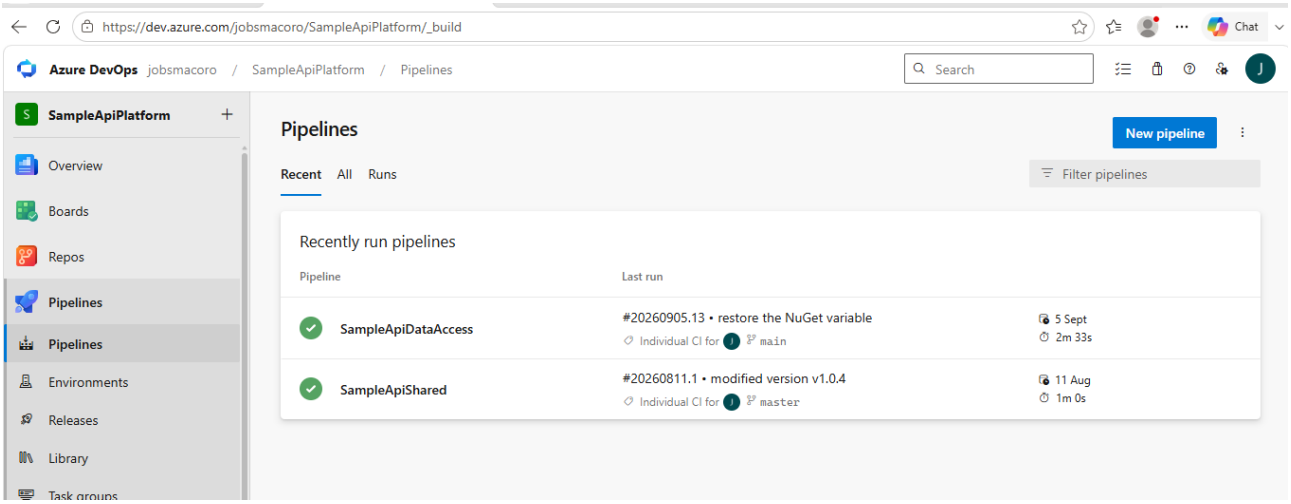
ENV ASPNETCORE_URLS=http://+:8080
EXPOSE 8080

ENTRYPOINT ["dotnet", "SampleDataAccessApi.dll"]
```

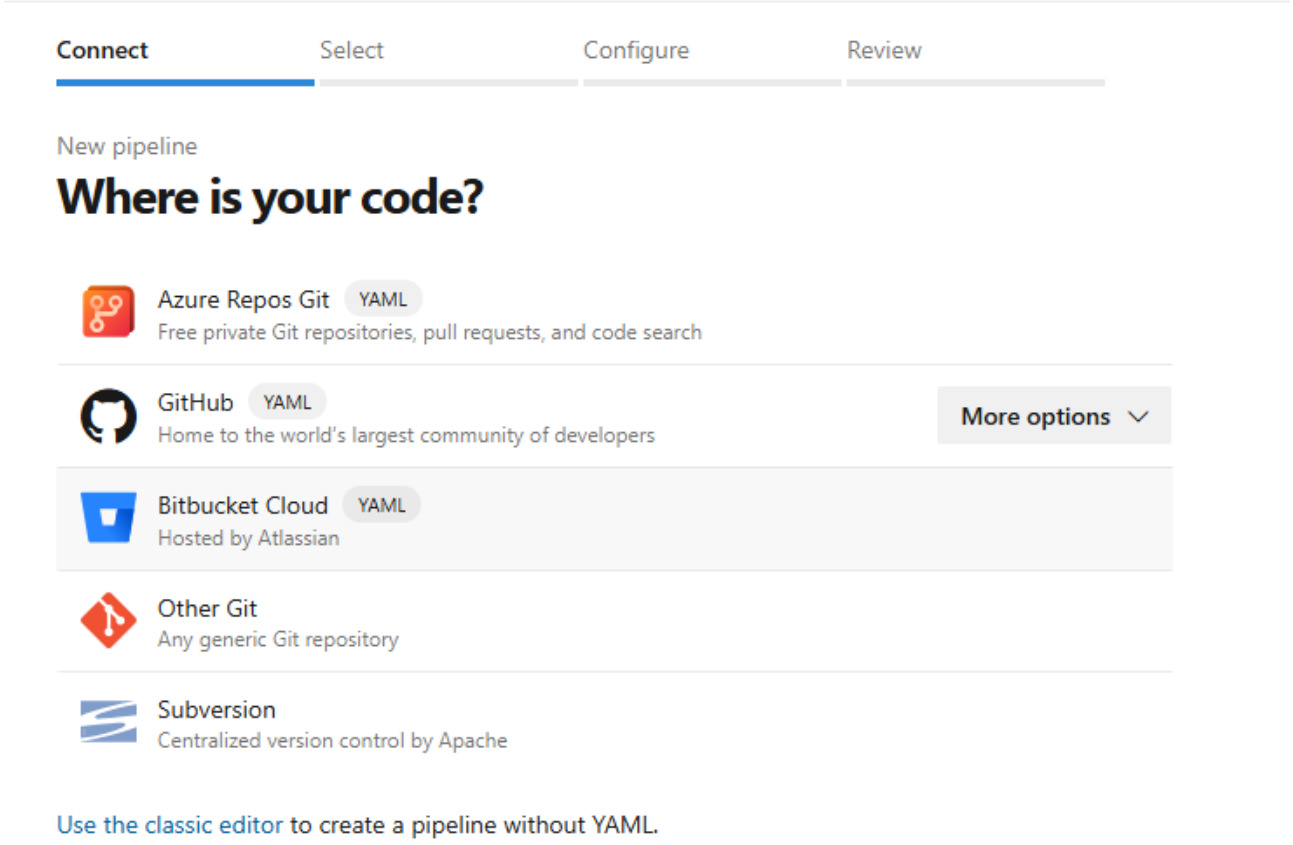
The **image and tag** for simplicity we chose the **latest**.

In this case we need to create the pipeline in Azure Devops:

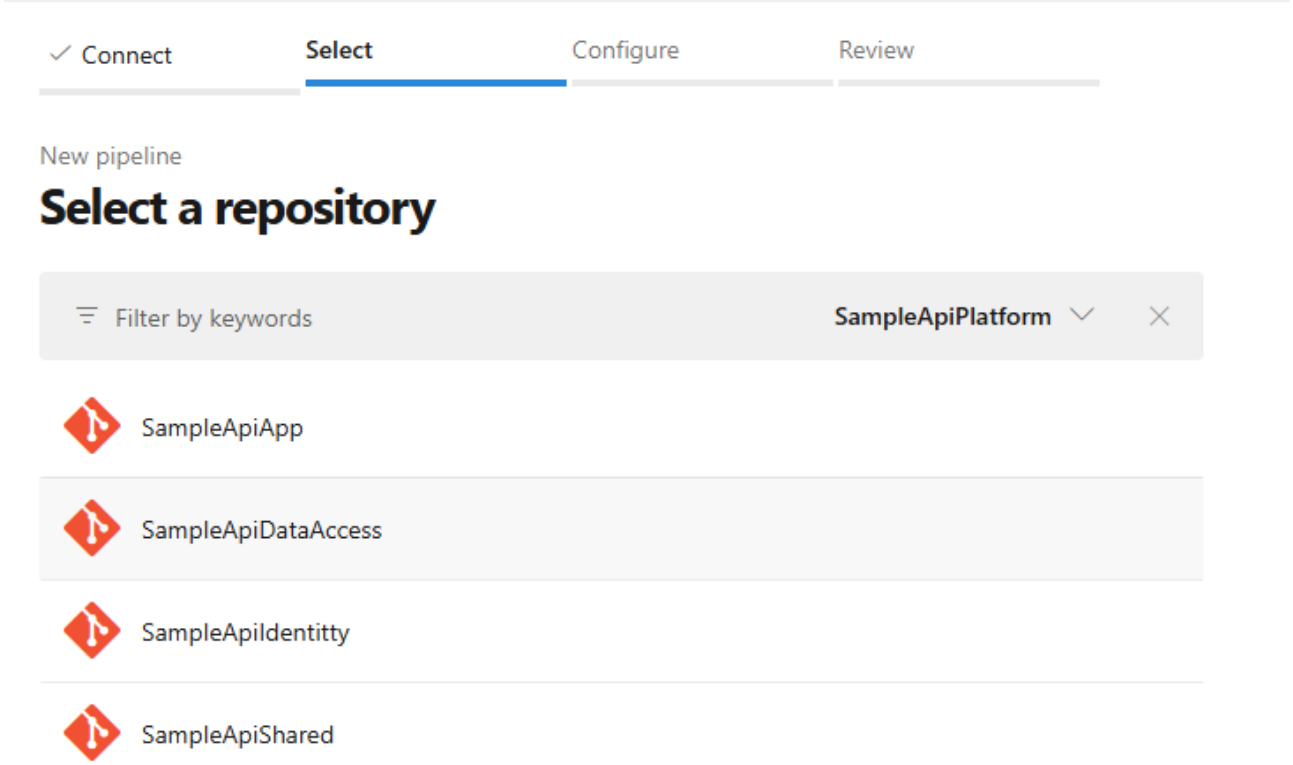
Click on New Pipeline button



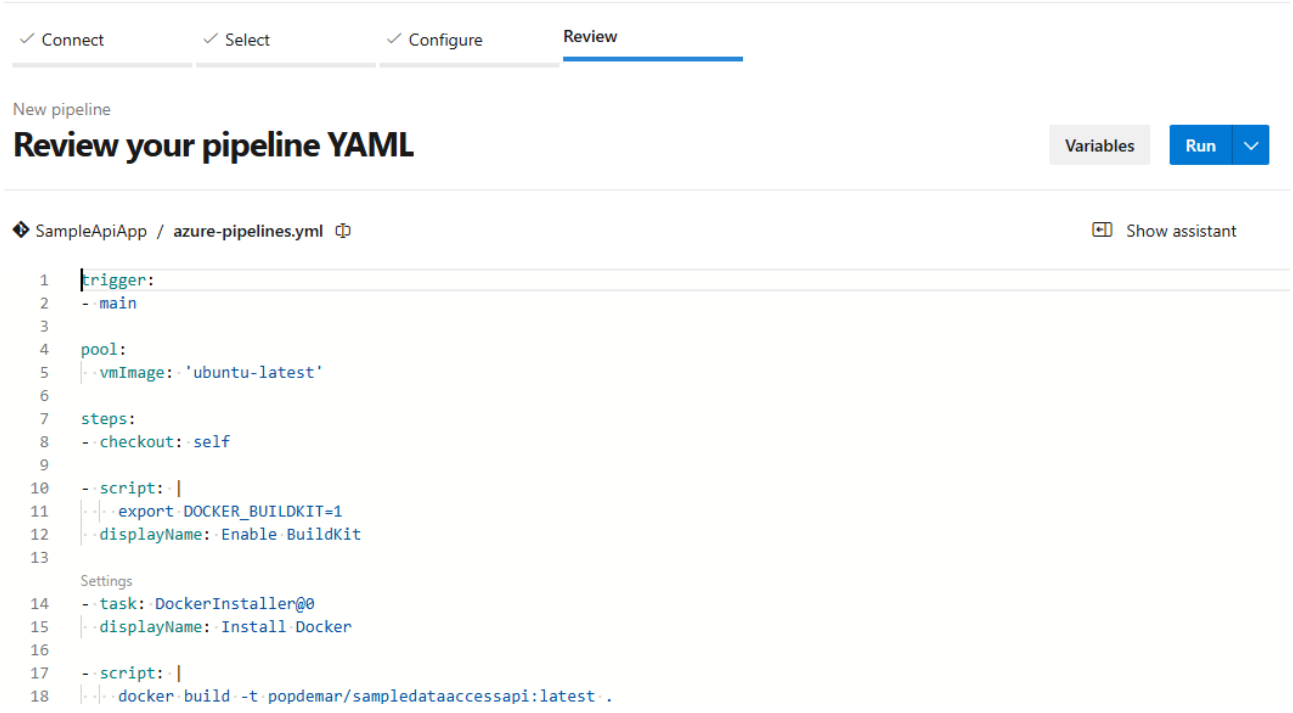
Then select the Azure Repos Git



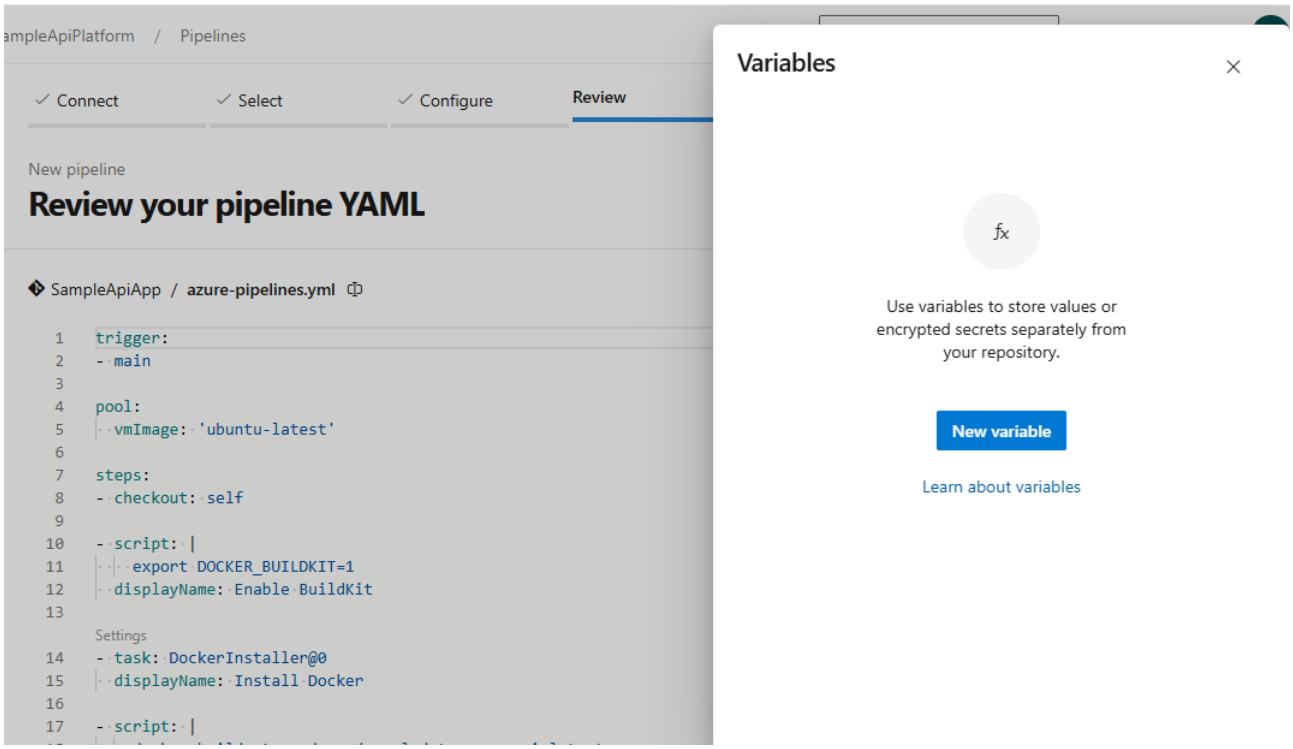
Select the SampleAppApi Repositiory



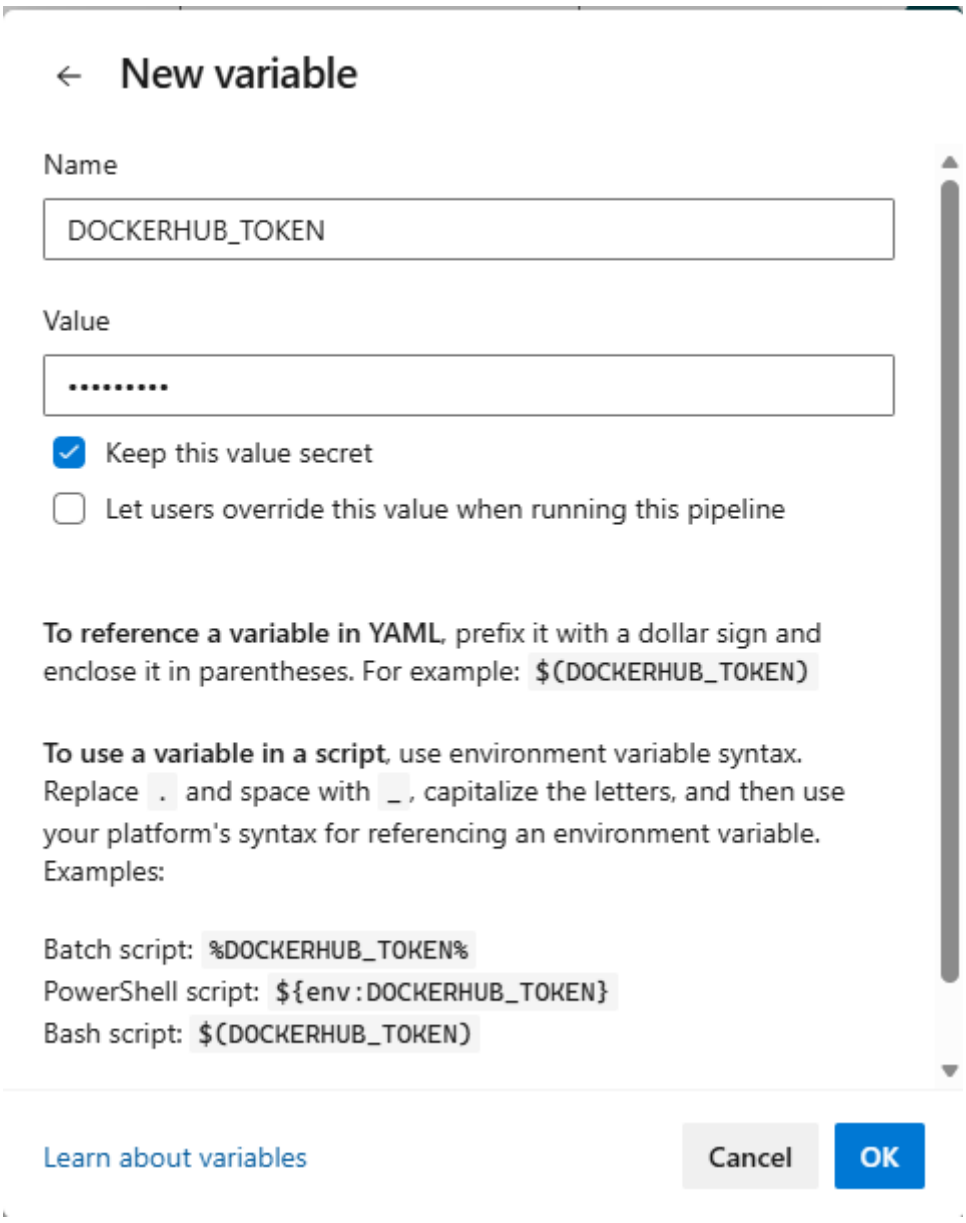
And then automatically it will load the pipeline that we have just created to create an image in Dockerhub.



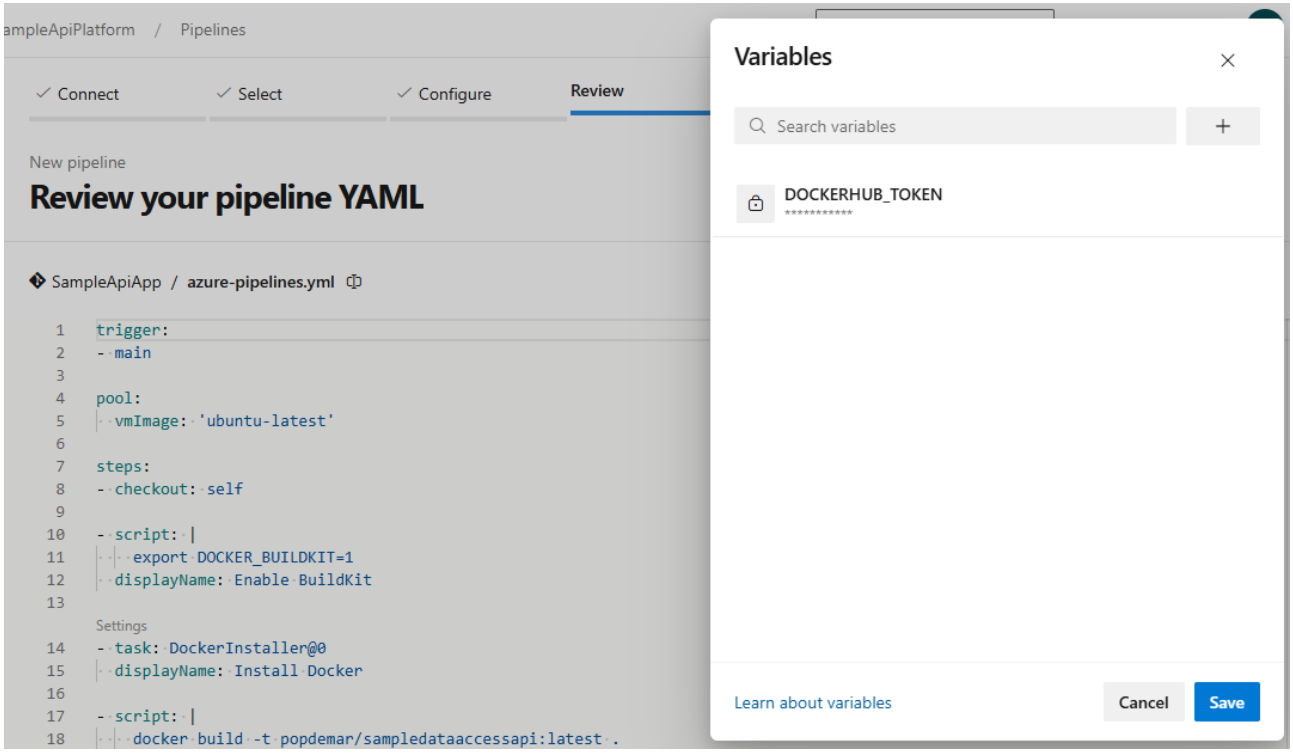
Then we need to create the environment variable in the pipeline to store the password of Dockerhub. Click on Variables:



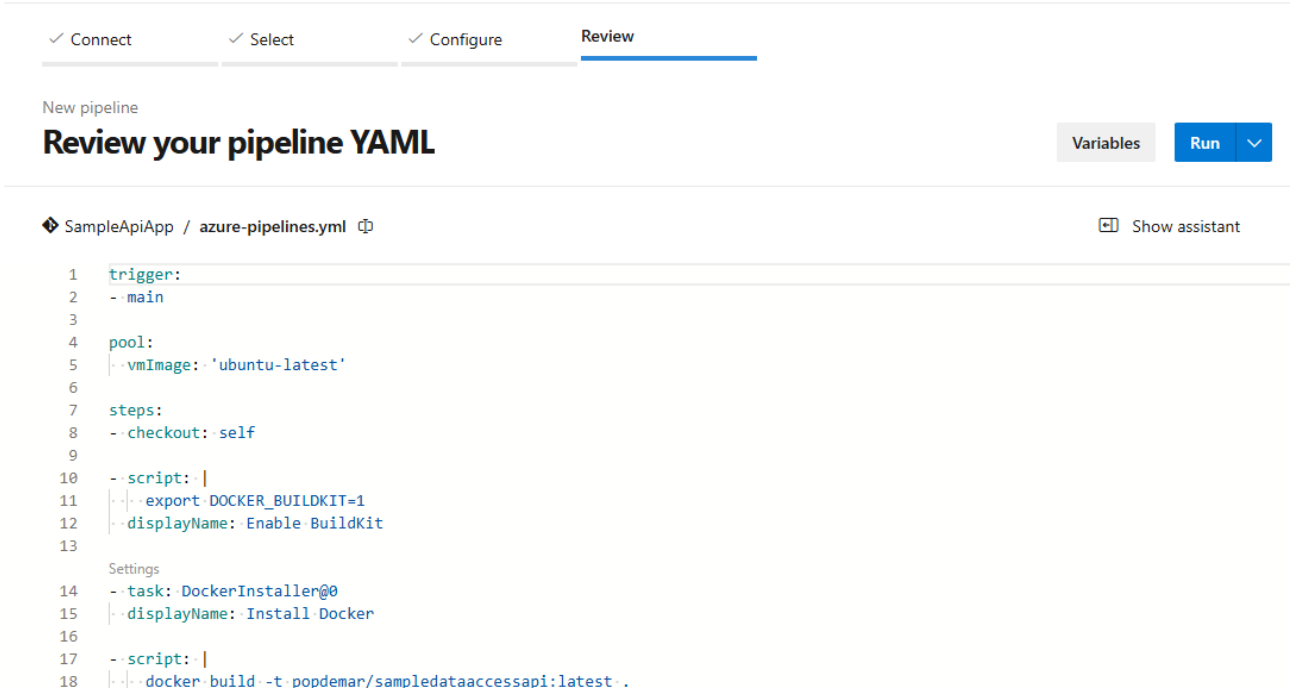
Then on New Variable, and add the DOCKERHIB_TOKEN variable, check it as Secret:



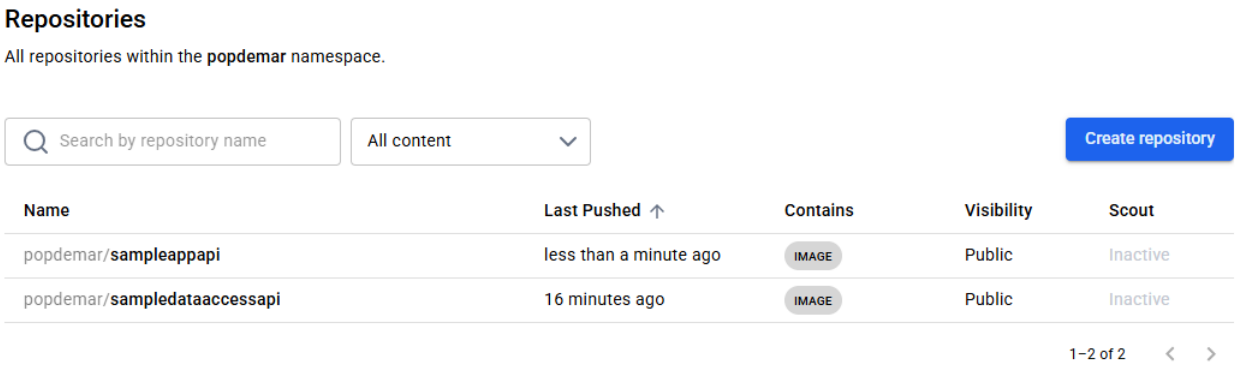
Click OK and then Save



And now we can run the pipeline: click on the Run button



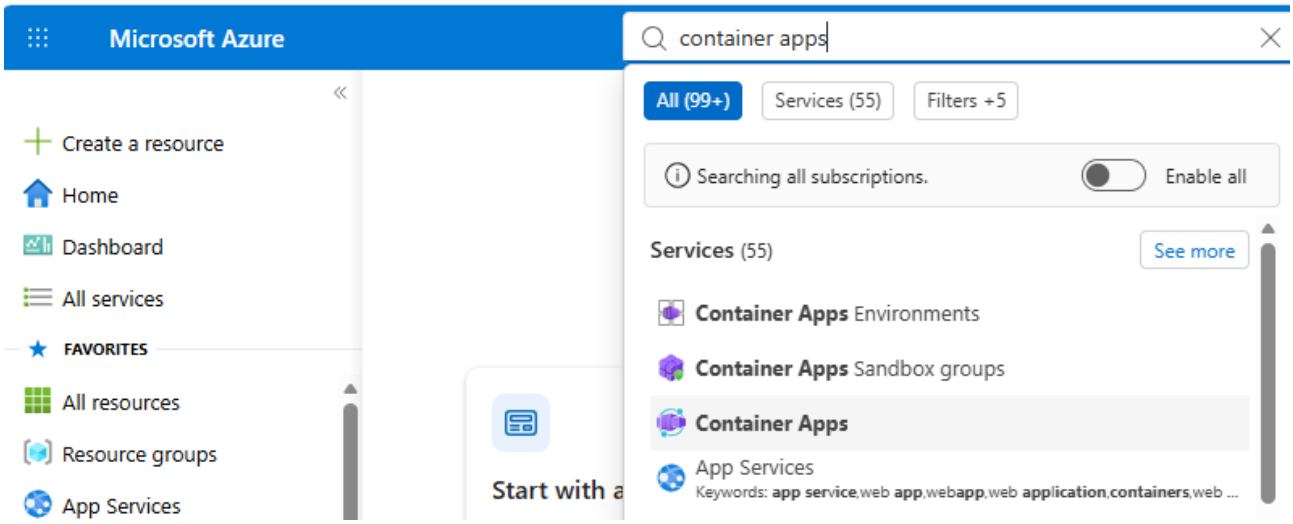
Once we complete this stage we will have our image, sampleaccessapi, uploaded in the Dockerhub and now we will be able to create a container in Azure.



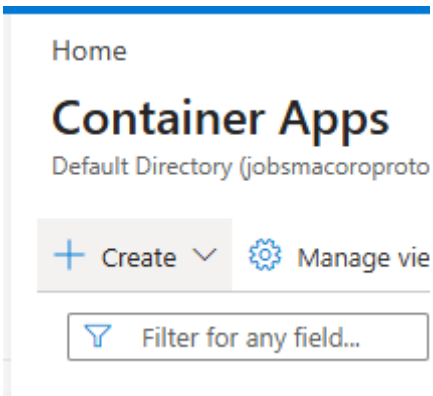
Create Container App in Azure

Before deploying and running our pipeline we need to create in Azure a container in the Container App section as follows:

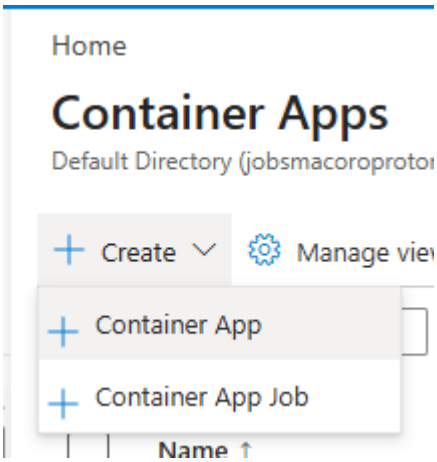
Navigate to Container Apps:



Click on Create:



Select the Container App option:



!!! The resource group should be shared with another container. Look at what we wrote about source groups!!!!

Fill up the form with the following values:

Azure Subscription = your subscription

Resource Group = SampleAppApi_rg (create a new resource group)

Container app name = sampleappapi

Optimize For Azure Functions = False (Unchecked)

Deployment Source = Container Image

Container Apps Environment:

- Show Environment In All Regions = False (Uncheked)
- Region= France Central (Your Region)
- Container Apps Environment= sampleappapi-env (create a new one)
- Zone Redudancy = Disabled

Create Container App:

- Container Details
  - Name= sampleappapi
  - Image Source = Dockerhub or other registries
  - Image type = public
  - Registry login server = docker.io
  - Image and tag = popdemar/sampleappapi:latest
  - Command override = empty
  - Arguments override = empty

Development Stack Specific Features:

Development stack = Unspecified

Container Resource Allocation:

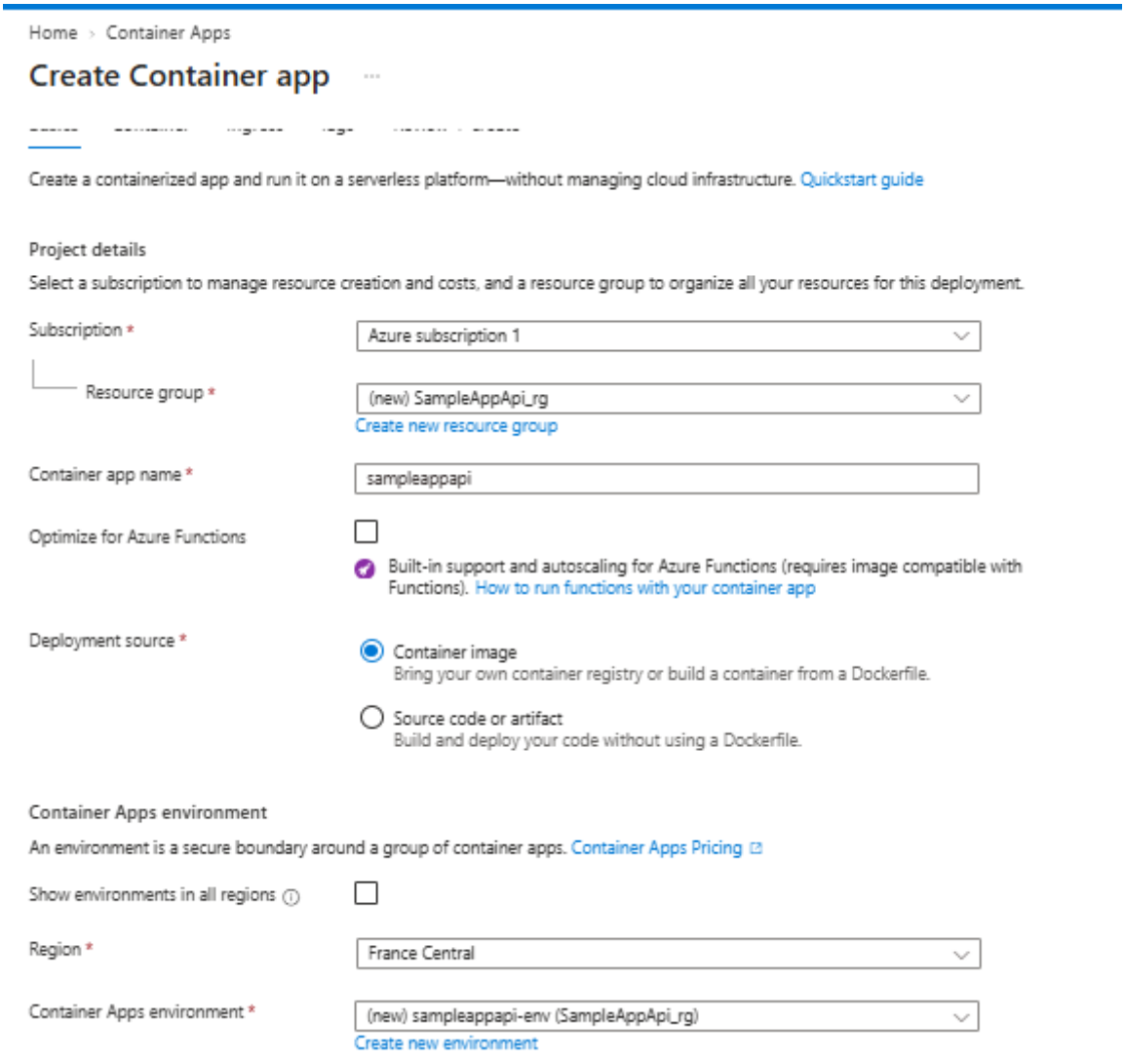
Workload Profile=Consumption-Up to 4 vCPU 8GB memory

GPU = False (Unchecked)

CPU and Memory = 0.5 CPU Cores , 1 Gi memory

Environment Variables:

Keine



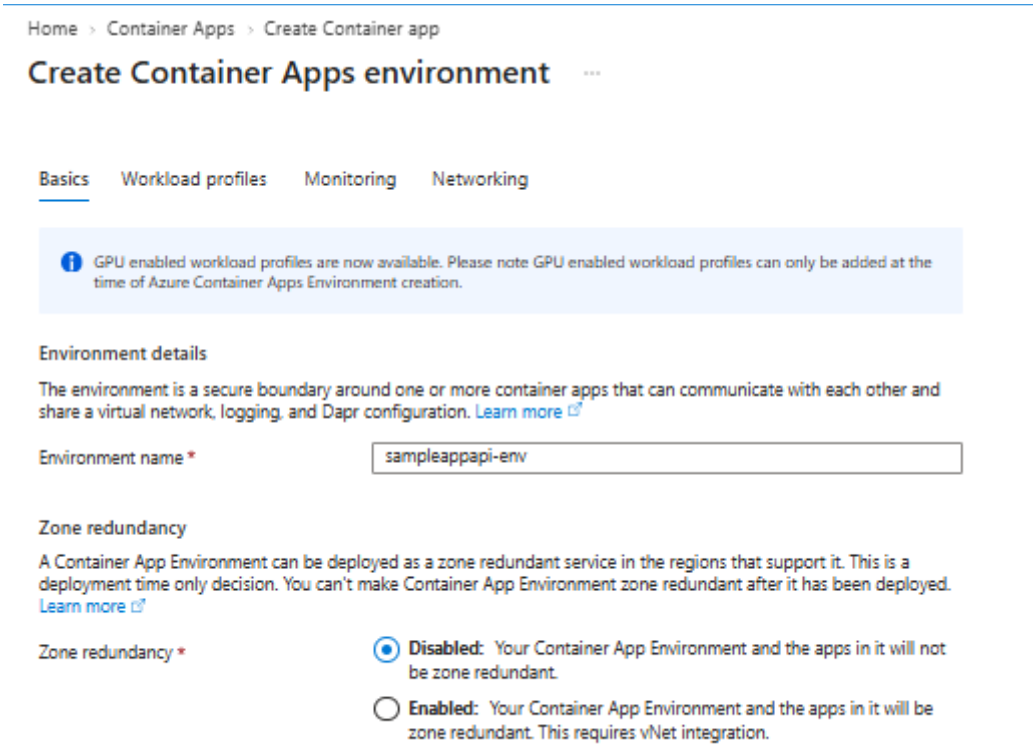
Notice that we have created:

- new resource = SampleAppApi_rg
- new environment = sampleappapi-env

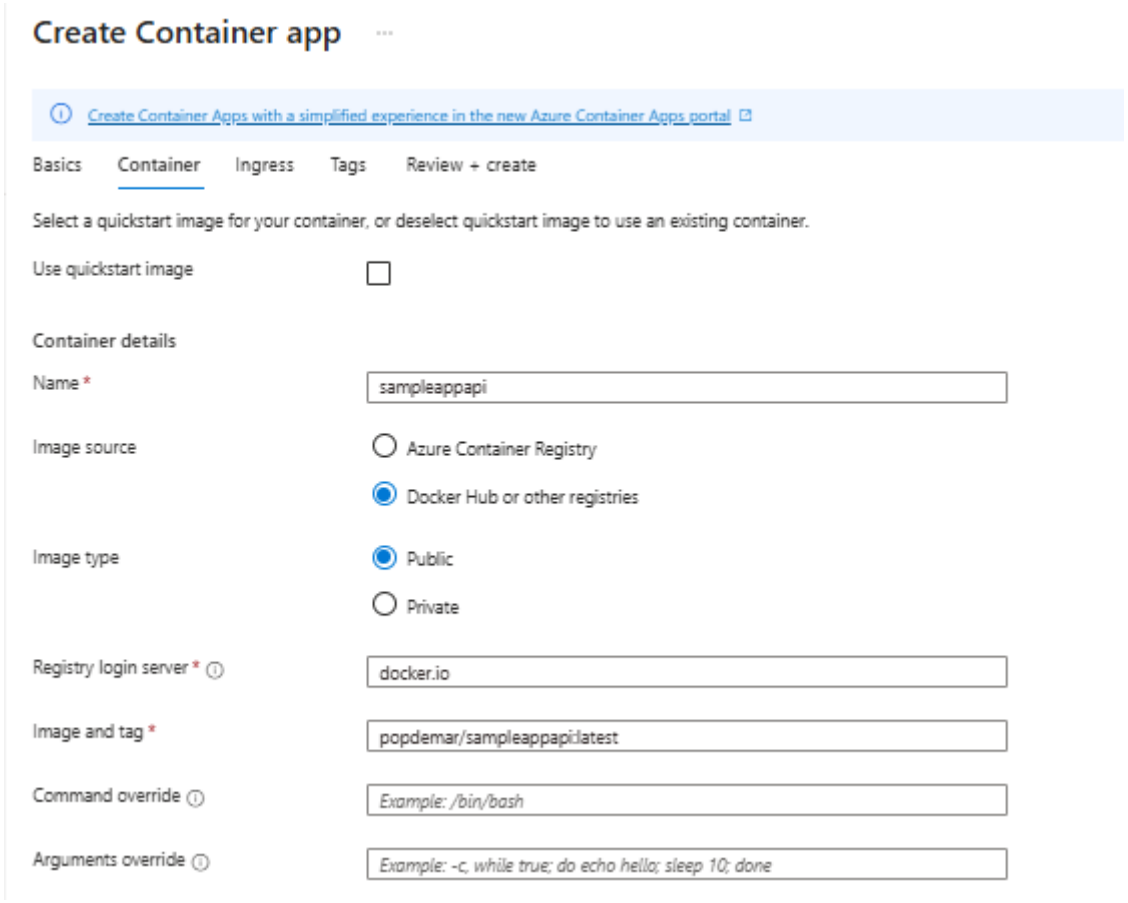
For this purpose fill out the form with the following values:

Environment Name= sampleappapi-env

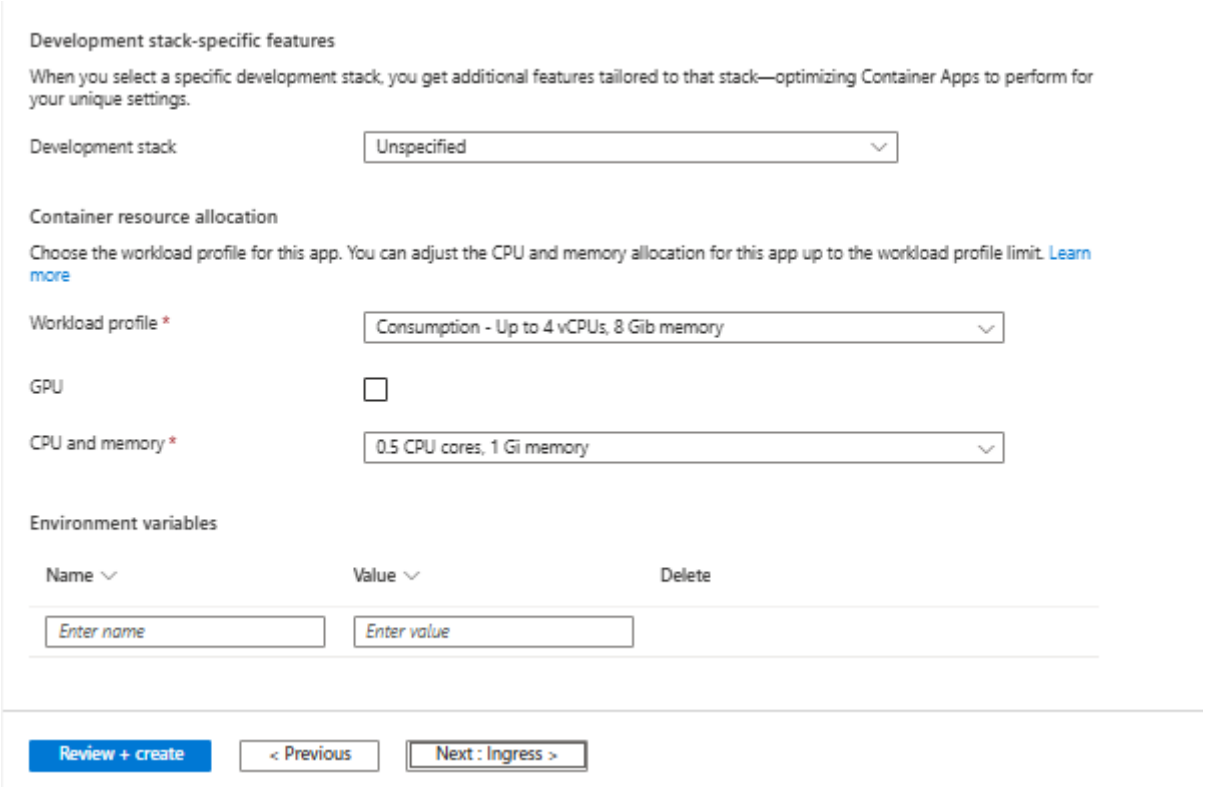
Zone Redundancy=disabled



Next we have to specify from which image and from which repository this container will be created.



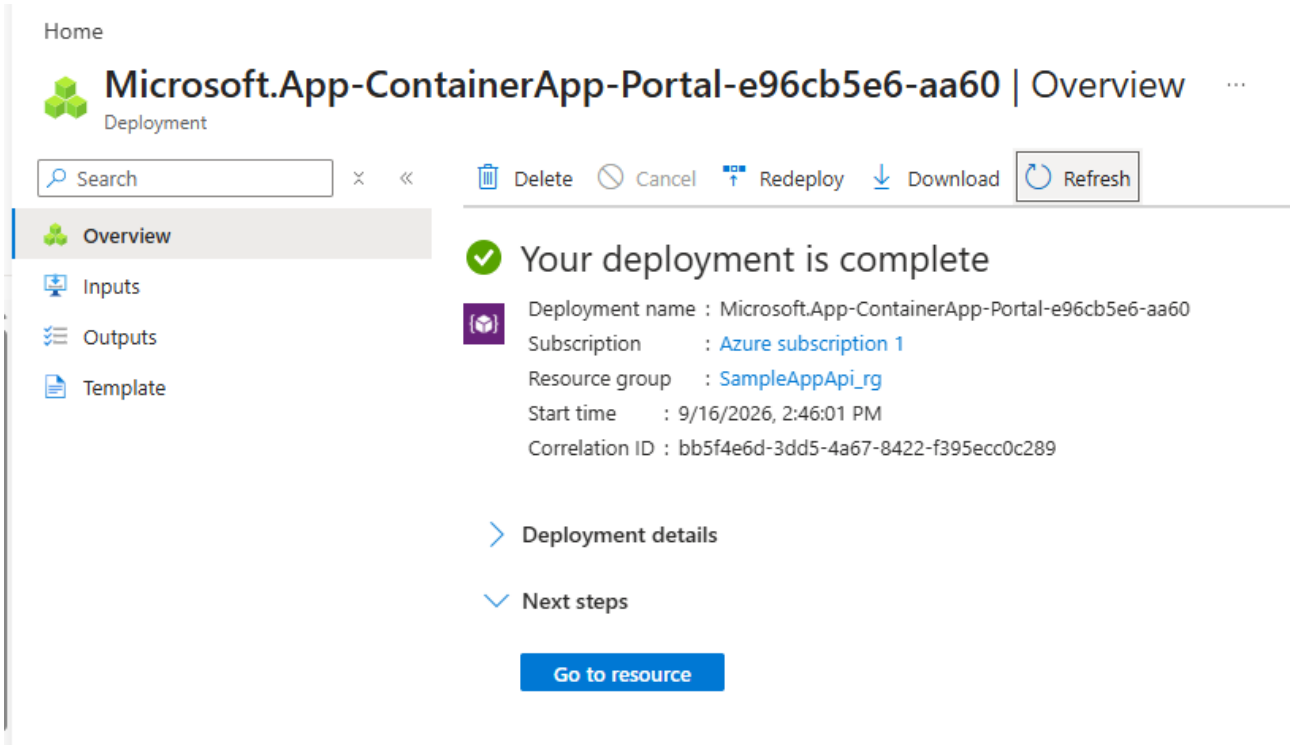
In our case we are storing our image in Dockerhub and our repositiory is marked in Docker as public.



The container resource allocation can be left as the default values provided.

In this case we do not need any Environment Varibles.

After that click on Review + Create > Create



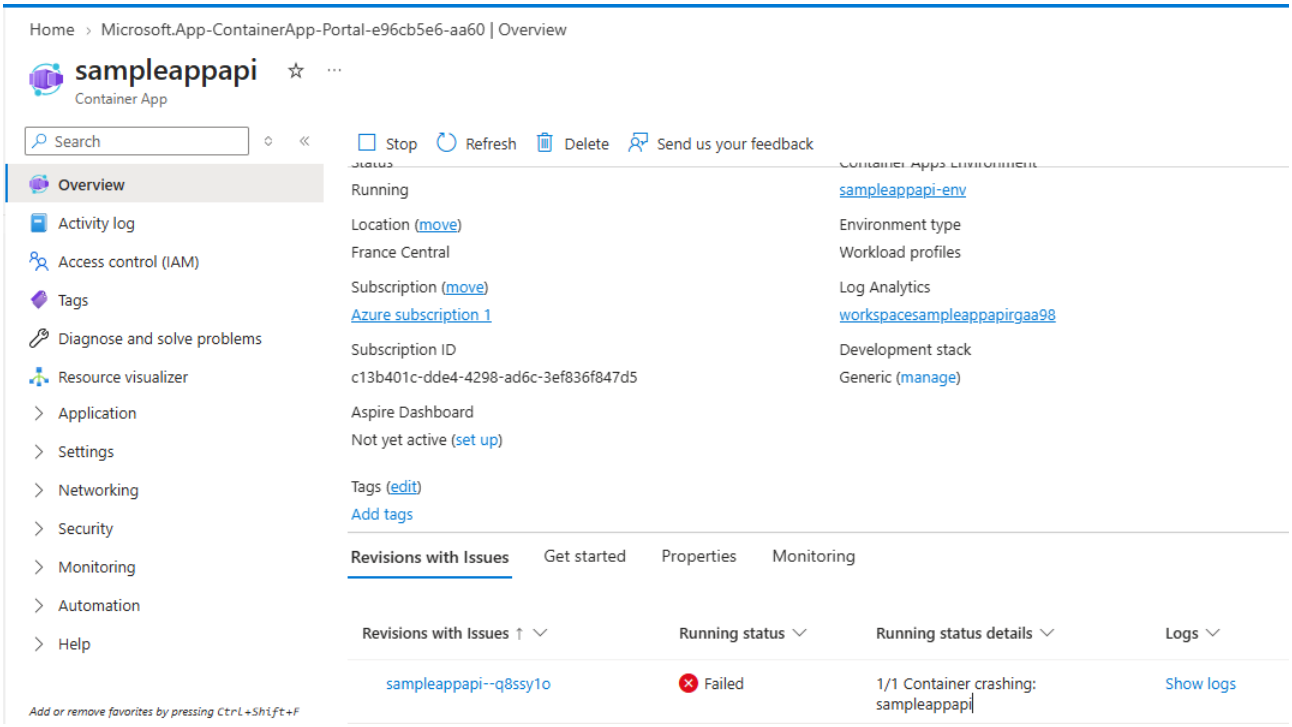
Once we receive the confirmation that Azure has created our container, we need to check if the container is running, and if there is a revision active, and if the app is reachable etc.

For this purpose in the Azure Portal go to: **Azure Portal → Container Apps → Deine App → Application → Revisions**

It should show something like this:

- **Active Revision: Yes**
- **Healthy: Yes**
- **Container: Running**
- **Image: popdemar/<repo>:latest**

When all is green then this means that the deployment was Successful.



However in our case it was not green , but red. When trying to get the logs did not give us too much information.

az containerapp logs show \

  --name sampleappapi \

  --resource-group SampleAppApi_rg \

--follow

We can also get the system logs:

az containerapp logs show \

  --name sampleappapi \

  --resource-group SampleAppApi_rg \

  --type system

After removing the all the code in Program.cs related to the JWT (the key was missing) and redeploying the image with the pipeline and updating the image from the command line, we managed to create the container.

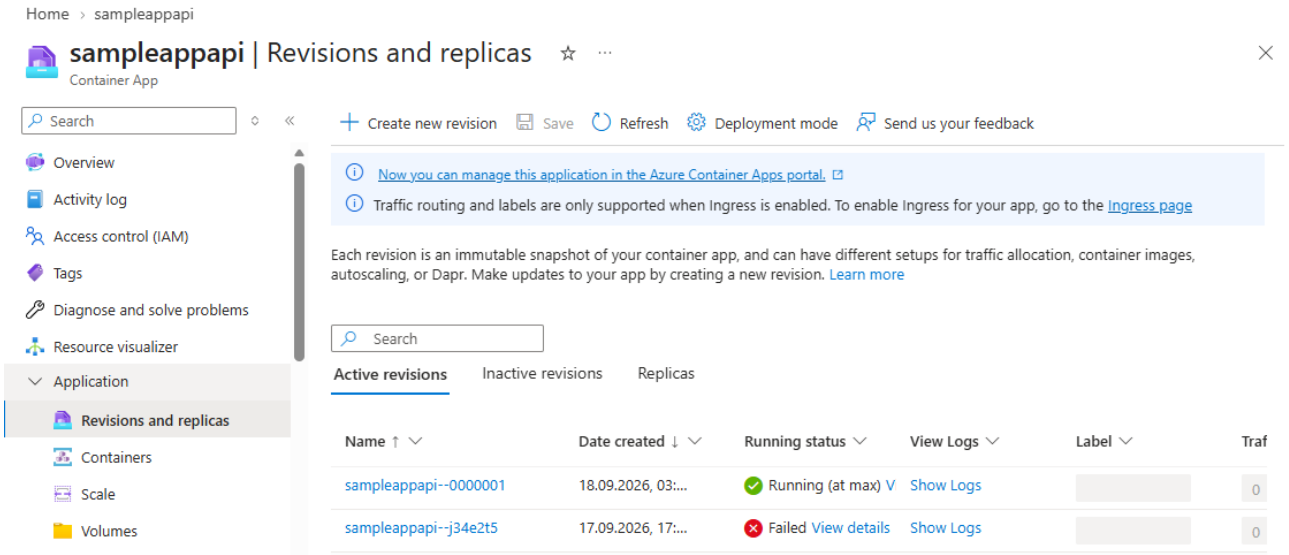
az containerapp update \

  --name sampleappapi \

  --resource-group SampleAppApi_rg \

--image popdemar/sampleappapi:latest

And after the problem has been solved and the container is up and running:



Finally now we will run the normal pipeline that deploys the code. Before we could not do it because the azure devops was complaining that the container did not exist:

Pipeline

Now we can deploy the normal pipeline that will deploy the code and update the image. Before we could not do it because the container did not exist yet in Azure, and then Azure Devops cannot complete the job because it cannot find the container:

```
trigger:
- master

pool:
  vmImage: ubuntu-latest

variables:

- name: solution
  value: '**/*.sln'

- name: buildPlatform
  value: 'Any CPU'

- name: buildConfiguration
  value: 'Release'

#this variable is necessary to allow the agent to give access to the agent to the NuGet packages.
- name: NUGET_PACKAGES
  value: $(Agent.TempDirectory)/.nuget/packages

steps:

# ------------------------------------------------------
# Restore
# ------------------------------------------------------
- task: DotNetCoreCLI@2
```

```
  displayName: Restore packages
  inputs:
    command: restore
    projects: '$(solution)'

# ----------------------------------------------------------
# Build
# ----------------------------------------------------------
- task: DotNetCoreCLI@2
  displayName: Build solution
  inputs:
    command: build
    projects: '$(solution)'
    arguments: '--configuration $(buildConfiguration)'

# ----------------------------------------------------------
# Test + Coverage
# ----------------------------------------------------------
- script: |
    dotnet test SampleAppApi.sln --configuration Release --collect:"XPlat Code Coverage"
  displayName: Run tests with coverage

# ----------------------------------------------------------
# Publish coverage
# ----------------------------------------------------------
- task: PublishCodeCoverageResults@1
  displayName: Publish coverage
  inputs:
    codeCoverageTool: Cobertura
    summaryFileLocation: '$(Agent.TempDirectory)/**/coverage.cobertura.xml'
    reportDirectory: '$(Agent.TempDirectory)'

# ----------------------------------------------------------
# Build Docker image
# ----------------------------------------------------------
- task: Docker@2
  displayName: Build Docker image
  inputs:
    command: build
    Dockerfile: Dockerfile
    repository: $(DOCKERHUB_USERNAME)/sampleappapi
    tags: |
      latest

- script: |
    echo "$(DOCKERHUB_TOKEN)" | docker login -u "$(DOCKERHUB_USERNAME)" --
password-stdin
  displayName: Docker Hub Login

# ----------------------------------------------------------
# Push Docker image
# ----------------------------------------------------------
```

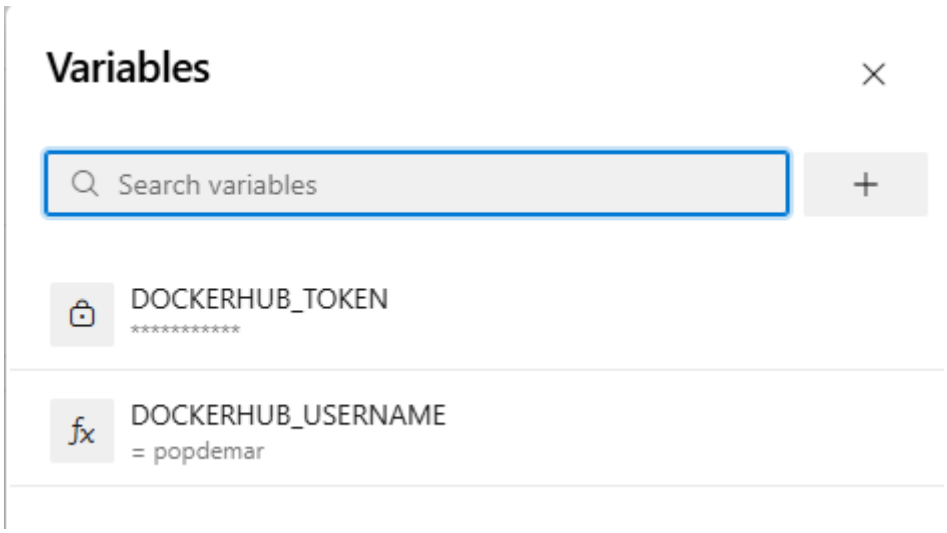
```
- script: |
    docker push $(DOCKERHUB_USERNAME)/sampleappapi:latest
  displayName: Push Docker image

# ----------------------------------------------------------
# Update ACA image
# ----------------------------------------------------------
- task: AzureCLI@2
  displayName: Update ACA image
  inputs:
    #we reuse the same service connection as in SampleDataAccessApi because it is only to login
into Azure
    azureSubscription: 'SampleAppApi_ServiceConnection'
    scriptType: bash
    scriptLocation: inlineScript
    inlineScript: |
      az containerapp update \
        --name sampleappapi \
        --resource-group SampleAppApi_rg \
        --image docker.io/$(DOCKERHUB_USERNAME)/sampleappapi:latest
```

We can reuse the service connection for Docker that we have previously created for the SampleDataAccessApi: DockerHub_ServiceConnection. However, we will need to create an new ARM Service Connection (SampleAppApi_ServiceConnection) as we did for the previous container app because the usage of that service connection is limited to the resource group that is not the same as the one that we are currently using: as well as the SampleDataAccessApi_ServiceConnection that we had used in the previous pipeline to allow Azure Devops to connect to Azure and execute the command to update the image. This reuse is valid, however the name should be more appropriate:

```
azureSubscription: 'SampleAppApi_ServiceConnection'
    scriptType: bash
    scriptLocation: inlineScript
    inlineScript: |
      az containerapp update \
        --name sampleappapi \
        --resource-group SampleAppApi_rg \
        --image docker.io/$(DOCKERHUB_USERNAME)/sampleappapi:latest
```

Do not forget to check that the following pipeline variables are still there: DOCKERHUB_TOKEN, DOCKERHUB_USERNAME.



Managed Identity

Now we go to the Container App and we will see this User Defined Identity in the list of available identities that we can use:

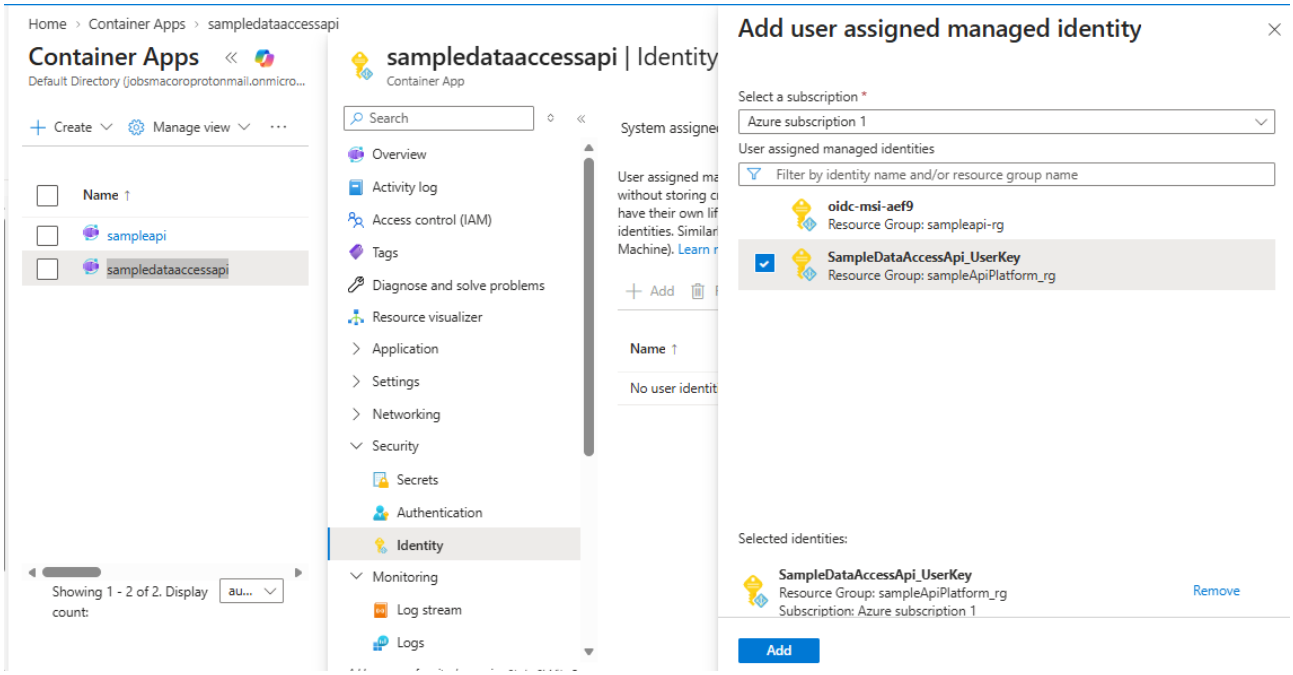


Figure 25: Adding The User Defined Identity To The Container App

After clicking on the Add button:

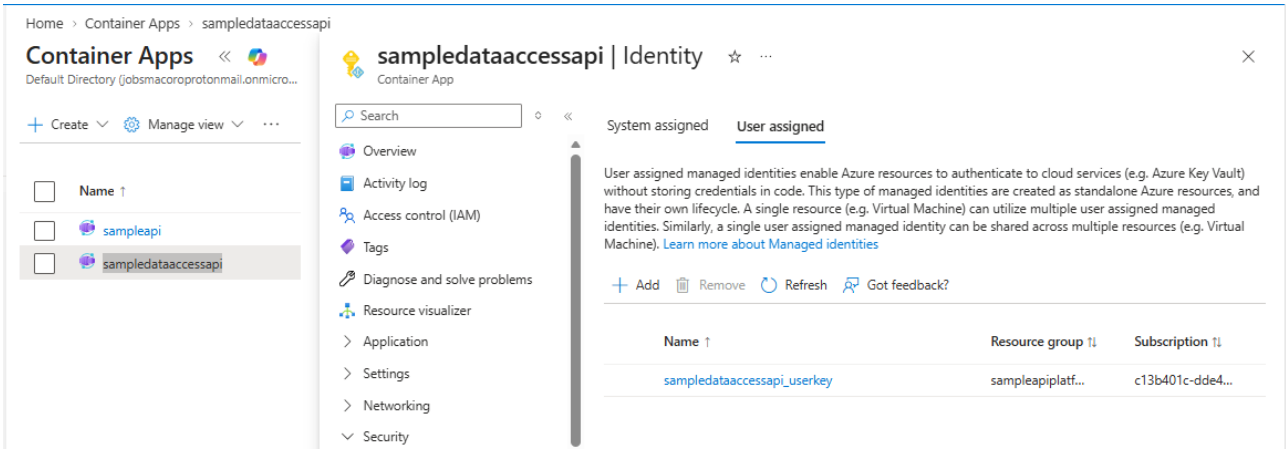


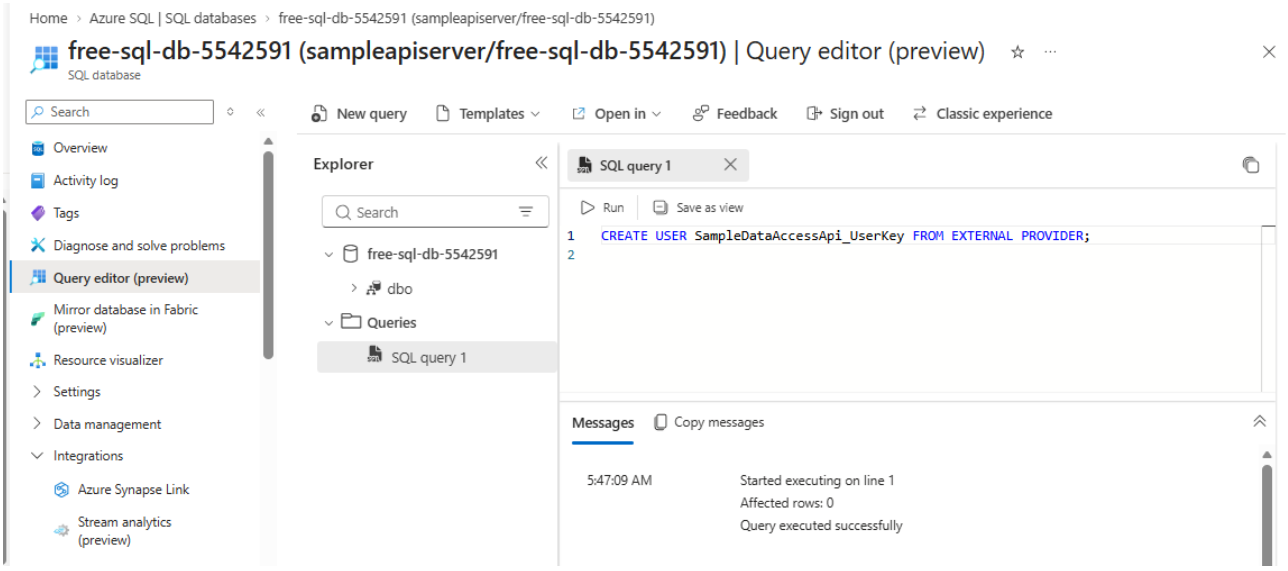
Figure 26: User Defined Identity Added To The Container App

Identity Roles

Now we need to add permissions to this User Identity: for instance to give access to our Azure SQL Database.

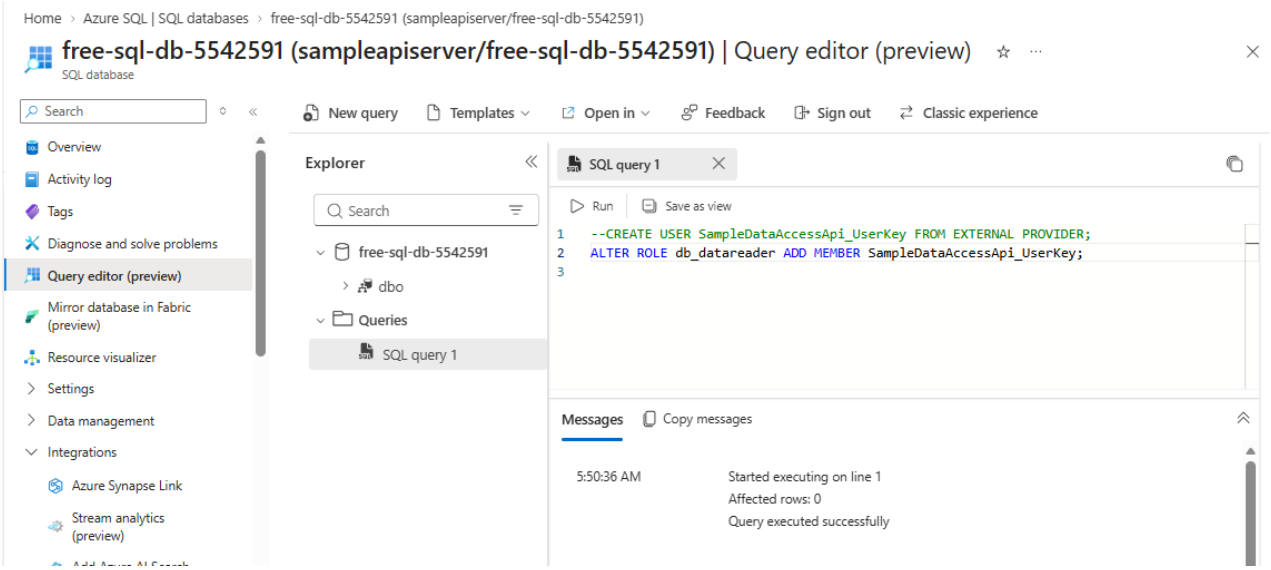
We need to login into our Azure Database with an Entra ID admin credential (in our case as jobs.macoro), not as a database admin (sampleapiserveradmin), and add our User Defined Identity , SampleDataAccessApi_UserKey, so that the database recognizes as a user and can login without a password (behind the scenes Azure will exchange a token between the database and the container).

CREATE USER `SampleDataAccessApi_UserKey` FROM EXTERNAL PROVIDER;

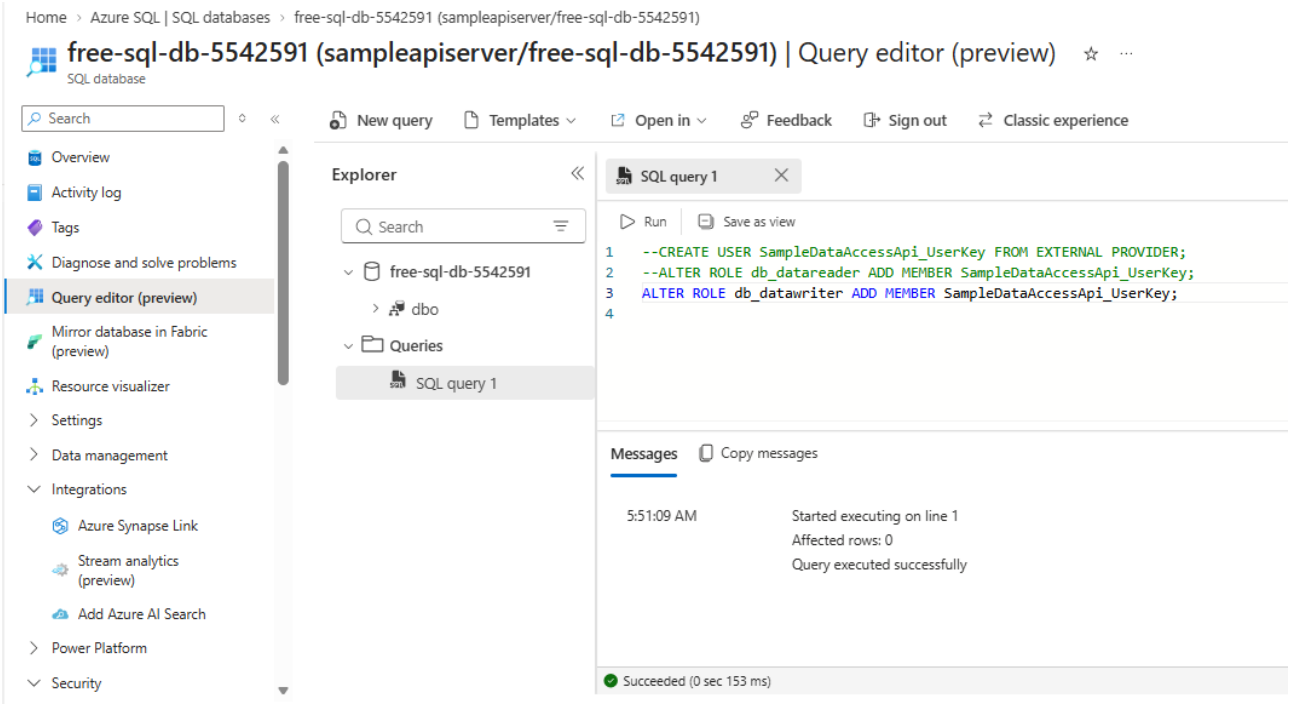


and that allows to login without password and to performed the operations.

ALTER ROLE db_datareader ADD MEMBER `SampleDataAccessApi_UserKey`;

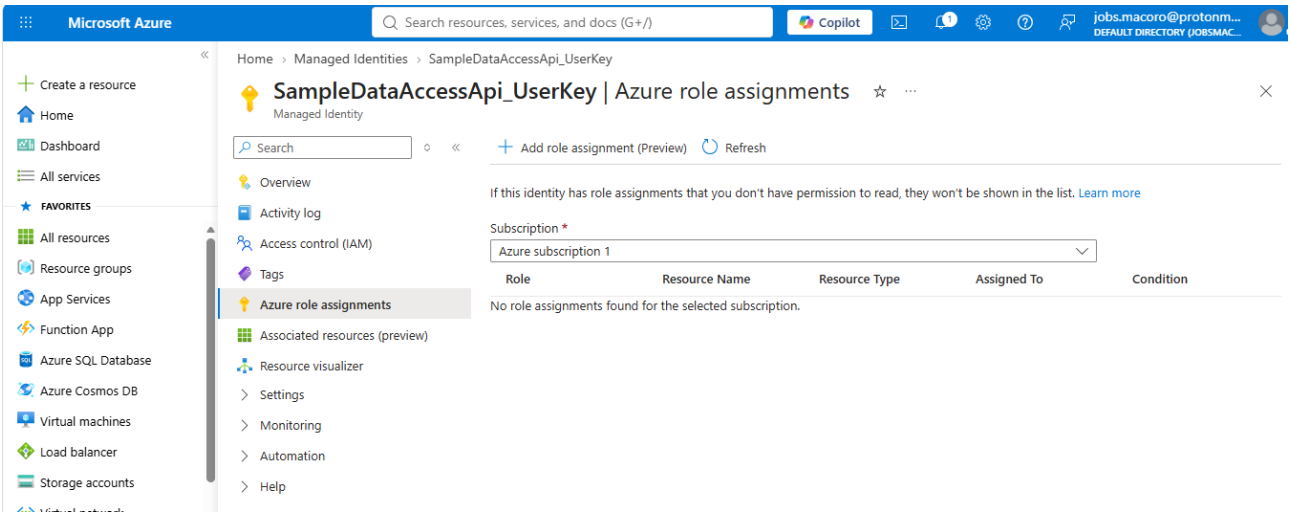


ALTER ROLE db_datawriter ADD MEMBER `SampleDataAccessApi_UserKey`;



Here are the tasks that we need to perform are similar to the ones that we did for the SampleDataAccessApi.

In order to define roles for our identity in Azure we can do it here: Azure Portal > Managed Identities > *SampleDataAccessApi_UserKey* > Azure role assignments



Then we click on the + Add Role Assigment button

A role can be defined at four different scopes: Subscription, Resource Group, Key Vault, Storage, SQL.

In our application currently we have the following defined.

Object	Object Type	Resource Group	Identity Roles
Subscription	Subscription		**Contributor**
sampleappapi	Container app	**SampleAppApi_rg**	**Reader**.
sampledataaccessapi	Container app	**SampleDataAccessApi_rg**	**Reader**
sampledataaccessapi-env2	Resource Container app Enviroment for sampledataaccessapi	sampleDataAccessApi_rg	Inherits from the resource group or subscription where the Container app has been defined. In this case will be Contributor or Reader.
sampleappapi-env5	Resource Container app Enviroment	SampleAppApi_rg	Inherits from the resource group or subscription where the

	for sampledataaccessapi		Container app has been defined. In this case will be Contributor or Reader.
free-sql-db-5542591 (sampleapiserver/free-sql-db-5542591)	SQL Database	DefaultResourceGroup-CCAN	**db_datareader** **db_datawriter**
Key Vault(in case we save secrets)			**Key Vault Secrets User**
Log Analytics Workspace			**Reader**
Storage			**Storage Blob Data Reader**

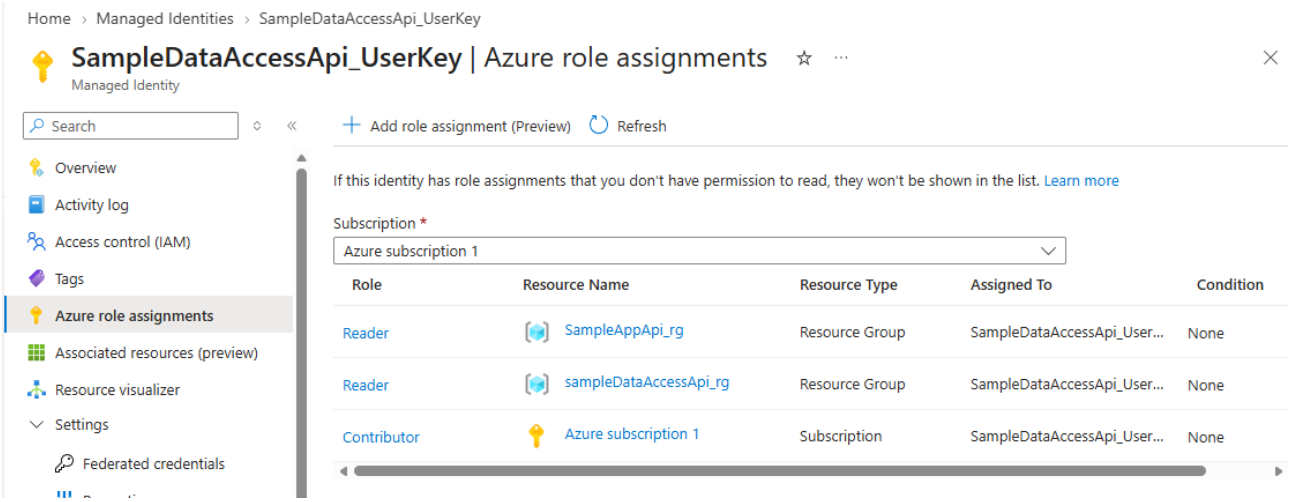
We will define the Identity Roles defined in the correspondent column for the resources / objects that are in bold.

For the ACA environments: sampledataaccessapi-env2, sampleappapi-env5, they will inherit the role from the Subscription or from the resource group of the container. As we have defined roles for both, the subscription and the resource groups of each container app, in this case the will take the role from the container app resource group as it overwrites the roles defined in at Subscription level.

We have defined this user key in the following resource group: sampleApiPlatform_rg

For the Log Analytics Workspace, Storage and Key Vault we will only be able to define a role once we have one.

At the end of this process we will get all the roles defined:



Communication between Containers

We can use : Inress/HTTPS or Dapr.

Dapr requires that all of the containers use the same ACA environment which is not the case. Now we have:

Container App	Environment
sampledataaccessapi	**Sampledataaccessapi-env2**
sampleappapi	**sampleappapi-env5**

For this reason we will need to recreate all of the container apps again and both having the same Environment. I have deleted the previous containers and created new ones

Container App	Environment
sampledataaccessapi	**sampleapi-env**
sampleappapi	**sampleapi-env**

We will follow the same steps as described in previous sections and expecting not to enounter the same issues as now we know that the images are have been into ACA successfully deployed and once created we will assign them the sampleDataAccessApi_UserKey User assigned role to the both of them.

We don not need to change the roles as the reource groups where these new containers are locared is the same as the previous ones, and the ACA environments will inherit the roles from these resource groups as we have mentioned before.

Dapr Architecture

Dapr consists of two parts:

Your Container (App)

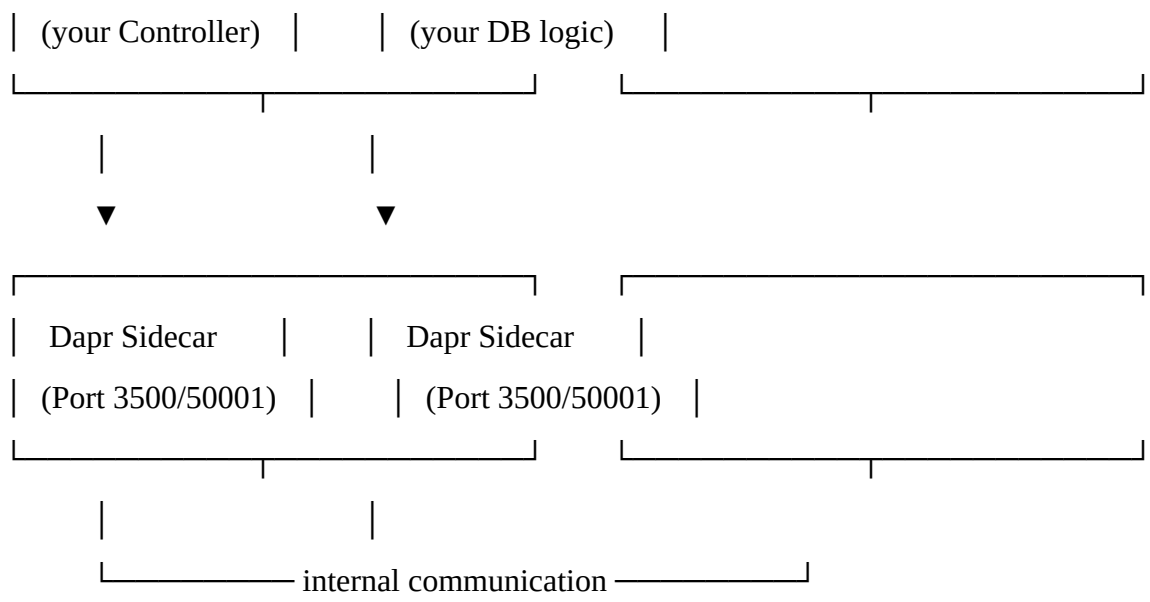
The Dapr Sidecar (always runs alongside it)

Together, both form a single unit.

1.Architecture: SampleAppApi → Dapr → SampleDataAccessApi This is how the communication looks:

Code





No URLs
No Ports
No Ingress
Only internal sidecar communication

2.Architecture: SampleDataAccessApi → Database Code SampleDataAccessApi → EF Core → SQL Server / Azure SQL

Dapr is NOT involved here.
Dapr does NOT replace EF Core.
Dapr is NOT a database proxy.

3.Why only SampleAppApi needs the Dapr Client **SampleAppApi** calls other container apps

needs service discovery

needs sidecar invocation

needs the Dapr client or sidecar HTTP

SampleDataAccessApi only receives requests

does NOT need to call other services

does NOT need to use the Dapr client

does NOT need to use Dapr components

does NOT need to use Dapr state store

It only needs the sidecar so that it is reachable.

4.Dapr Service Invocation Flow This is how a request works: SampleAppApi calls: http://localhost:3500/v1.0/invoke/dataaccess/method/movies

Dapr sidecar of SampleAppApi:

automatically finds the correct revision

automatically finds the correct container app

handles automatic retries

handles automatic circuit breakers

handles automatic telemetry

Dapr sidecar of SampleDataAccessApi:

receives the request

forwards it to your API

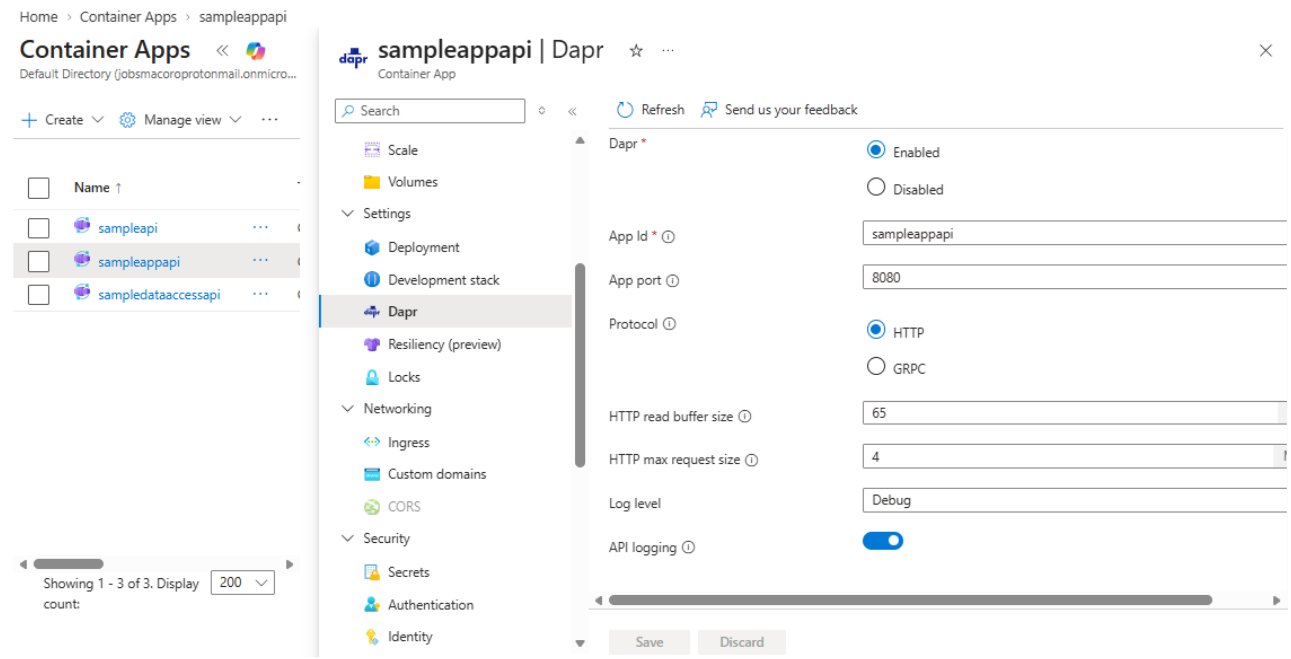
your API executes EF Core

the response goes back via the sidecar

You don't need to know any URLs.
You don't need to know any ports.
You don't need to know any ingress endpoints.

5.Why Dapr is perfect for your architecture Microservices without ingress Zero-trust communication No ports, no DNS issues Automatic service discovery Automatic retries Automatic circuit breakers Automatic observability Perfect for Azure Container Apps

Setting Dapr for SampleAppApi



Dapr Settings = Enabled

App Id = sampleappapi

App Port = 8080

Protocol = HTTP (because our APIs are HTTP Rest)

HTTP read buffer size = 65 KB (this is the maximum value value.)

HTTP max request size = 4 MB (This is the standard value. And it is the max size HTTP request that the sidecar accept)

Log Level = Debug (for test) / Info (for production)

API Logging = On for testing / Off for production.. It will log all Dapr operations. For production set it to off, to avoid to generate too many logs.

Setting Dapr for SampleDataAccessApi

Dapr Settings = Enabled

App Id = sampledataaccessapi

App Port = 8080

Protocol = HTTP (because our APIs are HTTP Rest)

HTTP read buffer size = 65 KB (this is the maximum value value.)

HTTP max request size = 4 MB (This is the standard value. And it is the max size HTTP request that the sidecar accept)

Log Level = Debug (for test) / Info (for production)

API Logging = On for testing / Off for production.. It will log all Dapr operations. For production set it to off, to avoid to generate too many logs.

Install the NuGet package for Dapr: dotnet add package Dapr.AspNetCore

Activate the dapr client in the SampleAppApi in Program.cs: builder.Services.AddDaprClient();

In Program.cs add the following code. The header gives the instruction to the sidecar to provide an answer in less thatn 30 seconds.

```
builder.Services.AddHttpClient<IDataAccessClient, DataAccessClientDapr>(client =>
{
    client.BaseAddress = new Uri("http://localhost:3500");
    //instruct the dapr sidecar to return an response in less than 504. Otherwise timeout.
    client.DefaultRequestHeaders.Add("Dapr-TimeoutInSeconds", "30");
});
```

Make sure that:

- ~~to remove [Authorize] from sampledataaccessapi movies controller as it is no needed anymore in the context of dapr are the connections between containers are managed by azure itself (trusted)~~
- make sure the url to connect to the sampledataccessapi container in dapr is correct, according to the path defined in the controller:

  [Route("api/movies")]

  `v1.0/invoke/sampledataaccessapi/method/api/movies`
- Make sure that all interfaces and the classes that implement that interface are declared in the Program.cs:

  builder.Services.AddScoped<IMovieRepository, MovieRepository>();
  builder.Services.AddScoped<IMovieService, MovieService>();
-

And then replace the class that queries that sampledataaccess container by one that is dapr compliant:

```
using NuGet.SampleSharedModels.DTO;
using NuGet.SampleSharedModels.Results;
using SampleAppApi.Interfaces.ExternalServices;
using Microsoft.Extensions.Logging;
using System.Net.Http.Json;
using NuGet.SampleSharedModels.Interfaces;

public class DataAccessClientDapr : IDataAccessClient
{
    private readonly HttpClient _httpClient;
    //private readonly ISharedServicesClient _sharedServicesClient;
    private readonly ILogger<DataAccessClientDapr> _logger;
    private readonly string _category = string.Empty;
```

```
    public DataAccessClientDapr(IHttpClientFactory httpClientFactory,
                                ILogger<DataAccessClientDapr> logger
                               //ISharedServicesClient sharedServicesClient
                               )
    {
        // Dapr Sidecar HTTP Port ist IMMER 3500 in Azure Container Apps
        _httpClient = httpClientFactory.CreateClient("dapr");
        //_sharedServicesClient = sharedServicesClient;
        _logger = logger;
        _category = this.GetType().Name;
    }

    public async Task<ServiceResult<IEnumerable<MovieDTORead>>> GetAll()
    {
        try
        {
            var url = "v1.0/invoke/dataaccess/method/movies";

            var result = await
_httpClient.GetFromJsonAsync<ServiceResult<IEnumerable<MovieDTORead>>>(url);

            return result!;
        }
        catch (Exception ex)
        {
            var message = $"DataAccessApi unreachable: {ex.Message}";
            _logger.LogError("GetAll Exception: {message}", message);
            //await _sharedServicesClient.LogAsync(_category, $"GetAll Exception: {message}",
LogLevel.Error);
            return ServiceResult<IEnumerable<MovieDTORead>>.Fail(new List<string> { message });
        }
    }

    public async Task<ServiceResult<MovieDTORead>> GetById(string id)
    {
        try
        {
            var url = $"v1.0/invoke/dataaccess/method/movies/{id}";

            var result = await _httpClient.GetFromJsonAsync<ServiceResult<MovieDTORead>>(url);

            return result!;
        }
        catch (Exception ex)
        {
            var message = $"DataAccessApi unreachable: {ex.Message}";
            _logger.LogError("GetById Exception: {message}", message);
            //await _sharedServicesClient.LogAsync(_category, $"GetById Exception: {message}",
LogLevel.Error);
            return ServiceResult<MovieDTORead>.Fail(new List<string> { message });
        }
```

```
    }

    public async Task<ServiceResult<MovieDTORead>> Add(MovieDTOAdd movieDTOAdd)
    {
        try
        {
            var url = "v1.0/invoke/dataaccess/method/movies";

            var response = await _httpClient.PostAsJsonAsync(url, movieDTOAdd);

            var result = await
response.Content.ReadFromJsonAsync<ServiceResult<MovieDTORead>>();

            return result!;
        }
        catch (Exception ex)
        {
            var message = $"DataAccessApi unreachable: {ex.Message}";
            _logger.LogError("Add Exception: {message}", message);
            //await _sharedServicesClient.LogAsync(_category, $"Add Exception: {message}",
LogLevel.Error);
            return ServiceResult<MovieDTORead>.Fail(new List<string> { message });
        }
    }

    public async Task<ServiceResult<MovieDTORead>> Update(string id, MovieDTOUpdate
movieDTOUpdate)
    {
        try
        {
            var url = $"v1.0/invoke/dataaccess/method/movies/{id}";

            var response = await _httpClient.PutAsJsonAsync(url, movieDTOUpdate);

            var result = await
response.Content.ReadFromJsonAsync<ServiceResult<MovieDTORead>>();

            return result!;
        }
        catch (Exception ex)
        {
            var message = $"DataAccessApi unreachable: {ex.Message}";
            _logger.LogError("Update Exception: {message}", message);
            //await _sharedServicesClient.LogAsync(_category, $"Update Exception: {message}",
LogLevel.Error);
            return ServiceResult<MovieDTORead>.Fail(new List<string> { message });
        }
    }

    public async Task<ServiceResult<bool>> Delete(string id)
    {
        try
```

```
        {
            var url = $"v1.0/invoke/dataaccess/method/movies/{id}";

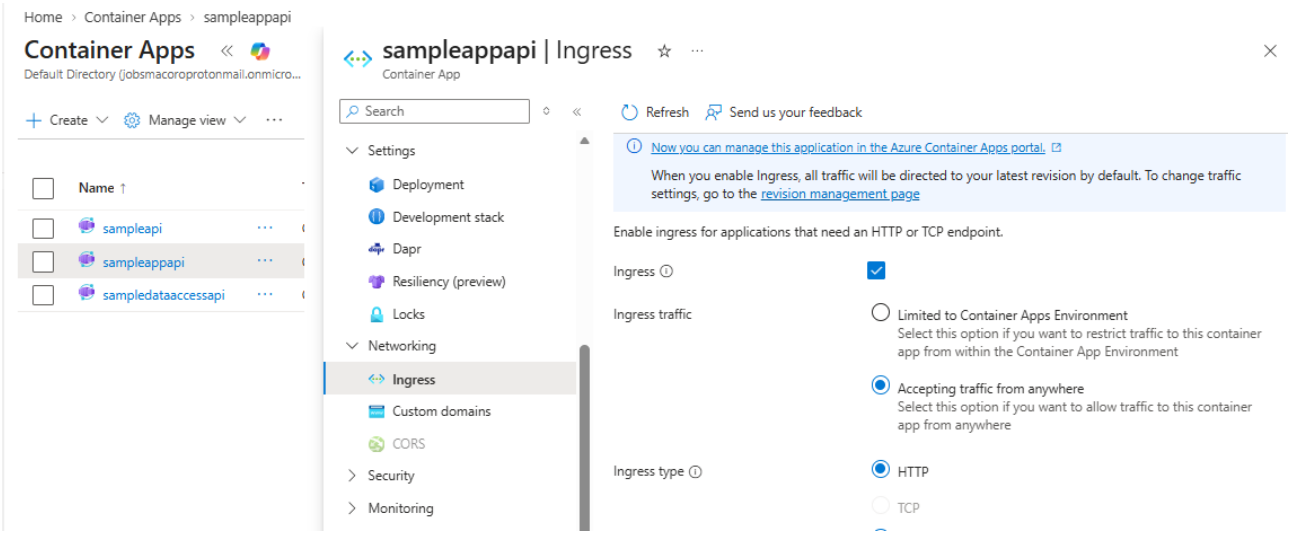
            var response = await _httpClient.DeleteAsync(url);

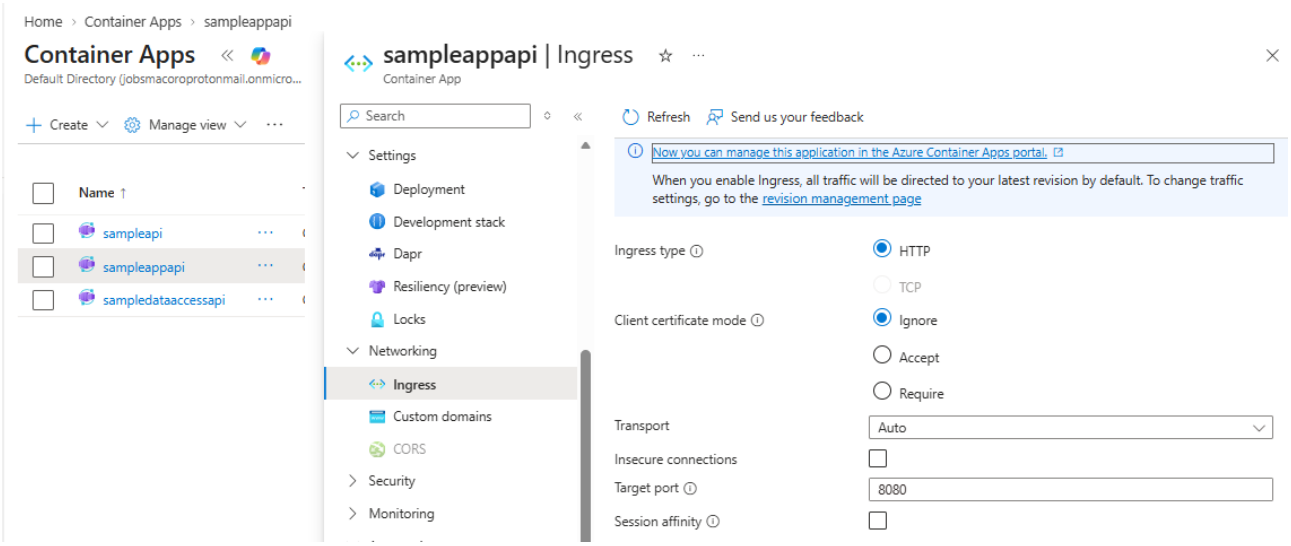
            var result = await response.Content.ReadFromJsonAsync<ServiceResult<bool>>();

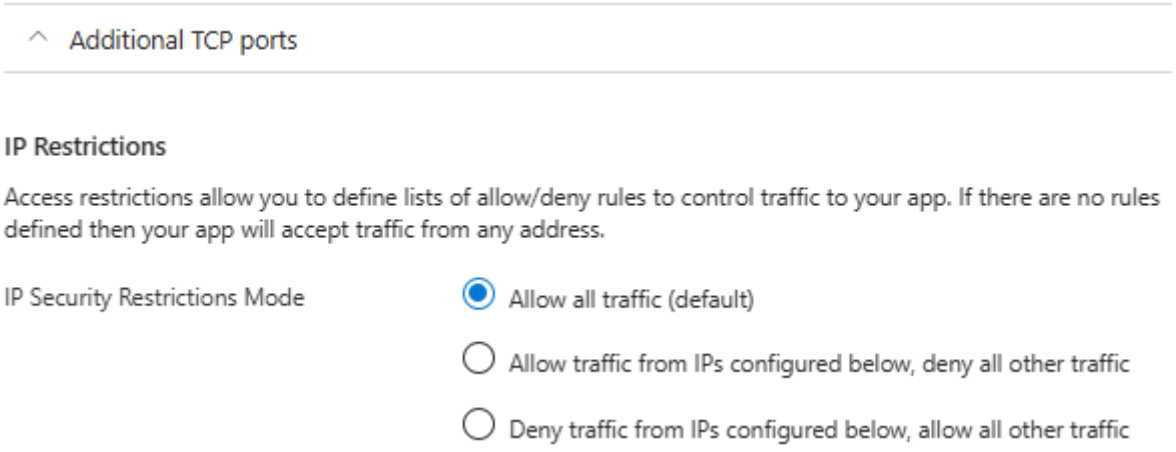
            return result!;
        }
        catch (Exception ex)
        {
            var message = $"DataAccessApi unreachable: {ex.Message}";
            _logger.LogError("Delete Exception: {message}", message);
            //await _sharedServicesClient.LogAsync(_category, $"Delete Exception: {message}",
LogLevel.Error);
            return ServiceResult<bool>.Fail(new List<string> { message });
        }
    }
}
```

Ingress activation for the SampleAppApi

This is necessary when our application needs to be called by cient applications: Angular UI, Postman, Browser, Mobile App, Swagger, Azure Devops Health Checks







Ingress traffic= Accepting Traffic from Anywhere

Ingress Type = HTTP (we are exposing a REST interface)

Client Certificate Mode: Ignore (we do not need any mTLS for an open API)

Transport: Auto (azure will decide between HTTP/1 or HTTP/2

Insecure connections=False (You want to enforce HTTPS)

Target port=8080

Session affinity= False (We do not have any Session based workloads)

IP Restrictions > IP Security Restrictions Mode = Allow all traffic (Default) (To make the API reachable from anywhere).

Machine-to-Machine JWT

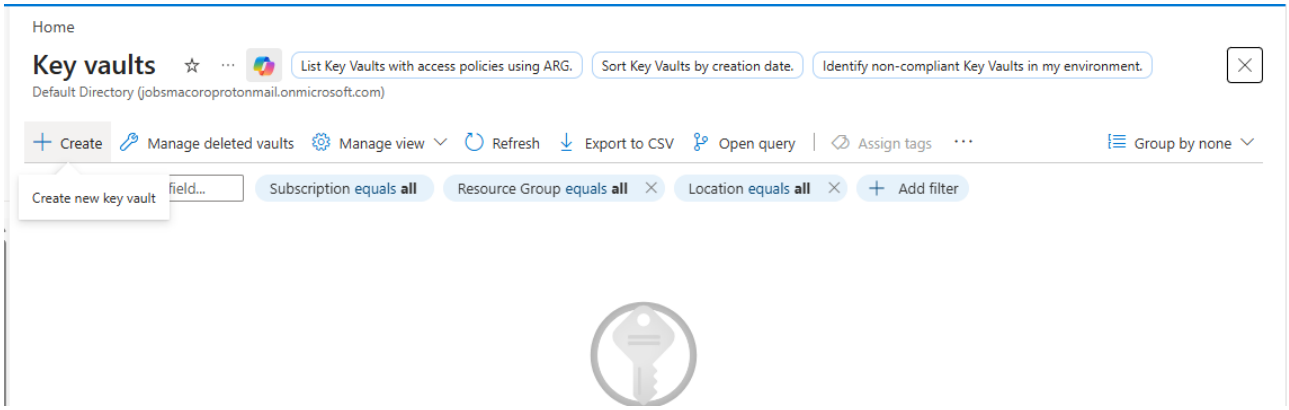
We will require all the applications that want to access the SampleAppApi to provide a Service Token.

This tokens will be stored in the Key Vault.

This requires the following steps

Create A Key Vault

Portal > Search: Key Vaults > Click on on the “+ Create” button



Resource Group = sampleApiPlatform_rg (because the Key Vault belongs logically to the whole platform)

Instance Details:

Key vault name = sampleapiplatform-kv

Region = France Central (in the same region where we have our containers for performance and compliance).

Pricing Tier = Standard (Premium is basically for HSM = Hardware Security Module which we do not need)

Recovery Option

Soft delete = enable (in case that we delete a secret that we should not have deleted)

Days to retain deleted vaults = 90 days

Purge Protection = Disable purge protection (allow key vault and objects to be purged during retention period) . For demos this is the right option. If enabled the vault will be not possible to delete for 90 days (days to retain deleted vaults).

Create a key vault

Basics | Access configuration | Networking | Tags | Review + create

Azure Key Vault is a cloud service used to manage keys, secrets, and certificates. Key Vault eliminates the need for developers to store security information in their code. It allows you to centralize the storage of your application secrets which greatly reduces the chances that secrets may be leaked. Key Vault also allows you to securely store secrets and keys backed by Hardware Security Modules or HSMs. The HSMs used are Federal Information Processing Standards (FIPS) 140-3 Level 3 validated. In addition, key vault provides logs of all access and usage attempts of your secrets so you have a complete audit trail for compliance.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * sampleApiPlatform_rg

Create new

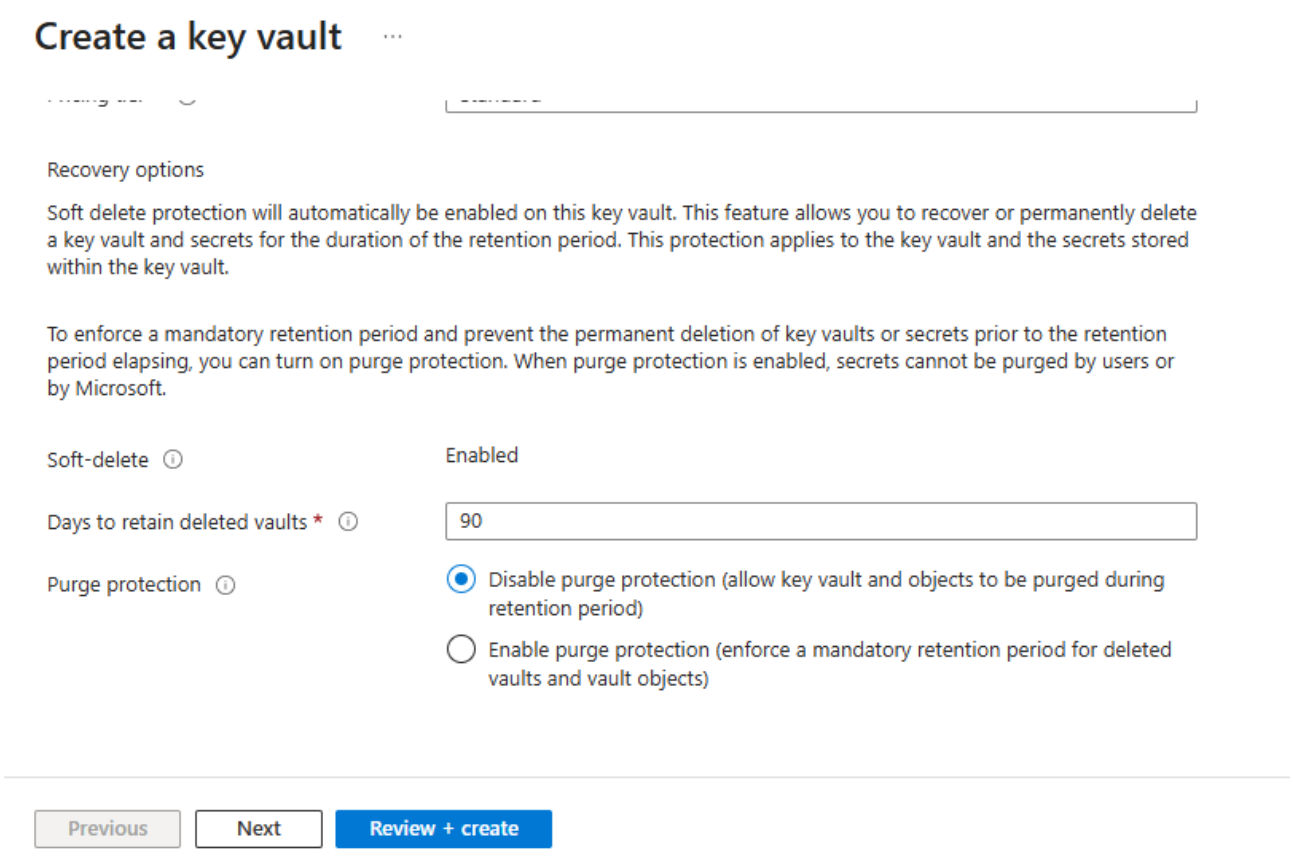
Create a key vault

Instance details

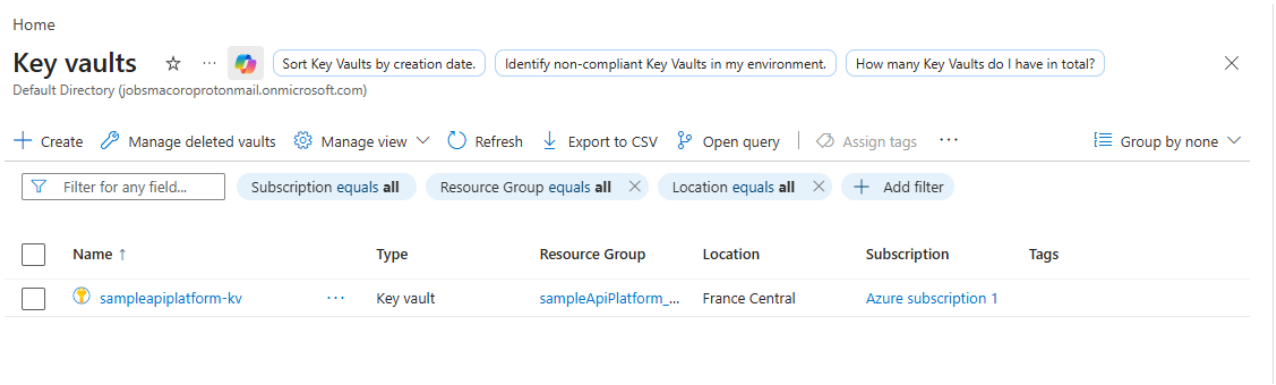
Key vault name * sampleapiplatform-kv

Region * France Central

Pricing tier * Standard



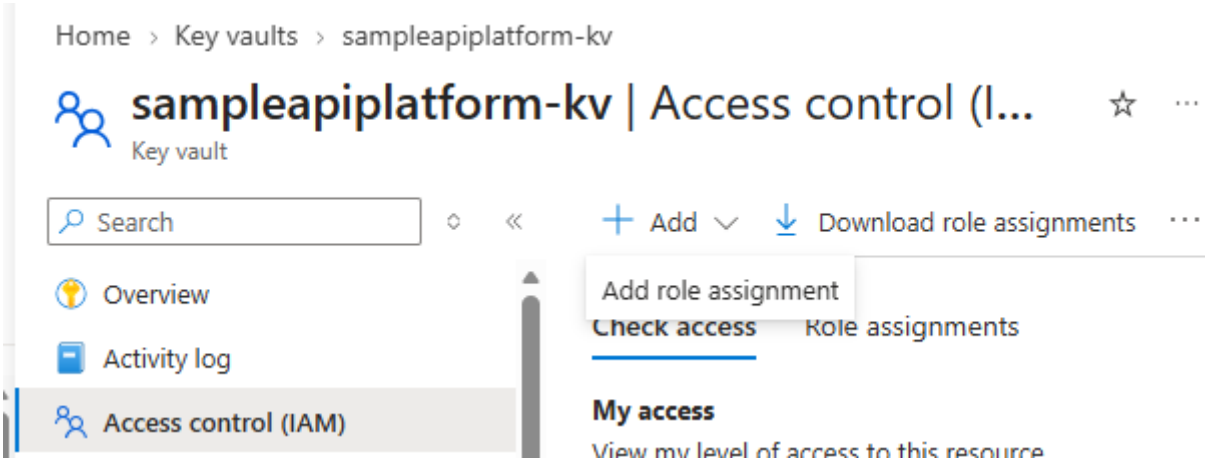
Click on Review and create and the Kay Vault will be created:



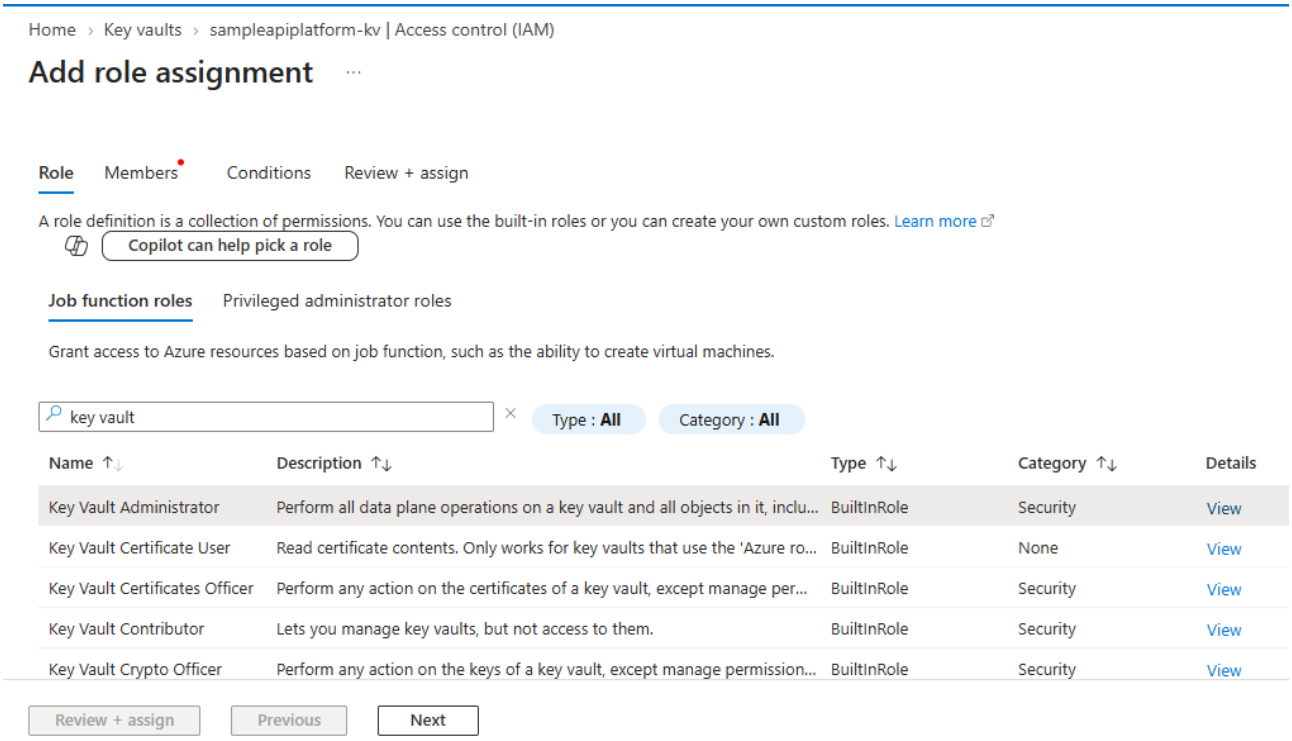
Generate Secrets In The Key Vault

Before creating a secret we need to grant the user that is creating the secret permission to do that:

Go to the sampleapiplatform-kv Key Vault > Access Control (IAM) > Click on “Add role assignment”:

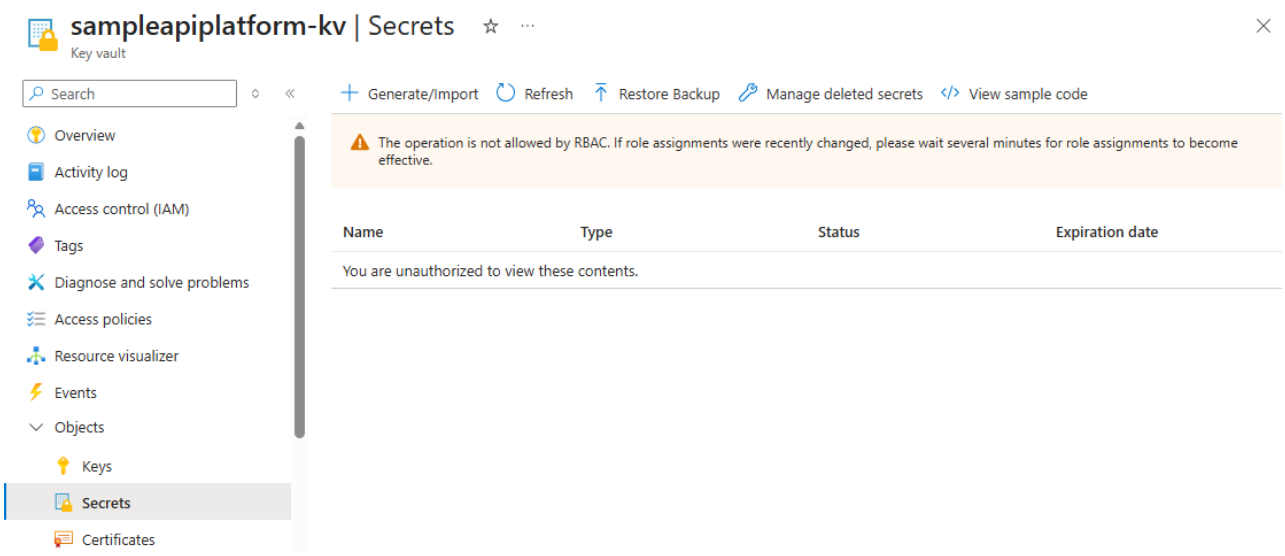


A list will be presented. Search for Key Vault Administrator (this role can create, delete, update secrets, manage keys and policies) > select it and click on the Next button:



An screen will be presented. Click on “Select members” and choose your account and click on Select and finally click on the “Review + assign” button.

Go to the sampleapiplatform-kv Key Vault > Objects > Secrets > click on the Generate / Import button :



Then we are presented with the Create a Secret screen:

Upload options = Manual

Name = sampleapiplatform-token-client1

Secret Value = 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a (a guid for instance. The value will be secret and only the clients get this value)

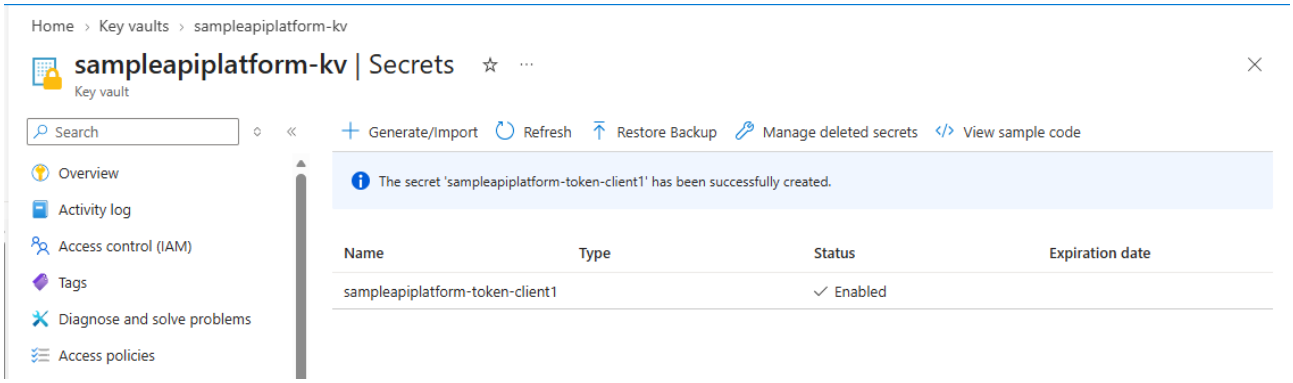
Content Type (optional) = empty (this is for metadata)

Set activation date = checkbox unchecked

Set expiration date = checkbox unchecked

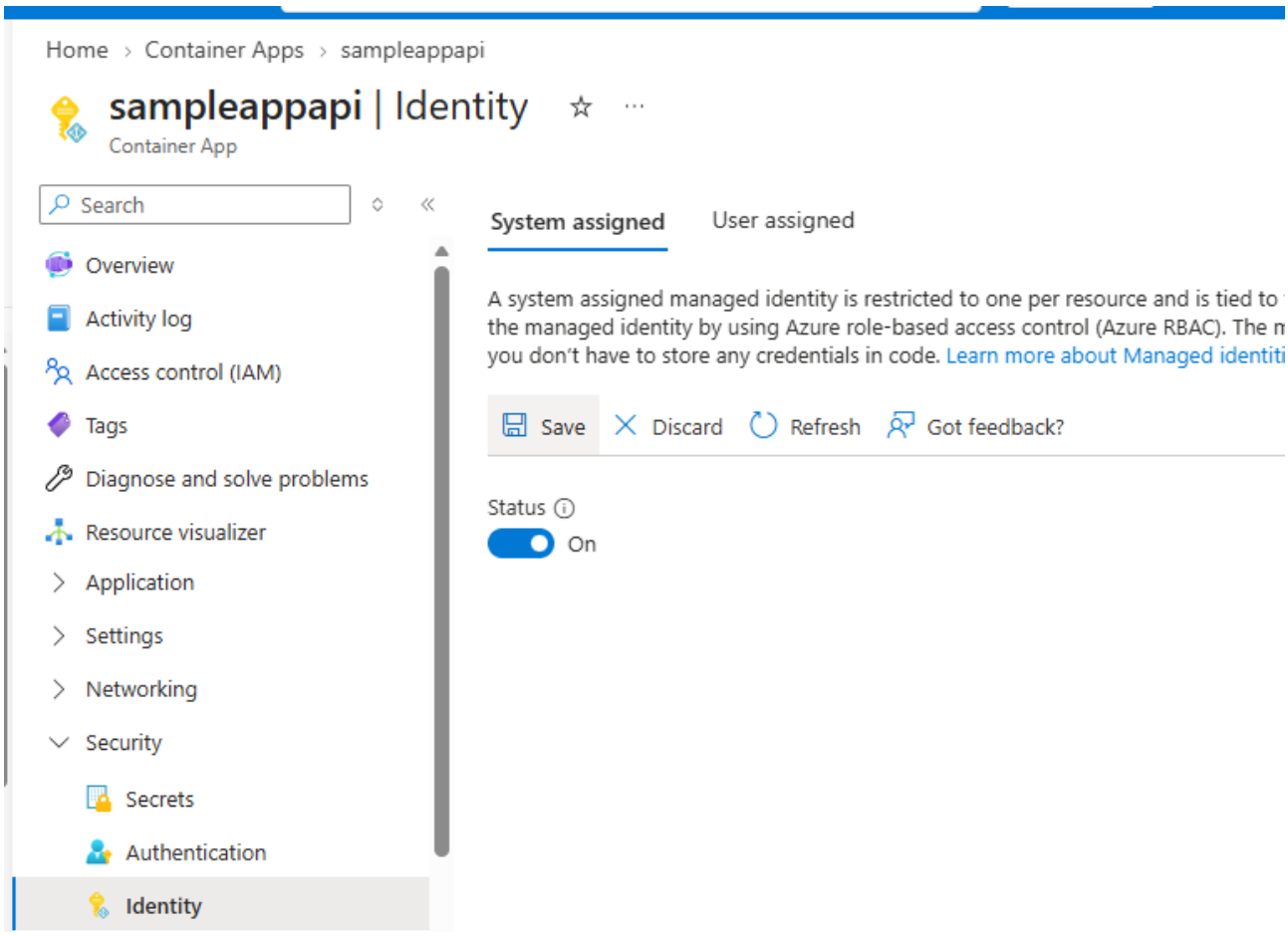
Enabled = Yes (it needs to be active)

Click on the Create Button and the secret will be created:

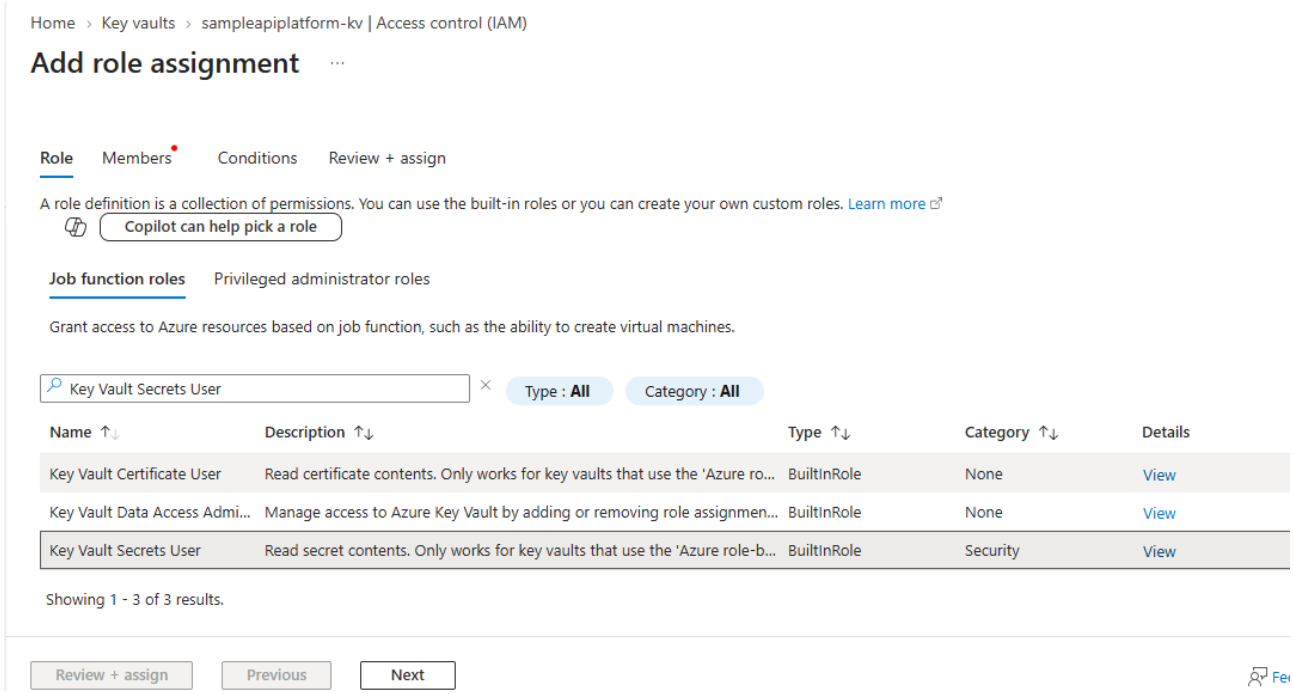


Connect the SampleAppApi with the Key Vault

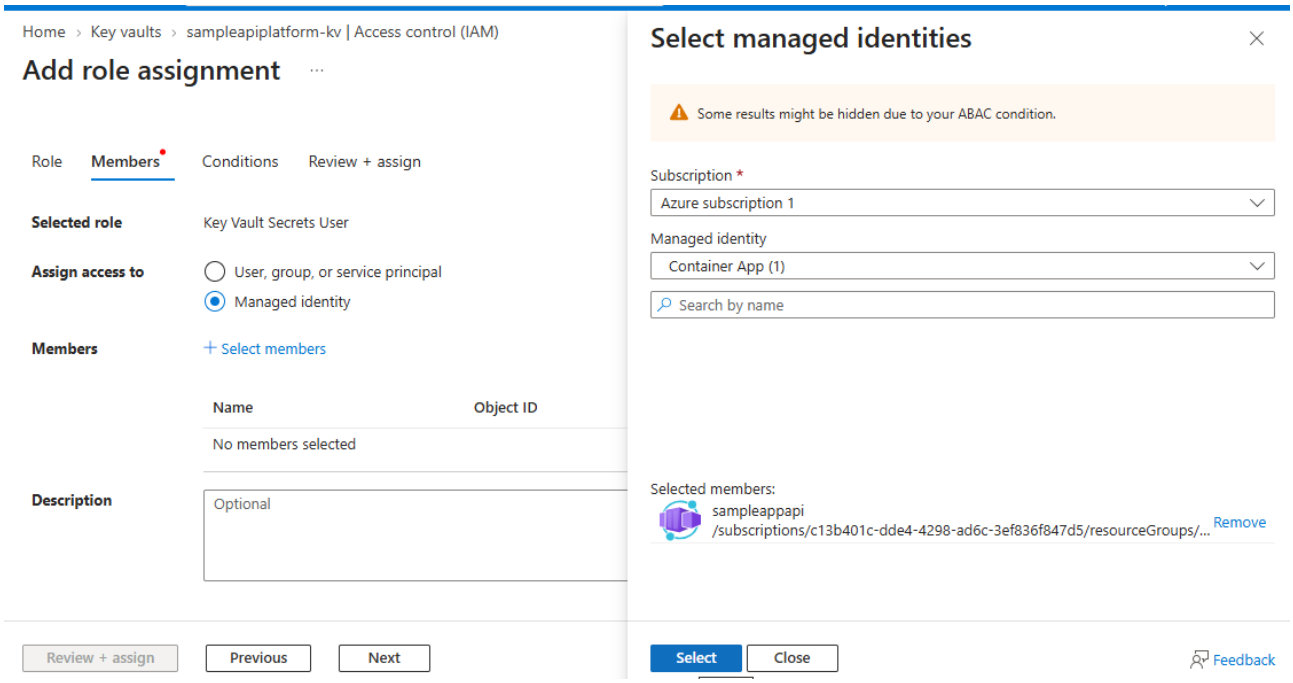
Activate the system assigned identity for the samppleappapi container app: Go to Container Apps > sampleappapi (contaner app) > Security > Identity > System Assigned = On and click on the Save button:



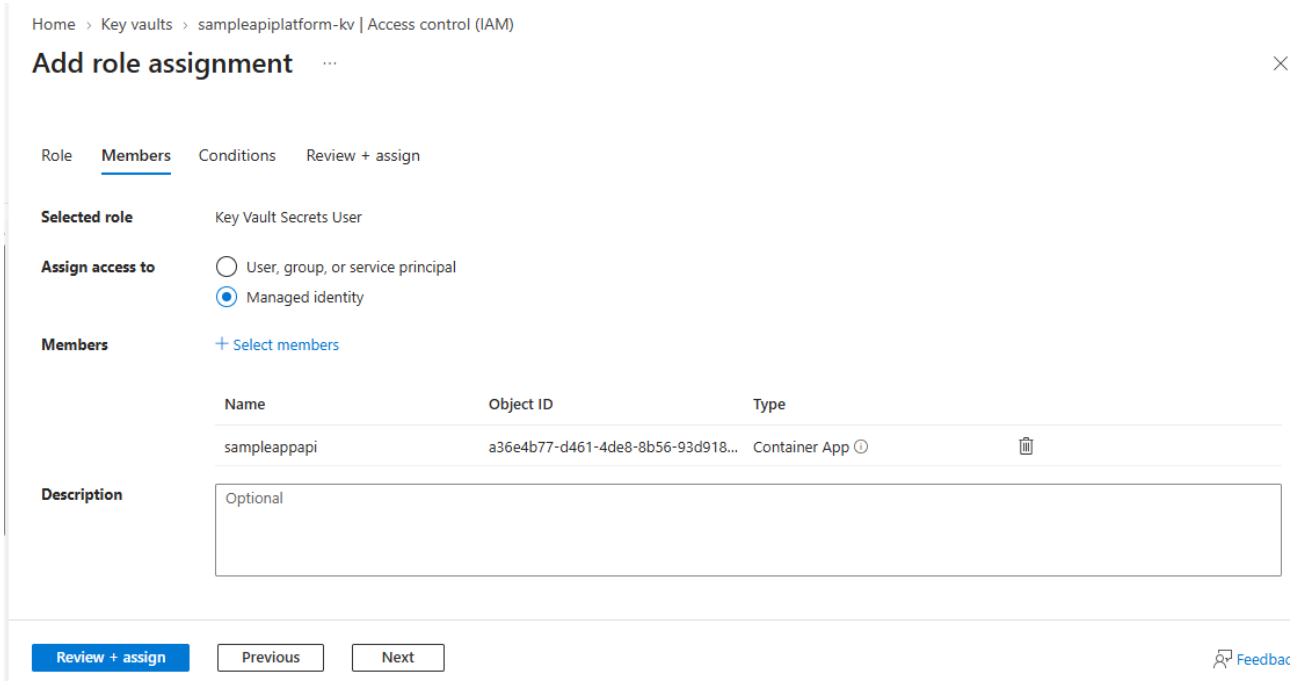
Go to the sampleapiplatform-kv > Access Control (IAM) > Add Role Assignment (similar to what we have been doin before to grant permission to a user to create a secret) > Look for the “Key Vault Secrets User”



And click on the “Next” button. Then on Assign Access to select Managed Identity and then click on “+ Select members” and choose the conatiner app and then the sampleappapi and click on “Select”.



Then on the “Review + assign” button:



Set the secret in the SampleAppApi Container App

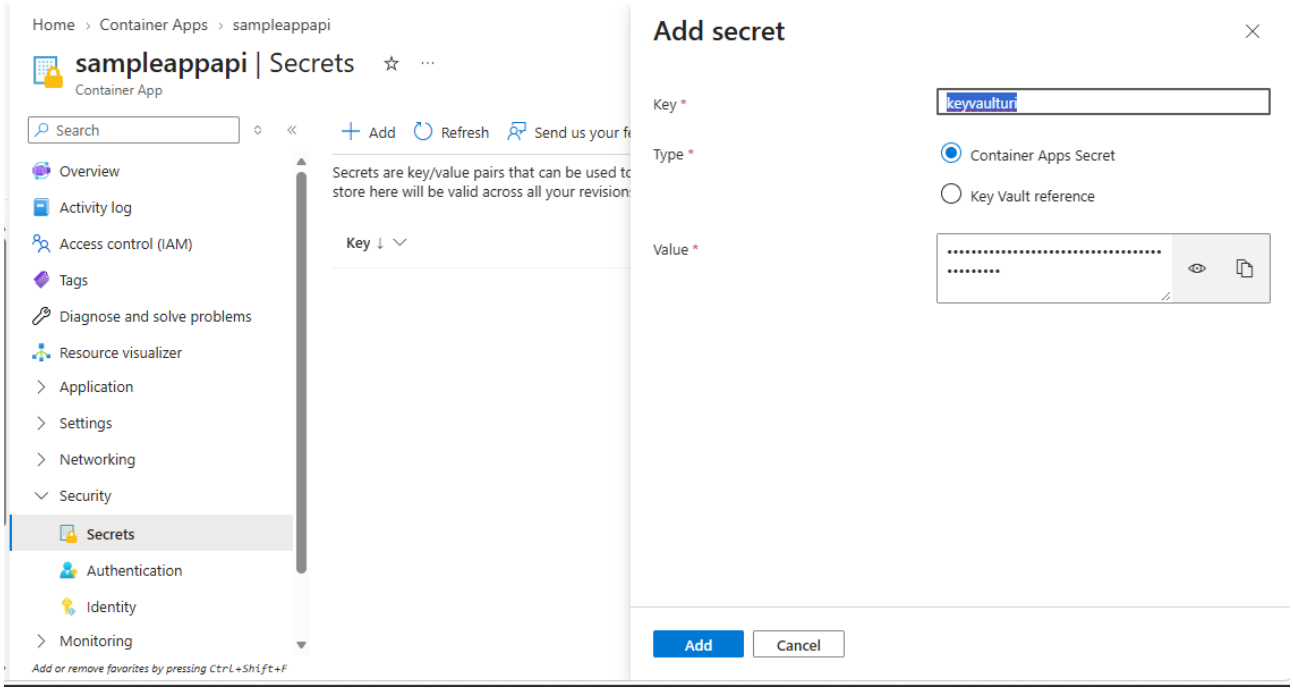
ContainerApps > sampleappapi > security > secrets > click on Add and fill out the the form with the following values:

Key = keyvaulturi

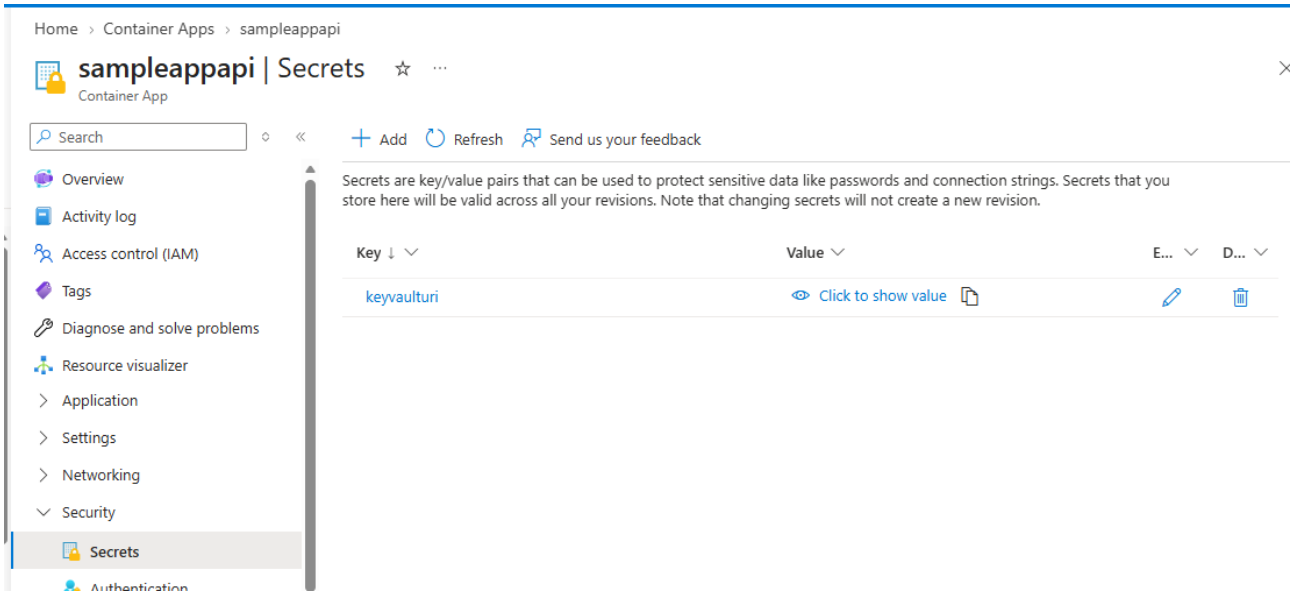
Type = Container App

Value = https://sampleapiplatform-kv.vault.azure.net/ (this is the url of our kay Vault that can be found in the Key Vault > Overview)

Click on the Add button:



If everything is successful then we should see the created secret:



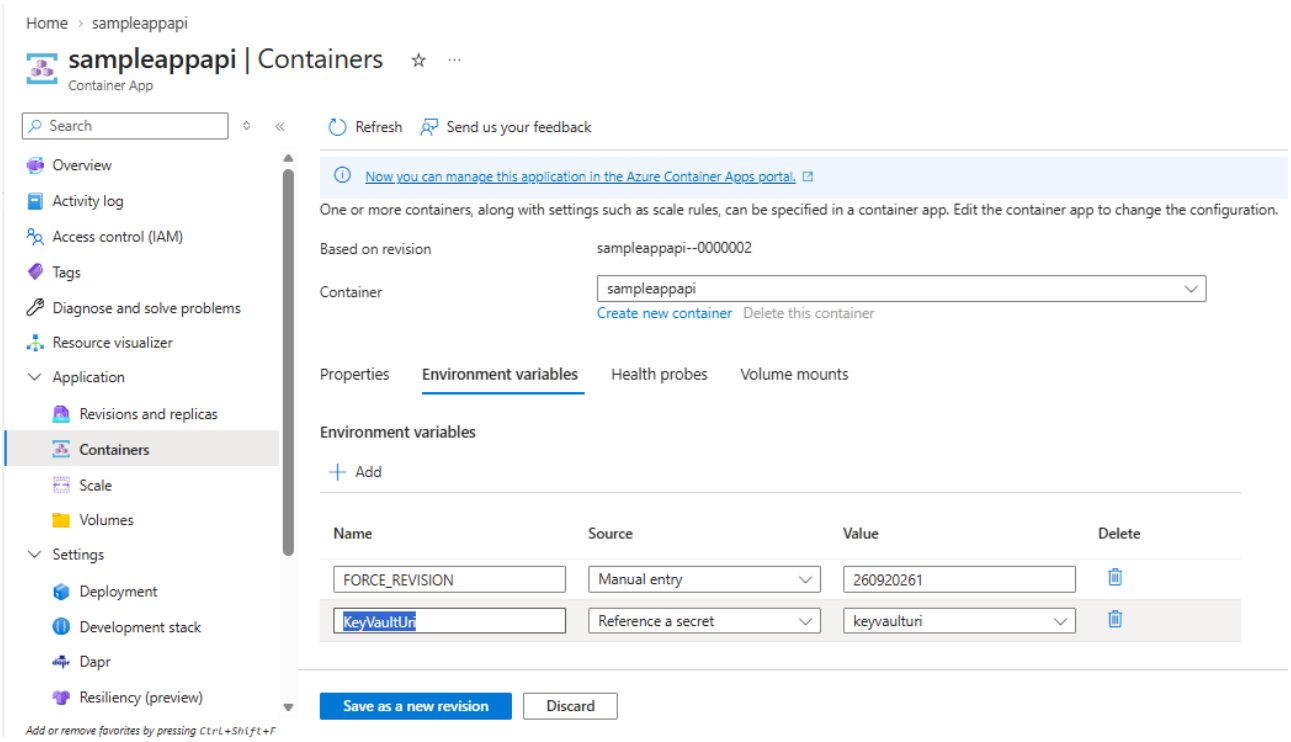
Create an environment variable

Go to the sampleappapi container app > Application > Containers > Environment Variables Tab. Create the following environment variable:

Name=KeyVaultUri

Source=Reference a secret

value= keyvaulturi (this value will be available automatically in the drop down menu)



Program.cs: integrate the key vault

Install the following NuGet packages:

dotnet add package Azure.Extensions.AspNetCore.Configuration.Secrets

dotnet remove package Azure.Core

dotnet remove package Azure.Identity

dotnet add package Azure.Identity --version 1.11.0

dotnet add package Azure.Extensions.AspNetCore.Configuration.Secrets --version 1.2.2

Add the lines in bold to Program.cs in the SampleAppApi code:

using Azure.Identity;
using Azure.Extensions.AspNetCore.Configuration.Secrets;
using SampleAppApi.Extensions;
using SampleAppApi.Interfaces.ExternalServices;
using SampleApi.Options;

var builder = WebApplication.CreateBuilder(args);
// --
// 1 KeyVaultUri loaded from the Container App Secret
// --

```
var keyVaultUri = builder.Configuration["KeyVaultUri"];

if (string.IsNullOrWhiteSpace(keyVaultUri))
{
    throw new InvalidOperationException("KeyVaultUri configuration value is missing.");
}
// ------------------------------------------------------
// [2] Key Vault integrate
// ------------------------------------------------------
builder.Configuration.AddAzureKeyVault(
    new Uri(keyVaultUri),
    new DefaultAzureCredential());


builder.Services.AddScoped<IDataAccessClient, DataAccessClientDapr>();

//Swagger
builder.Services.AddSwaggerDocumentation();

// Controllers
builder.Services.AddControllers();

// ------------------------------------------------------
// [3] ServiceTokenOptions load (Tokens from the Key Vault)
// ------------------------------------------------------
builder.Services.Configure<ServiceTokenOptions>(options =>
{
    var token1 = builder.Configuration["sampleapiplatform-token-client1"];

    if (string.IsNullOrWhiteSpace(token1))
    {
        throw new InvalidOperationException("Token 'sampleapiplatform-token-client1' is
missing from Key Vault.");
    }

    options.ValidTokens.Add(token1);
});

// ------------------------------------------------------
// [4] Token-Middleware registrieren
// ------------------------------------------------------
builder.Services.AddSingleton<TokenValidationMiddleware>();

// Authorization
builder.Services.AddAuthorization();
// Dapr Client aktivieren (GENAU HIER)
builder.Services.AddDaprClient();
builder.Services.AddHttpClient("dapr", c =>
{
    c.BaseAddress = new Uri("http://localhost:3500/");
});
```

```
var app = builder.Build();
// this is not necessary:  GlobalExceptionMiddleware
app.UseMiddleware<GlobalExceptionMiddleware>();

if (app.Environment.IsDevelopment() || app.Environment.IsStaging())
{
   app.UseSwagger();
   app.UseSwaggerUI();
}

//Here we check if the token is valid or not. It will check if the request header has a valid X-API-SECRET. If not will return 401
//put it before the authentication and Authorization
//blocks bots/scanners before they hit the auth logic
app.UseMiddleware<TokenValidationMiddleware>();


// ⭐ Authentication + Authorization middleware
app.UseAuthentication();
app.UseAuthorization();

app.MapControllers();

app.Run();
```

Create the following classes:

SampleApiApp > Options > ServiceTokenOptions.cs

```
namespace SampleApi.Options;

public class ServiceTokenOptions
{
   public List<string> ValidTokens { get; set; } = new();
}
```

And SampleApiApp > Middlewares > TokenValidationMiddleware.cs

```
using Microsoft.Extensions.Options;
using SampleApi.Options;

public class TokenValidationMiddleware
{
   private readonly RequestDelegate _next;
   private readonly ServiceTokenOptions _options;

   public TokenValidationMiddleware(RequestDelegate next, IOptions<ServiceTokenOptions> options)
   {
```

```
        _next = next;
        _options = options.Value;
    }

    public async Task InvokeAsync(HttpContext context)
    {
        var token = context.Request.Headers["X-Service-Token"].FirstOrDefault();

        if (string.IsNullOrWhiteSpace(token) || !_options.ValidTokens.Contains(token))
        {
            context.Response.StatusCode = StatusCodes.Status401Unauthorized;
            await context.Response.WriteAsync("Invalid or missing service token.");
            return;
        }

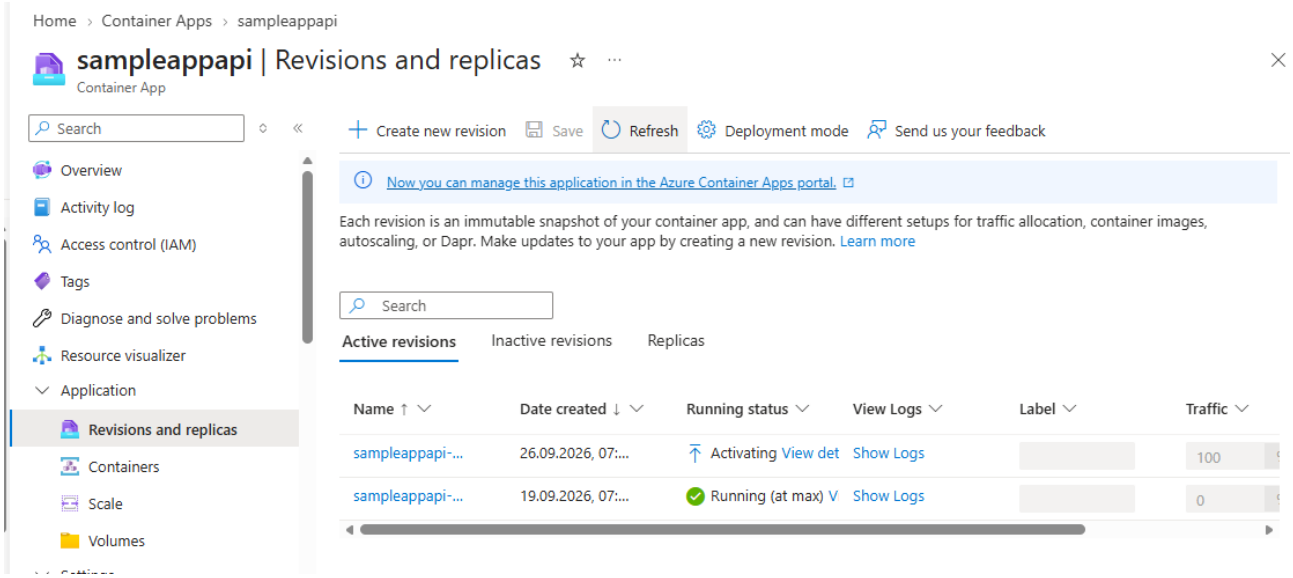
        await _next(context);
    }
}
```

Deploy the new code and make sure that the pipeline finishes successfully.

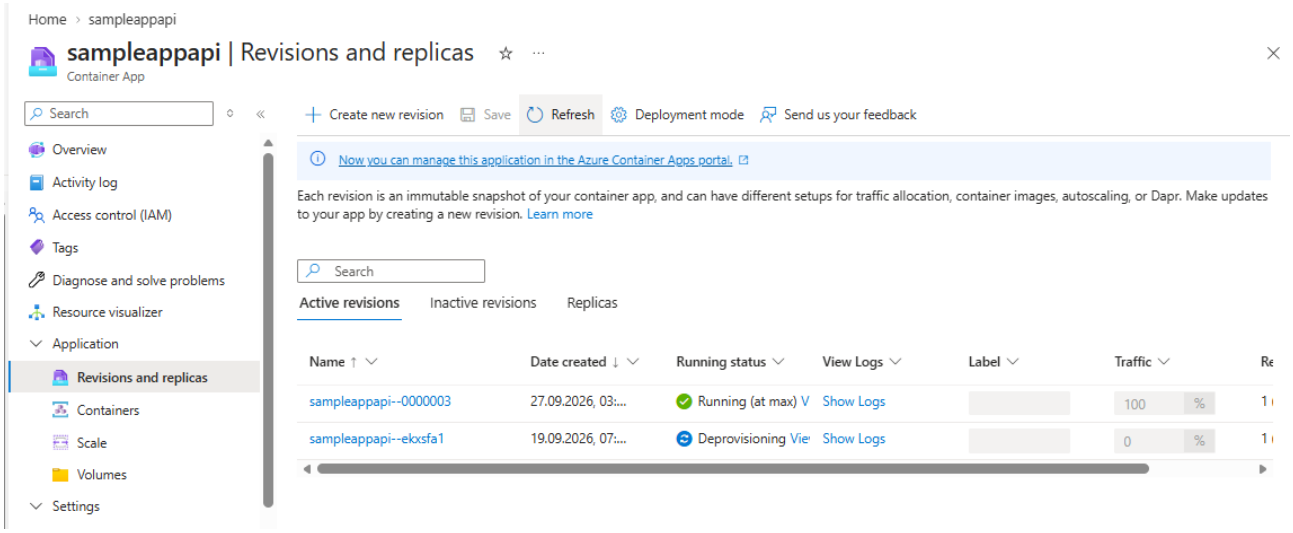
Restart the sampleappapi Container App

This is a necessary because contaner apps only load secrets when they start. We can do it from the shell:

```
az containerapp update \
  --name sampleappapi \
  --resource-group SampleAppApi_rg \
  --set-env FORCE_REVISION=26092026
```



So check that the new replica activates successfully



Check if the application is working

Sampleappapi Container App > overview > copy the url and add api/movies:

curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies -H "X-API-SECRET: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a"

into the command line.

Check the logs and correct the errors.

If we correct an error and we change the code we should force a new replica to be generated. So run the following command from the Azure shell, by changing everytime the vale:

```
az containerapp update \
  --name sampleappapi \
  --resource-group SampleAppApi_rg \
  --set-env FORCE_REVISION=202609260540
```

Troubleshooting Errors

For instance after typing the following url in the browser:
sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies

I get the following error in the browser:

```
{"status":500,"title":"Internal Server Error","detail":"An unexpected error occurred.","traceId":"0HNOS8V3U6ILA:00000002"}
```

This message was generated by the line in Program.cs:

app.UseMiddleware<GlobalExceptionMiddleware>();

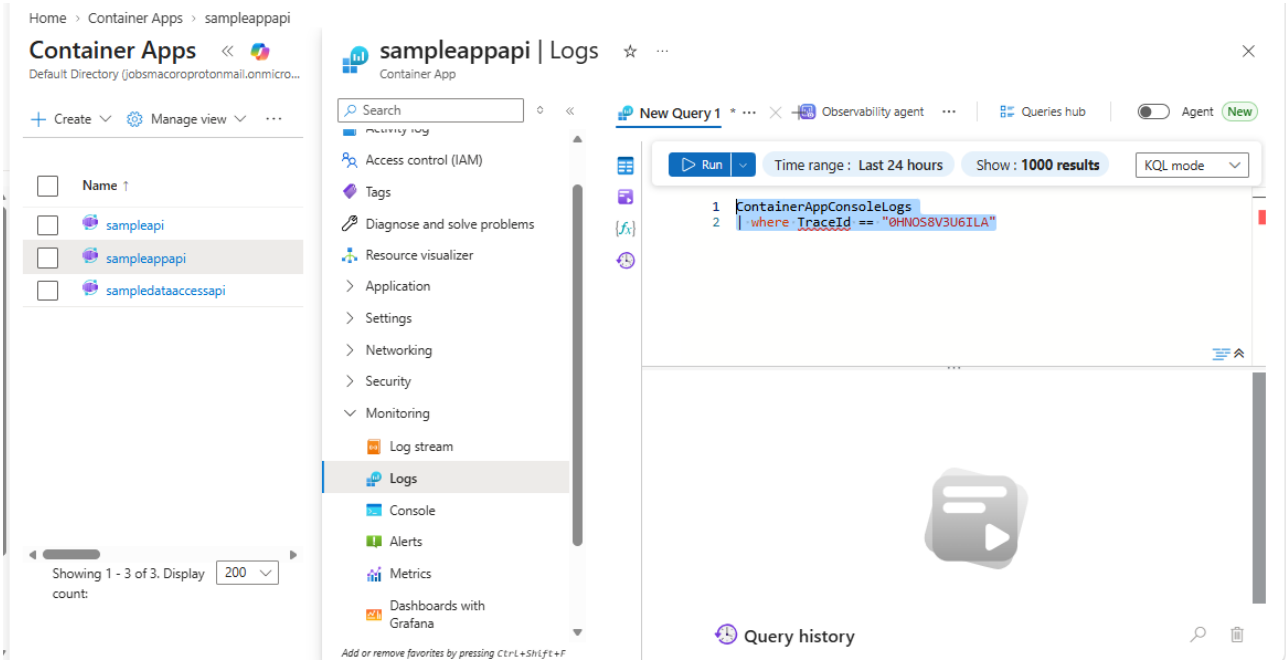
The class GlobalExceptionMiddleware genrates this message. If this line would not be here then the error would be exposed in the browser.

In this case the error is pointing to a certain traceId. We can try to find out more in the Logs.

Go to the sampleappapi Container App > Monitoring > Logs > Disable the Agent Toggle and use the KQL Mode (Kusto Query Language) and type the following query:

```
ContainerAppConsoleLogs
| where TraceId == "0HNOS8V3U6ILA"
```

And click on the Run button



However this query did not give any result. So I uncommitted the line in Program.cs and now I was able to see the errors in the logs and the browser does not display any information only that a 500 error happened.

To Correct

The user Assigned identity is only needed with App1 App2 App3 wanted to access a resource. But with dapr in place in our situation is not necessary. Only for the database we need this identity but we can implement a System Assigned Identity instead.

I keep the User Assigned Identity to show to the employer

To access to the database: create a secret in the container sampledataaccessapi container with the following properties:

Key=moviesdbsecret

Type=Container Apps Secret

Value=Server=tcp:sampleapiserver.database.windows.net,1433;Database=free-sql-db-5542591;Authentication=Active Directory Default;Encrypt=True;

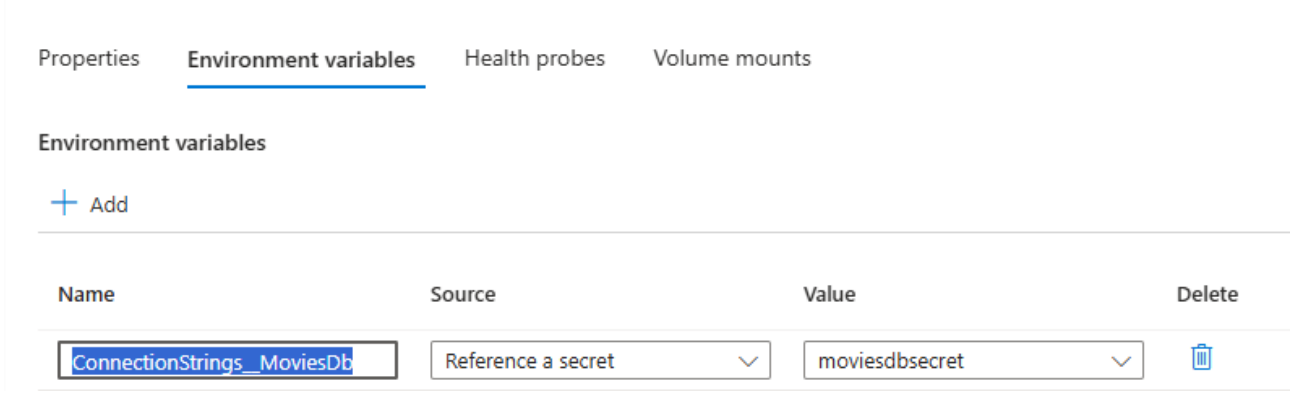
Notice that the value does not provide any password or user id because we are using the user defined identity.

Then create in the same container , sampledataaccessapi, an environment varibale :

Name=ConnectionStrings__MoviesDb

Source=Reference a secret

Value= moviesdbsecret



Created a new environment variable in sampledataaccessapi with the application id in order to connect to the database:

`ClientConnectionId:c93a55b4-29c9-439c-9fdf-f66f5e9eeeb9`

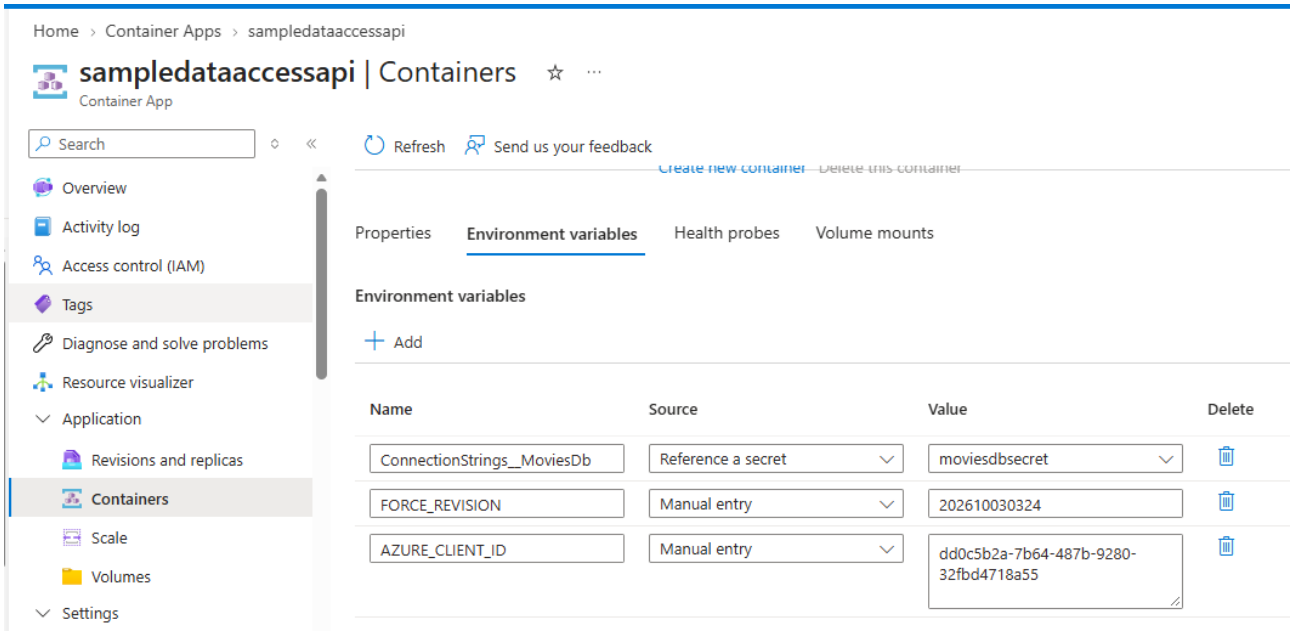
```
TState state, CancellationToken cancellationToken)
2026-10-04T04:44:55.8146669Z stdout F          at Microsoft.EntityFrameworkCore.Storage.ExecutionStrategy.ExecuteAsync[TState,TResult](TState
state, Func`4 operation, Func`4 verifySucceeded, CancellationToken cancellationToken)
2026-10-04T04:44:55.8146765Z stdout F          at
Microsoft.EntityFrameworkCore.Query.Internal.SingleQueryingEnumerable`1.AsyncEnumerator.MoveNextAsync()
2026-10-04T04:44:55.8146801Z stdout F       ClientConnectionId:c93a55b4-29c9-439c-9fdf-f66f5e9eeeb9
2026-10-04T04:44:55.8146826Z stdout F       Error Number:-2,State:0,Class:11
2026-10-04T04:44:55.8147510Z stdout F fail: SampleDataAccessApi.Controllers.MoviesController[0]
2026-10-04T04:44:55.8147565Z stdout F       MovieController.GetAll Exception: Connection Timeout Expired.  The timeout period elapsed during
the post-login phase.  The connection could have timed out while waiting for server to complete the login process and respond; Or it could
have timed out while attempting to create multiple active connections.  The duration spent while attempting to connect to this server was -
[Pre-Login] initialization=32; handshake=384; [Login] initialization=0; authentication=0; [Post-Login] complete=29161;
2026-10-04T04:46:15.87931  No logs since last 60 seconds
```

This clientConnectionId changes with each replica and needs to be updated. This value is only displayed when I issue an HTTP GET and not a POST.

Name = AZURE_CLIENT_ID

Source= Manual Entry

Value = `c93a55b4-29c9-439c-9fdf-f66f5e9eeeb9`



Register Postman in EntraID

This is the part where we created a secret

Test Postman / curl

This has been covered with curl

User Registration / Login with Entra ID

one quick decision point, because it changes everything:

"Internal or external" is the fork in the road.

•**Internal only** (work/school accounts) → single Entra ID tenant, simplest setup.

•**External/public users who self-register** → you need **Entra External IDs** (or B2C), which is a separate tenant and more setup.

My recommendation: **start with a single-tenant app registration now**, get login + DB access working end-to-end, then bolt on external users later. Switching from single-tenant to multi-tenant/external later is easy; starting over is not.

App registration

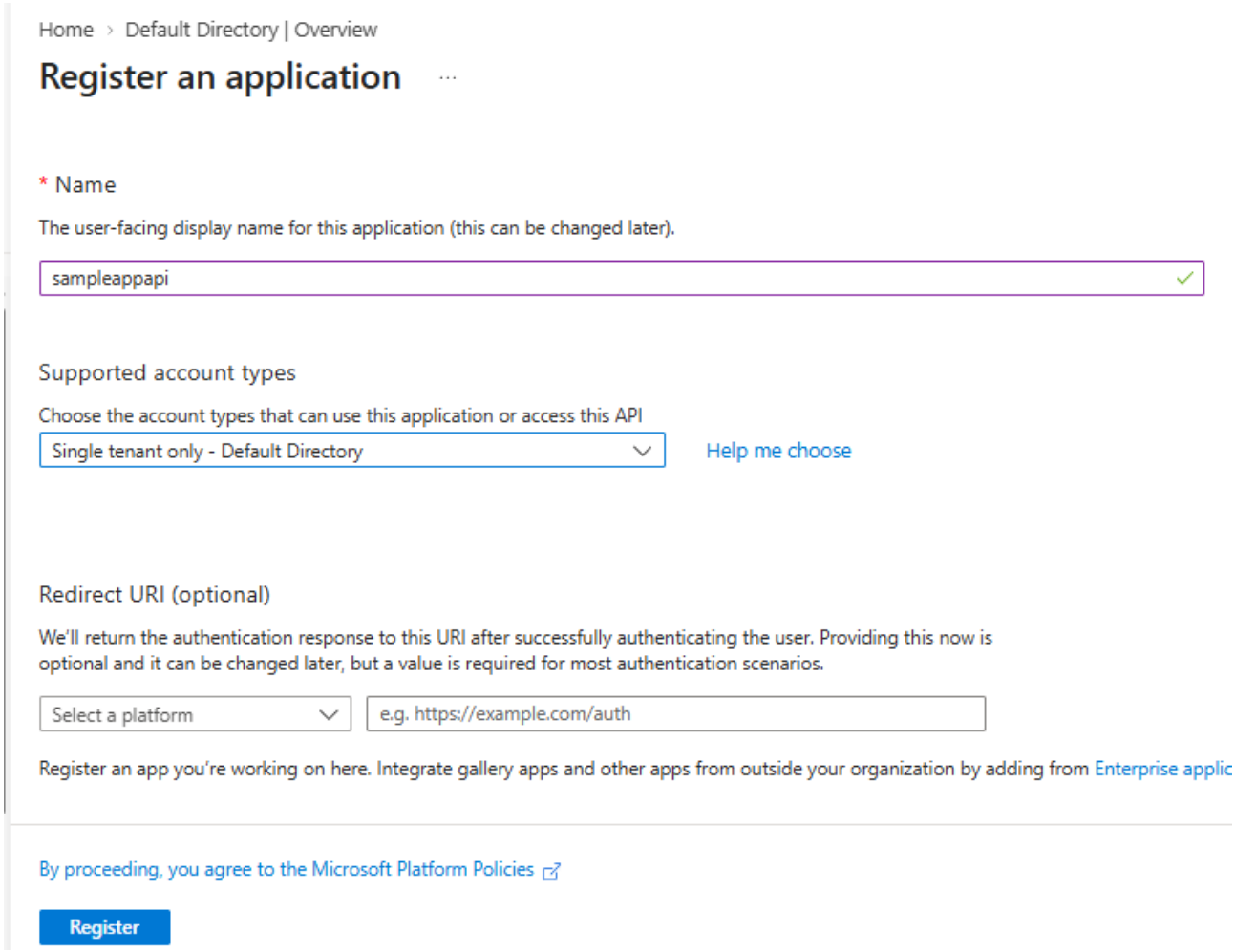
Go to EntraID > Add > Choose App Registration

Name = sampleappapi (like the container)

Supported Account Types = "Accounts in this organizational directory only" (Single tenant)

Redirect URI = blank

Click **Register**.



Application (client) ID = f9a3d163-e6e5-481b-9ea9-869076084bc7

Directory (tenant) ID = 9ca003dc-da5d-4b31-808b-9d52337fbf6a

In the SampleAppApi from the directory where the Program.cs is located , download the following packages:

```
dotnet add package Azure.Identity --version 1.12.0
dotnet add package Microsoft.Identity.Web --version 2.7.0
```

In appsettings.json, add the following entry:

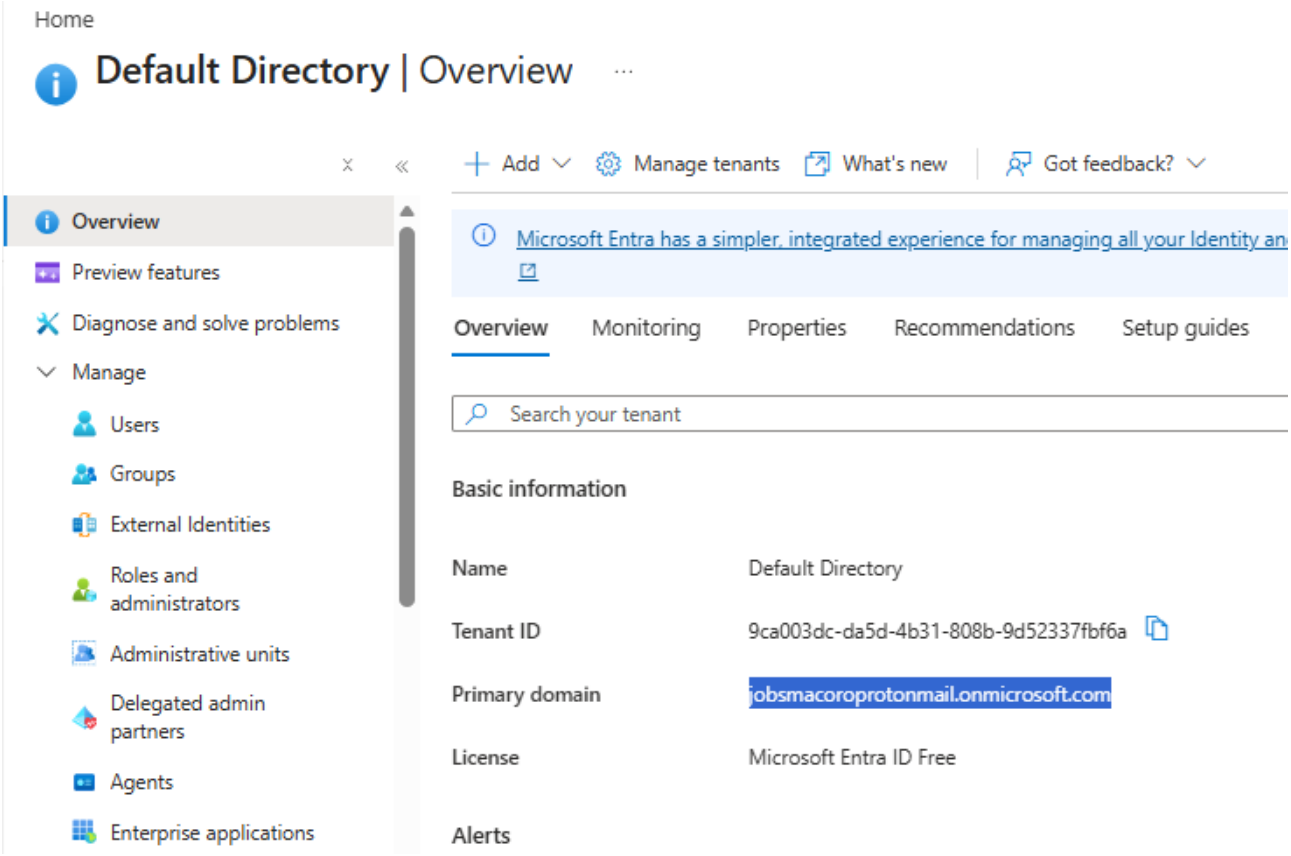
```
{
  "AzureAd": {
   "Instance": "https://login.microsoftonline.com/",
   "Domain": "jobsmacoroprotonmail.onmicrosoft.com",
```

```
  "TenantId": "9ca003dc-da5d-4b31-808b-9d52337fbf6a",
  "ClientId": "f9a3d163-e6e5-481b-9ea9-869076084bc7",
  "Audience": "api://f9a3d163-e6e5-481b-9ea9-869076084bc7",
  "Scope": "access_as_user"
 },

 "Jwt": {
  "Issuer": "MovieApi",
  "Audience": "MovieApiClient",
  "ExpiresInMinutes": 60
 },
 "ConnectionStrings": {
  "sampleApi": ""
 },
 "AllowedHosts": "*"
}
```

~~Domain = localhost . For a demo is fine. Otherwise should be something like "**yourdomain**.onmicrosoft.com". YourDomain can be found in azure.~~

The domain can be found in EntraId > Default Directory > Overview
Primary Domain = jobsmacoroprotonmail.onmicrosoft.com



Modify Program.cs with the following code:

before `builder.Build()`:

```
using Microsoft.AspNetCore.Authentication.OpenIdConnect;
using Microsoft.Identity.Web;
using Microsoft.Identity.Web.UI;

// 1. Register authentication with Microsoft Identity
builder.Services.AddAuthentication(JwtBearerDefaults.Authenticatio
nScheme)
    .AddMicrosoftIdentityWebApi(builder.Configuration.GetSection("
AzureAd"));


builder.Services.AddControllersWithViews();

// 2. Accept BOTH app ID URI (v1) and client ID (v2) as valid
audiences
builder.Services.Configure<JwtBearerOptions>(
    JwtBearerDefaults.AuthenticationScheme,
    options =>
    {
        var clientId = builder.Configuration["AzureAd:ClientId"];
        var appIdUri = builder.Configuration["AzureAd:Audience"];

        options.TokenValidationParameters.ValidAudiences = new[]
        {
            appIdUri,   // api://f9a3d163-e6e5-481b-9ea9-
869076084bc7
            clientId    // f9a3d163-e6e5-481b-9ea9-869076084bc7
        };

        options.TokenValidationParameters.ClockSkew =
TimeSpan.FromMinutes(2);
    });


// 3. Register the scope policy
builder.Services.AddAuthorization(options =>
{
    options.AddPolicy("AccessAsUser", policy =>
        policy.RequireAssertion(ctx =>
            ctx.User.HasClaim(c => c.Type == "scp" &&
c.Value.Split(' ').Contains("access_as_user")) ||
            ctx.User.HasClaim(c => c.Type == "scp2" &&
c.Value.Split(' ').Contains("access_as_user"))));
});
```

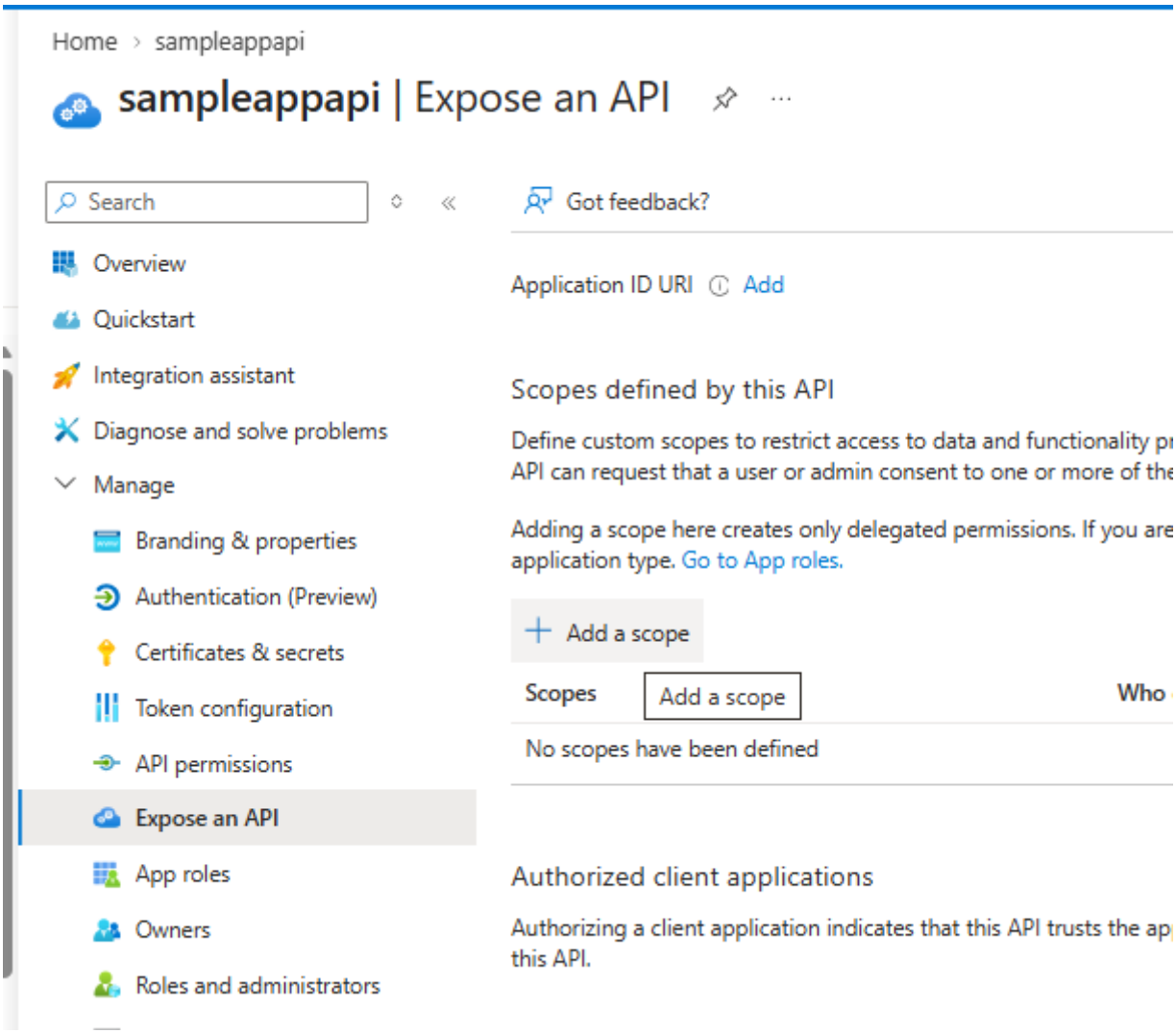
after `var app = builder.Build();`

```
app.UseAuthentication();
app.UseAuthorization();
app.MapControllers();//MapRazorPages()/MapDefaultControllerRoute()
```

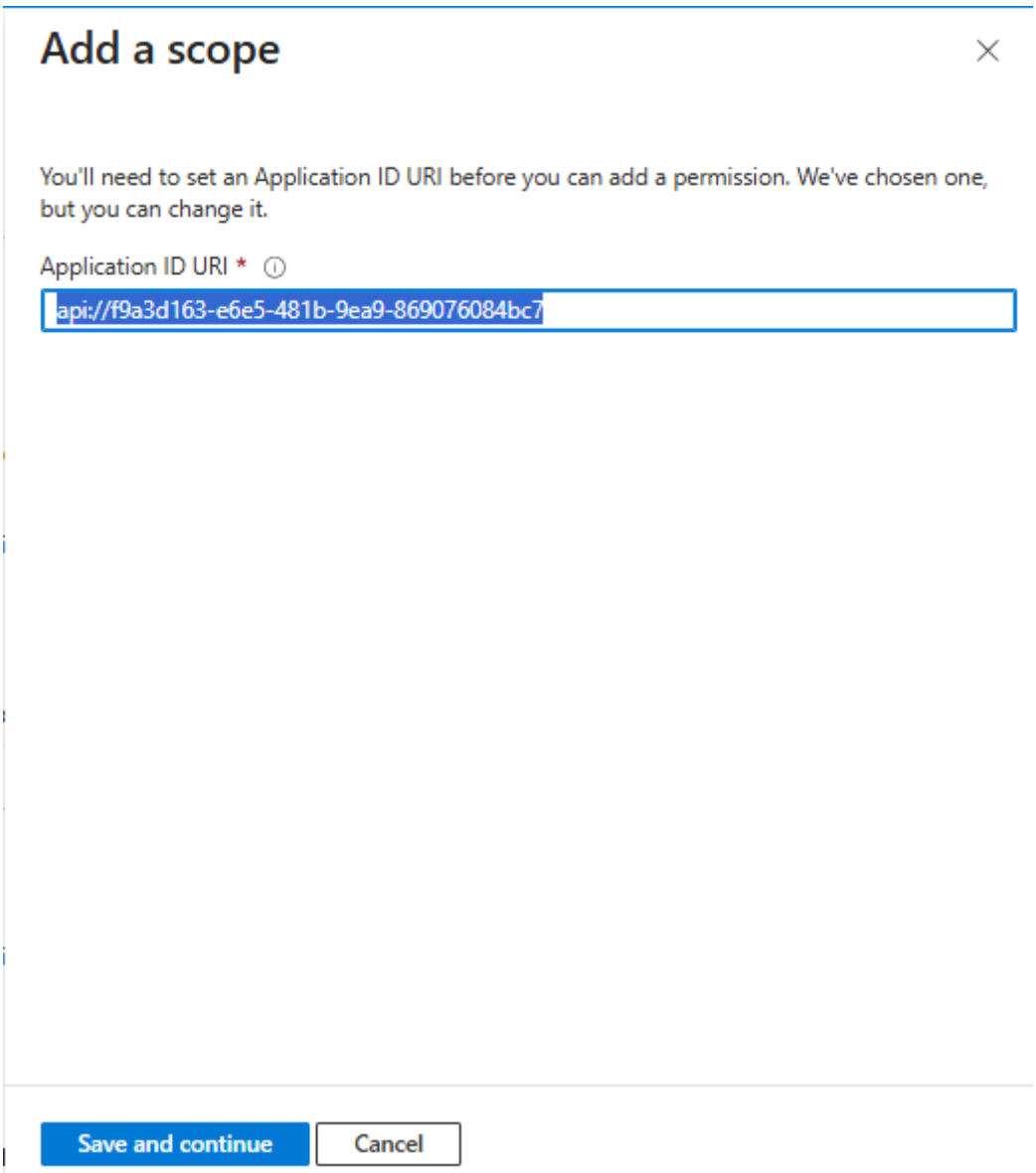
Modify the MoviesController and add thAuthorize as follows:

```
[Authorize(Policy = "AccessAsUser")]
```

Go to EntraID > sampleappapi > Expose An Api > Add a scope



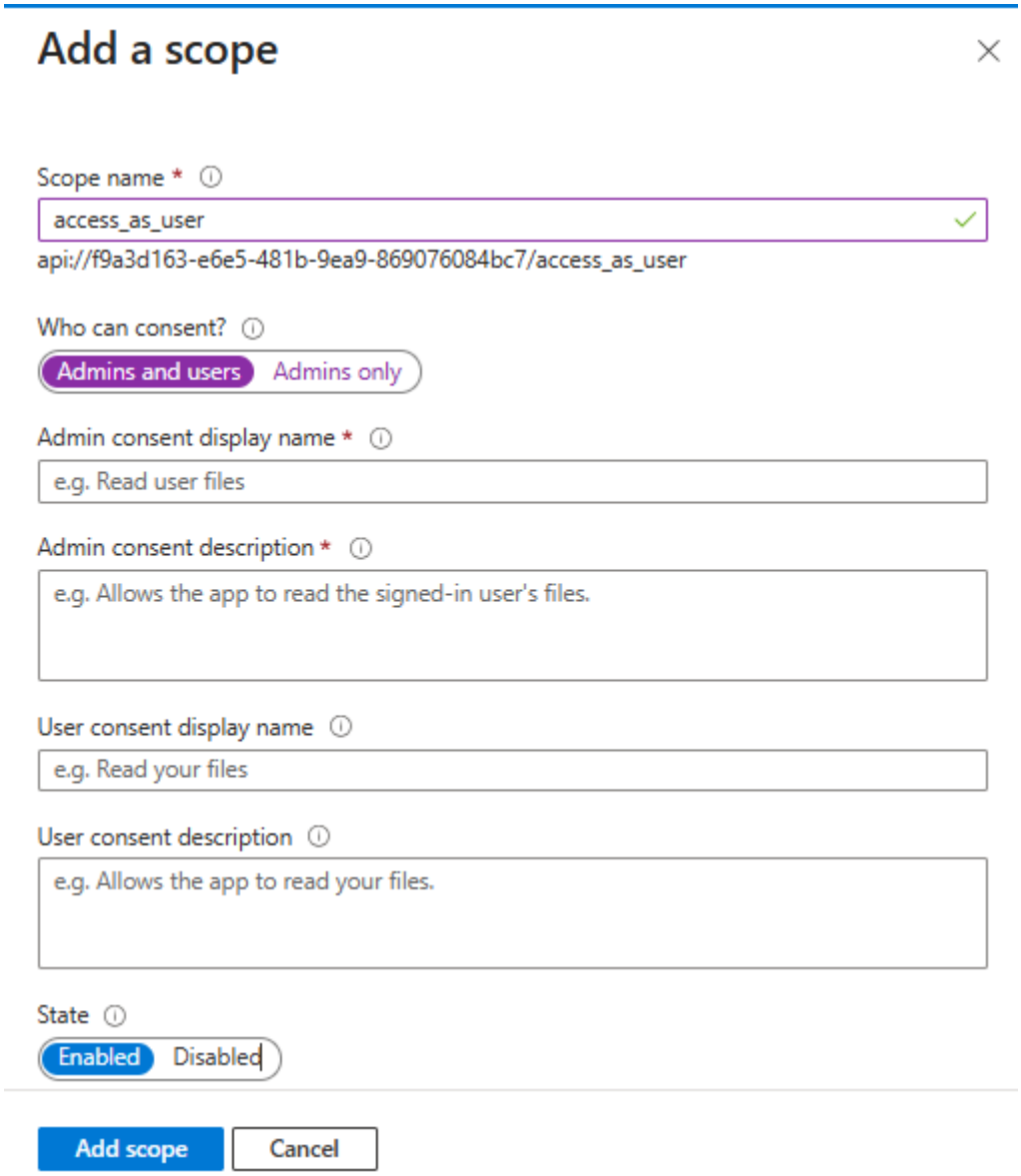
We get Application ID Uri = api://f9a3d163-e6e5-481b-9ea9-869076084bc7



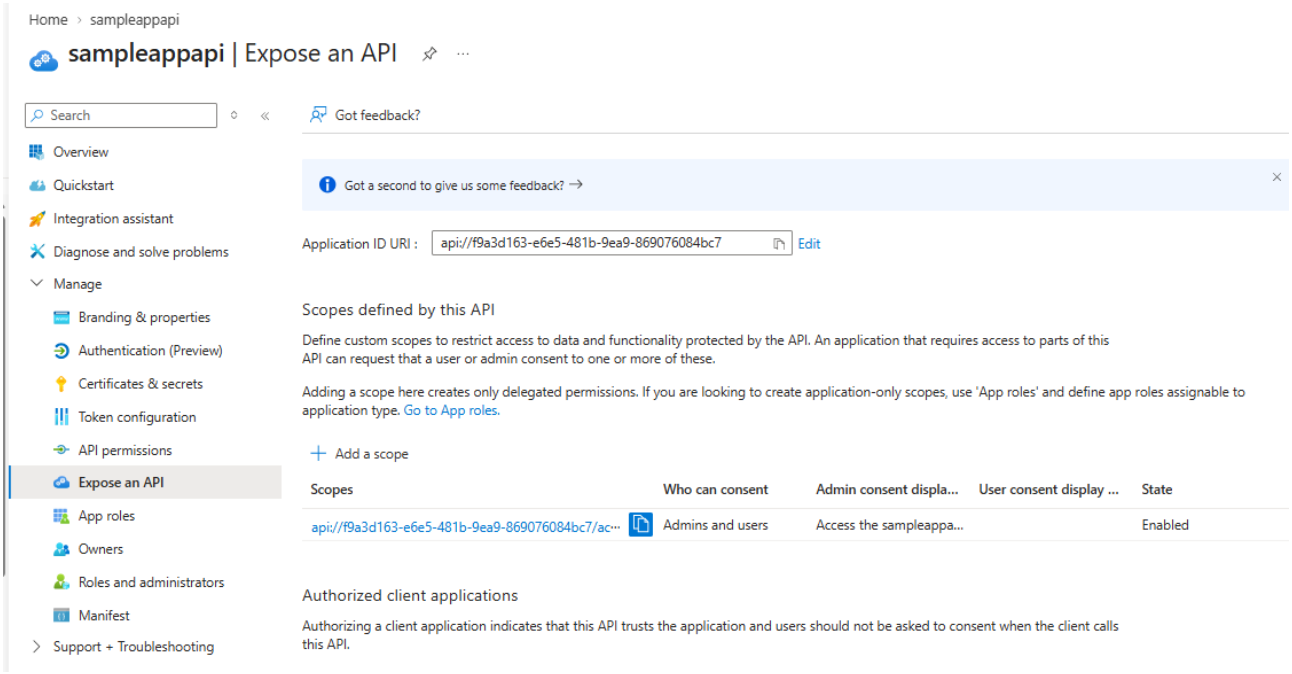
Then we need to add the scope

scope name = access_as_user
Who can Consent? = Admins and users
Admin consent display name = Access the sampleappapi application
Admin consent description= Allow the application to access sampleappapi on behalf of the signed-in user
State = Enabled

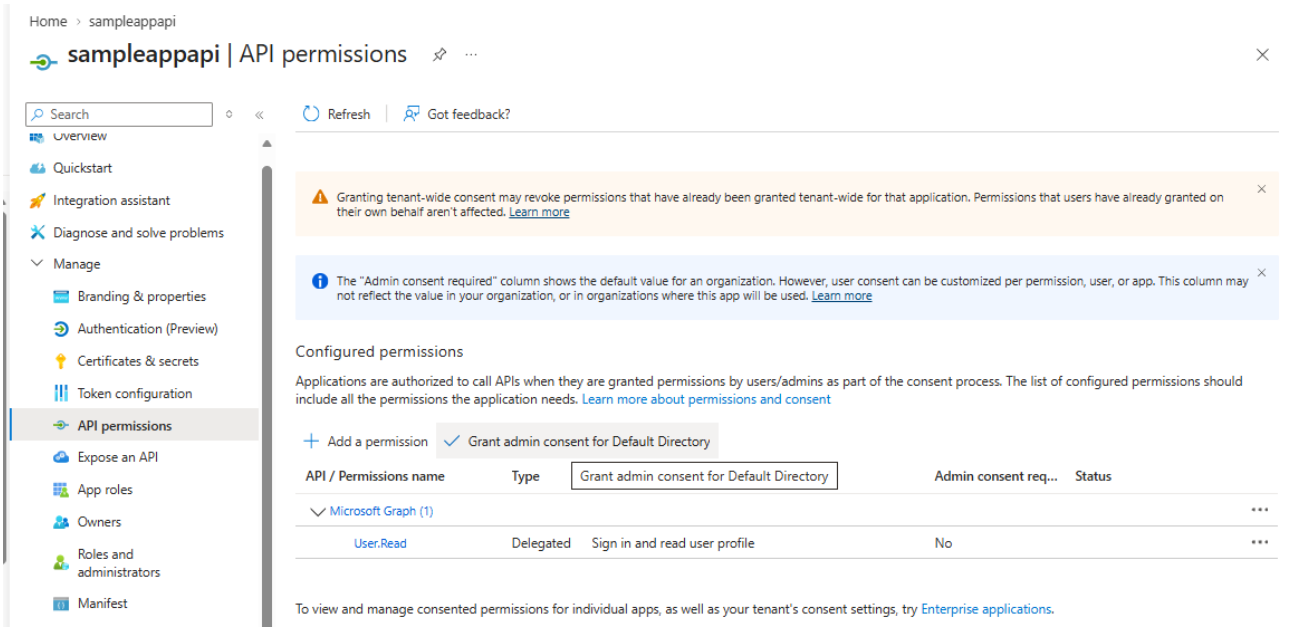
After exposing the Api we will have the following screen:



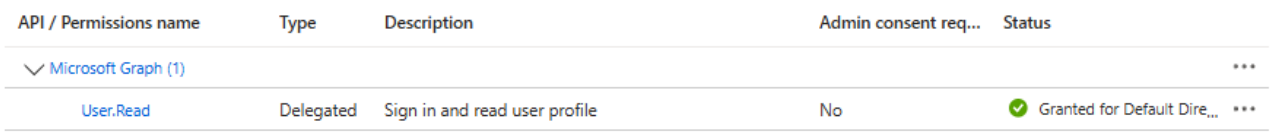
After that we will get the following screen:



For the next steps do the following: API Permissions > Grant Admin Consent for the Default Directory (in this case we have only 1 so it is not an issue to choose this option).



After doing that we will see that the status column has changed:

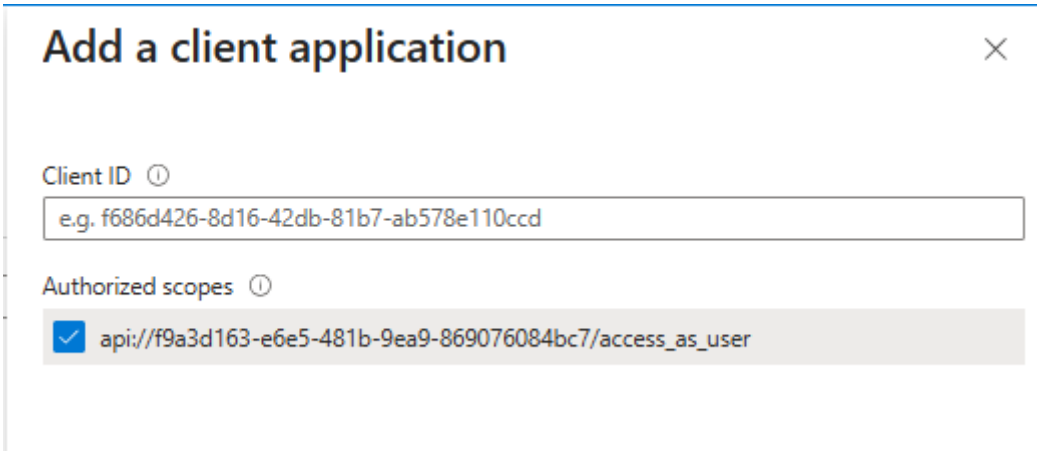


Then click on + Add Client Application

Client ID = 04b07795-8ddb-461a-bbee-02f9e1bf7b46

Check the checkbox :

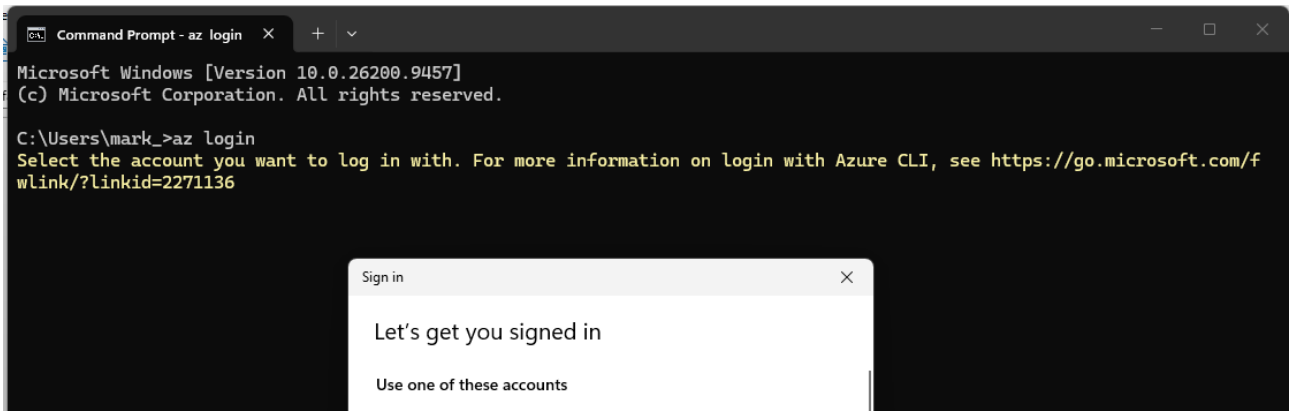
api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user



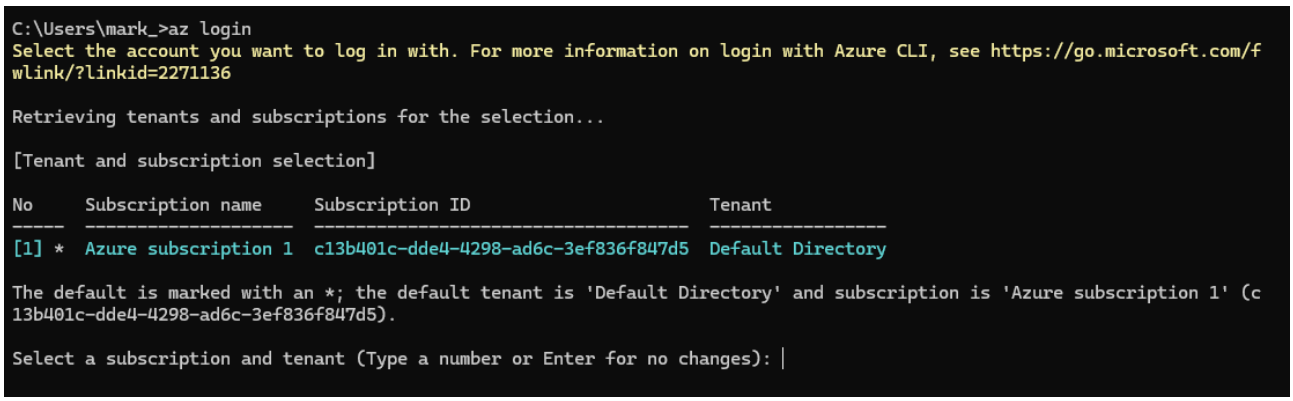
We need to add tupgrade some packages in sampleappapi.csproj and in the sampleappapi.test.csproj:
<PackageReference Include="Azure.Identity" Version="1.12.0" />
<PackageReference Include="Microsoft.Identity.Web" Version="4.16.0" />

dotnet restore
dotnet build

Now login insot azure from the command prompt and choose the microsoft account that you want to use:



The following screen is presented:



And is asking to select the subscription and tenant. As we have only 1, type 1.

Now type in the command prompt the following command:

```
az logout
az login --tenant 9ca003dc-da5d-4b31-808b-9d52337fbf6a --scope "api://f9a3d163-e6e5-481b-9ea9-869076084bc7/.default"
```

And choose the account and the type 1 again as we did before.

```
az account get-access-token --resource "api://f9a3d163-e6e5-481b-9ea9-869076084bc7"
```

```
az account get-access-token --scope "api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user" --query accessToken -o tsv
```

Then we get the access token:


```
az account get-access-token --scope "api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user" --query accessToken -o tsv eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJhcGk6Ly9mOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRiYzciLCJpc3MiOiJodHRwczovL3N0cy53aW5kb3dzLm5ldC85Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEvIiwiaWF0IjoxNzkxNDI3OTcwLCJuYmYiOjE3OTE0Mjc5NzAsImV4cCI6MTc5MTQzMzA1MCwiYWNyIjoiMSIsImFpbyI6IkFjUUFPLzhlQUFBQVkrQ2hrVmFhM0VSelRrcTBaSWpMaGQ4UkhiNDBmVWVZWXBHMjVnbWxRZXdxbkEyL1A1b0V3bzZLeWdnaDVhYkx4NmkzV1IrY0JWd2d6bzU4WWRjRDJ2clpQTzBxbllHeStNeHM0RGg2NmU3Y2FpSHI4MnZLQ0UzbnEvTi8ybi9ZZFBuVG5BbStMRStWekphNUVnWFBMTUtVOG1KUk5nLzhadWlTRmhEc21QSENxV0crMmZFK1ZXMW90MlA1cncydE14a1lWVGJrYmVlMk1mZGJaaWN0OFhZdTV4WlkzSFZ6RFRSUWFndWNaVzNLZk04Z0IzOXNybTJDTjJkMWVFbnAiLCJhbXIiOlsicHdkIiwi
```

~~bWZhIl0sImFwcGlkIjoiMDRiMDc3OTUtOGRkYi00NjFhLWJiZWUtMDJmOWUxYmY3YjQ2IiwiYXBwaWRhY3IiOiIwIiwiZW1haWwiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsImZhbWlseV9uYW1lIjoiQ29zdGEgUm9kcmlndWVzIiwiZ2l2ZW5fbmFtZSI6Ik1hcmNvcyIsImlkcCI6ImxpdmUuY29tIiwiaXBhZGRyIjoiNjIuODQuMTQ4LjExNCIsIm5hbWUiOiJNYXJjb3MgQ29zdGEgUm9kcmlndWVzIiwib2lkIjoiYjczMmZiNTQtYjI0Ny00YmNmLTk3NTYtMzg2ODhjYWM4NTE5IiwicmgiOiIxLkFhOEEzQU9nbkYzYU1VdUFpNTFTTTMtX2FtUFJvX25sNWh0SW5xbUdrSFlJUzhjQUFMaXZBQS4iLCJzY3AiOiJhY2Nlc3NfYXNfdXNlciIsInNpZCI6IjAwOTlhZmRhLThiODAtNGQyNS0wODJhLWRmY2QyOTE3ZDJlMSIsInN1YiI6Illh6R0JGWjUyLTFWemFjRTVWb0g1OUNLX0pMQkVKVVhOYlBKeGxPUk5LMEUiLCJ0aWQiOiI5Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEiLCJ1bmlxdWVfbmFtZSI6ImxpdmUuY29tI2pvYnMubWFjb3JvQHByb3Rvbm1haWwuY29tIiwidXRpIjoicFNEcG8yakJTMHVISGpXSzIzLVpBQSIsInZlciI6IjEuMCIsInhtc19mdGQiOiI1TUs2Ty1UYnNJQU0zLXM1ZVQ1bTRYeTNhamYzTi1ETVdCU21UdEE5QTRZQlpYVnliM0JsZDJWemRDMWtjMjF6In0.Dqiwnjf LTMQ3IIaHJOXzunBxTWXO29CROF0JV5O42GAwFBW4hjPlKvSxkhCjquFAl5hgLuBw2h7DEDBt8xgwZHuehV8GyLypbQqvf0BIMX16D4gGKIl5Ep6zIBNxJgbQQXvwmbtoPyX_yCcXQEMFlw9UkhlmrISWacxD_A55FWsNzzWWgMir3-dedPGYL3qj3_EjHi6U2wa54cvWjA7tKqJm-JEz3vqknyE4RgnTPwkJh369sLcHoZ9R1DO3WD03dc8w9PWHAYTBqY7Kn-WRMpcWxYzENuYjEYDygxL-H1WLONBK4EGzat9RmfCqIQIrwt9UKiJN-uWQzzE9ohLNnw~~

From powershell (type powershell in the command prompt to start issuing powershell commands)

az account get-access-token --scope "api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user" --query accessToken -o tsv

Token:

eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJhcGk6Ly9mOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRiYzciLCJpc3MiOiJodHRwczovL3N0cy53aW5kb3dzLm5ldC85Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEvIiwiaWF0IjoxNzkxNDM1MTMxLCJuYmYiOjE3OTE0MzUxMzEsImV4cCI6MTc5MTQ0MDgxNCwiYWNyIjoiMSIsImFpbyI6IkFjUUFPLzhlQUFBQVdFY0tvdUF1UXRTRUVUakFRYWhxaUlyM1c3TnJRSDZZUnQ0Y1l0TmNud0NySGdUaW1FU1F6c2NweVB1V1NkcTRjRTFWaEUybWgySFQrVWZ0TnBXS2VNbEFMTFFWNUY2aHBpUVRvTkt3amtBRzh6NUoyK0tRQldCd3JldURpWkN0T3FCMEJhMU00bm9Ed3ROObnZYTDNOVXVYUytpbDJtenhhVTJsWXhVVnBTRDdXUlNtUHBHenF3VThhUTdGOVVwdlE1SlRzQzdZMUZWQmxxVkd5OGl3T1NIMWdqV0YzdGtzUklxTDI2bkQvWCtmb2U3alZWcG9LTXh3alBxeFIya1MiLCJhbXIiOlsicHdkIiwibWZhIl0sImFwcGlkIjoiMDRiMDc3OTUtOGRkYi00NjFhLWJiZWUtMDJmOWUxYmY3YjQ2IiwiYXBwaWRhY3IiOiIwIiwiZW1haWwiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsImZhbWlseV9uYW1lIjoiQ29zdGEgUm9kcmlndWVzIiwiZ2l2ZW5fbmFtZSI6Ik1hcmNvcyIsImlkcCI6ImxpdmUuY29tIiwiaXBhZGRyIjoiNjIuODQuMTQ4LjExNCIsIm5hbWUiOiJNYXJjb3MgQ29zdGEgUm9kcmlndWVzIiwib2lkIjoiYjczMmZiNTQtYjI0Ny00YmNmLTk3NTYtMzg2ODhjYWM4NTE5IiwicmgiOiIxLkFhOEEzQU9nbkYzYU1VdUFpNTFTTTMtX2FtUFJvX25sNWh0SW5xbUdrSFlJUzhjQUFMaXZBQS4iLCJzY3AiOiJhY2Nlc3NfYXNfdXNlciIsInNpZCI6IjAwOTlkNmVhLWIxMTQtNDBiMi00NWViLTVjMTljM2ZiMTg0MyIsInN1YiI6Illh6R0JGWjUyLTFWemFjRTVWb0g1OUNLX0pMQkVKVVhOYlBKeGxPUk5LMEUiLCJ0aWQiOiI5Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEiLCJ1bmlxdWVfbmFtZSI6ImxpdmUuY29tI2pvYnMubWFjb3JvQHByb3Rvbm1haWwuY29tIiwidXRpIjoiZnZidlVXRUhTazZpTnAwRktrRm5BQSIs

InZlciI6IjEuMCIsInhtc19mdGQiOiJ1WDl3MUdRa202eExDd1NqVEdrdTZjSGRTY2xseUJfeVVV
bXcyY0ZFYzh3QlpuSmhibU5sWXkxa2MyMXoifQ.cdypeMzRVquMsfsFwlHYTiri7NSYZ7P-
z6Gi3tVvOJaax1OiPfOblxneHpA32KVjOIYiQI5zRPL33ozGVfOrr9fPbFvv3Z3kBIRGdRfLF6k8v
0MVddBN4Bq36jlrxs2aRdUepz8PKFT4uvT8VhQvG0TAiNusU1VYm7Ks3p7zgKYLPbH8zztdY
2HykzV8p0KqK0NNMI7sgIbYb060RjdV32bxA9hhVDwR7kU5HlJ2PVSuWowSvncyVuwKW0F
k4LNZ1oIv5gEeW3--
uF1DnxT801v7sT17J_gW_RfRDdGjvHR4PjVInrtimi7mVV916yyOu8xAxDr9akfuC-DMiI6-Rg

Testing

Re-new the token, we issue the same command from powershell.

az account get-access-token --scope
"api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user" --query accessToken -o tsv

It works:

GetAll:

curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies -H
"X-Service-Token:4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer
eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1R
YRDNDQSIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJhc
Gk6Ly9mOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRiYzciLCJpc3MiOiJod
HRwczovL3N0cy53aW5kb3dzLm5ldC85Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMz
M3ZmJmNmEvIiwiaWF0IjoxNzkxNDM1MTMxLCJuYmYiOjE3OTE0MzUxMzEsImV4cCI6MT
c5MTQ0MDgxNCwiYWNyIjoiMSIsImFpbyI6IkFjUUFPLzhlQUFBQVdFY0tvdUF1UXRTRUVU
akFRYWhxaUlyM1c3TnJRSDZZUnQ0Y1l0TmNud0NySGdUaW1FU1F6c2NweVB1V1NkcTRjR
TFWaEUybWgySFQrVWZ0TnBXS2VNbEFMTFFWNUY2aHBpUVRvTkt3amtBRzh6NUoyK0t
RQldCd3JldURpWkN0T3FCMEJhMU00bm9Ed3RObnZYTDNOVXVYUytpbDJtenhhVTJsWXh
VVnBTRDdXUlNtUHBHenF3VThhUTdGOVVwdlE1SlRzQzdZMUZWQmxxVkd5OGl3T1NIM
WdqV0YzdGtzUklxTDI2bkQvWCtmb2U3alZWcG9LTXh3alBxeFIya1MiLCJhbXIiOlsicHdkIiwib
WZhIl0sImFwcGlkIjoiMDRiMDc3OTUtOGRkYi00NjFhLWJiZWUtMDJmOWUxYmY3YjQ2Iiw
iYXBwaWRhY3IiOiIwIiwiZW1haWwiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsImZh
bWlseV9uYW1lIjoiQ29zdGEgUm9kcmlndWVzIiwiZ2l2ZW5fbmFtZSI6Ik1hcmNvcyIsImlkcCI6I
mxpdmUuY29tIiwiaXBhZGRyIjoiNjIuODQuMTQ4LjExNCIsIm5hbWUiOiJNYXJjb3MgQ29zdG
EgUm9kcmlndWVzIiwib2lkIjoiYjczMmZiNTQtYjI0Ny00YmNmLTk3NTYtMzg2ODhjYWM4N
TE5IiwicmgiOiIxLkFhOEEzQU9nbkYzYU1VdUFpNTFTTTMtX2FtUFJvX25sNWh0SW5xbUdr
SFlJUzhjQUFMaXZBQS4iLCJzY3AiOiJhY2Nlc3NfYXNfdXNlciIsInNpZCI6IjAwOTlkNmVhL
WIxMTQtNDBiMi00NWViLTVjMTljM2ZiMTg0MyIsInN1YiI6Ilh6R0JGWjUyLTFWemFjRTV
Wb0g1OUNLX0pMQkVKVVhOYlBKeGxPUk5LMEUiLCJ0aWQiOiI5Y2EwMDNkYy1kYTVk
LTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEiLCJ1bmlxdWVfbmFtZSI6ImxpdmUuY29tI2pvYn
MubWFjb3JvQHByb3Rvbm1haWwuY29tIiwidXRpIjoiZnZidlVXRUhTazZpTnAwRktrRm5BQSIs
InZlciI6IjEuMCIsInhtc19mdGQiOiJ1WDl3MUdRa202eExDd1NqVEdrdTZjSGRTY2xseUJfeVVV

bXcyY0ZFYzh3QlpuSmhibU5sWXkxa2MyMXoifQ.cdypeMzRVquMsfsFwlHYTiri7NSYZ7P-z6Gi3tVvOJaax1OiPfOblxneHpA32KVjOIYiQI5zRPL33ozGVfOrr9fPbFvv3Z3kBIRGdRfLF6k8v0MVddBN4Bq36jlrxs2aRdUepz8PKFT4uvT8VhQvG0TAiNusU1VYm7Ks3p7zgKYLPbH8zztdY2HykzV8p0KqK0NNMI7sgIbYb060RjdV32bxA9hhVDwR7kU5HlJ2PVSuWowSvncyVuwKW0Fk4LNZ1oIv5gEeW3--uF1DnxT801v7sT17J_gW_RfRDdGjvHR4PjVInrtimi7mVV916yyOu8xAxDr9akfuC-DMiI6-Rg"

[{"id":"f3e36699-dbe5-4a6e-9ab2-28adeb559490","title":"The Matrix Reloaded","year":2003}]

GetById:

curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies/f3e36699-dbe5-4a6e-9ab2-28adeb559490 -H "X-Service-Token:4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJhcGk6Ly9mOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRiYzciLCJpc3MiOiJodHRwczovL3N0cy53aW5kb3dzLm5ldC85Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEvIiwiaWF0IjoxNzkxNDM1MTMxLCJuYmYiOjE3OTE0MzUxMzEsImV4cCI6MTc5MTQ0MDgxNCwiYWNyIjoiMSIsImFpbyI6IkFjUUFPLzhlQUFBQVdFY0tvdUF1UXRTRUVUakFRYWhxaUlyM1c3TnJRSDZZUnQ0Y1l0TmNud0NySGdUaW1FU1F6c2NweVB1V1NkcTRjRTFWaEUybWgySFQrVWZ0TnBXS2VNbEFMTFFWNUY2aHBpUVRvTkt3amtBRzh6NUoyK0tRQldCd3JldURpWkN0T3FCMEJhMU00bm9Ed3ROObnZYTDNOVXVYUytpbDJtenhhVTJsWXhVVnBTRDdXUlNtUHBHenF3VThhUTdGOVVwdlE1SlRzQzdZMUZWQmxxVkd5OGl3T1NIMWdqV0YzdGtzUklxTDI2bkQvWCtmb2U3alZWcG9LTXh3alBxeFIya1MiLCJhbXIiOlsicHdkIiwibWZhIl0sImFwcGlkIjoiMDRiMDc3OTUtOGRkYi00NjFhLWJiZWUtMDJmOWUxYmY3YjQ2IiwiYXBwaWRhY3IiOiIwIiwiZW1haWwiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsImZhbWlseV9uYW1lIjoiQ29zdGEgUm9kcmlndWVzIiwiZ2l2ZW5fbmFtZSI6Ik1hcmNvcyIsImlkcCI6ImxpdmUuY29tIiwiaXBhZGRyIjoiNjIuODQuMTQ4LjExNCIsIm5hbWUiOiJNYXJjb3MgQ29zdGEgUm9kcmlndWVzIiwib2lkIjoiYjczMmZiNTQtYjI0Ny00YmNmLTk3NTYtMzg2ODhjYWM4NTE5IiwicmgiOiIxLkFhOEEzQU9nbkYzYU1VdUFpNTFTTTMtX2FtUFJvX25sNWh0SW5xbUdrSFlJUzhjQUFMaXZBQS4iLCJzY3AiOiJhY2Nlc3NfYXNfdXNlciIsInNpZCI6IjAwOTlkNmVhLWIxMTQtNDBiMi00NWViLTVjMTljM2ZiMTg0MyIsInN1YiI6Ilh6R0JGWjUyLTFWemFjRTVWb0g1OUNLX0pMQkVKVVhOYlBKeGxPUk5LMEUiLCJ0aWQiOiI5Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEiLCJ1bmlxdWVfbmFtZSI6ImxpdmUuY29tI2pvYnMubWFjb3JvQHByb3Rvbm1haWwuY29tIiwidXRpIjoiZnZidlVXRUhTazZpTnAwRktrRm5BQSIsInZlciI6IjEuMCIsInhtc19mdGQiOiJ1WDl3MUdRa202eExDd1NqVEdrdTZjSGRTY2xseUJfeVVVbXcyY0ZFYzh3QlpuSmhibU5sWXkxa2MyMXoifQ.cdypeMzRVquMsfsFwlHYTiri7NSYZ7P-z6Gi3tVvOJaax1OiPfOblxneHpA32KVjOIYiQI5zRPL33ozGVfOrr9fPbFvv3Z3kBIRGdRfLF6k8v0MVddBN4Bq36jlrxs2aRdUepz8PKFT4uvT8VhQvG0TAiNusU1VYm7Ks3p7zgKYLPbH8zztdY2HykzV8p0KqK0NNMI7sgIbYb060RjdV32bxA9hhVDwR7kU5HlJ2PVSuWowSvncyVuwKW0Fk4LNZ1oIv5gEeW3—uF1DnxT801v7sT17J_gW_RfRDdGjvHR4PjVInrtimi7mVV916yyOu8xAxDr9akfuC-DMiI6-Rg"

{"id":"f3e36699-dbe5-4a6e-9ab2-28adeb559490","title":"The Matrix Reloaded","year":2003}

Add (new token)

```
curl -X POST
https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontain
erapps.io/api/movies -H "Content-Type: application/json" -H "X-
Service-Token:4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H
"Authorization: Bearer
eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6ImRndlNEdks4QTVLeUt5cH
B3MWRBd1RYRDNDQSIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9
.eyJhdWQiOiJhcGk6Ly9mOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRiY
zciLCJpc3MiOiJodHRwczovL3N0cy53aW5kb3dzLm5ldC85Y2EwMDNkYy1kYTVkLTR
iMzEtODA4Yi05ZDUyMzM3ZmJmNmEvIiwiaWF0IjoxNzkxNDQwODc0LCJuYmYiOjE3O
TE0NDA4NzQsImV4cCI6MTc5MTQ0NTIxNywiYWNyIjoiMSIsImFpbyI6IkFjUUFPLzh
lQUFBQUFsRFlsR0hTSUk4akxaTzIzNkFnL3RYYm9Ra1pscldHNCtoSFM0ZlZyZVR2d
Vo1ME9NREhnSTk3SVlid0NQWWo2MzNiaFZzcUdFdlFQRUkrc0h0M2pVeU4vWlY4WU5
rNTNEQTY3bDY1Q2ZtWms5Y2lKRTh5UmxKQmdUN05mY3Z0WmNuUUdkLzVHK2hRT0wrN
0FjUStQeTk2TU9NeEtacmxkRFZtK1l1ZTNYdytuOTlDWHN5aW1jcnBSanU1VUd3Mk9
TL2RDRG43OTdmK0NqaDZQU3ZaU3dxK3lacjl5WGdRSU84SE1EbE54M1RHRnlscVlhY
0xwZ3VaRVl6YWk2RzEiLCJhbXIiOlsicHdkIiwibWZhIl0sImFwcGlkIjoiMDRiMDc
3OTUtOGRkYi00NjFhLWJiZWUtMDJmOWUxYmY3YjQ2IiwiYXBwaWRhY3IiOiIwIiwiZ
W1haWwiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsImZhbWlseV9uYW1lIjo
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Login

This is an enterprise solution:

We do not provide any form to login and this will be delegated to Microsoft.

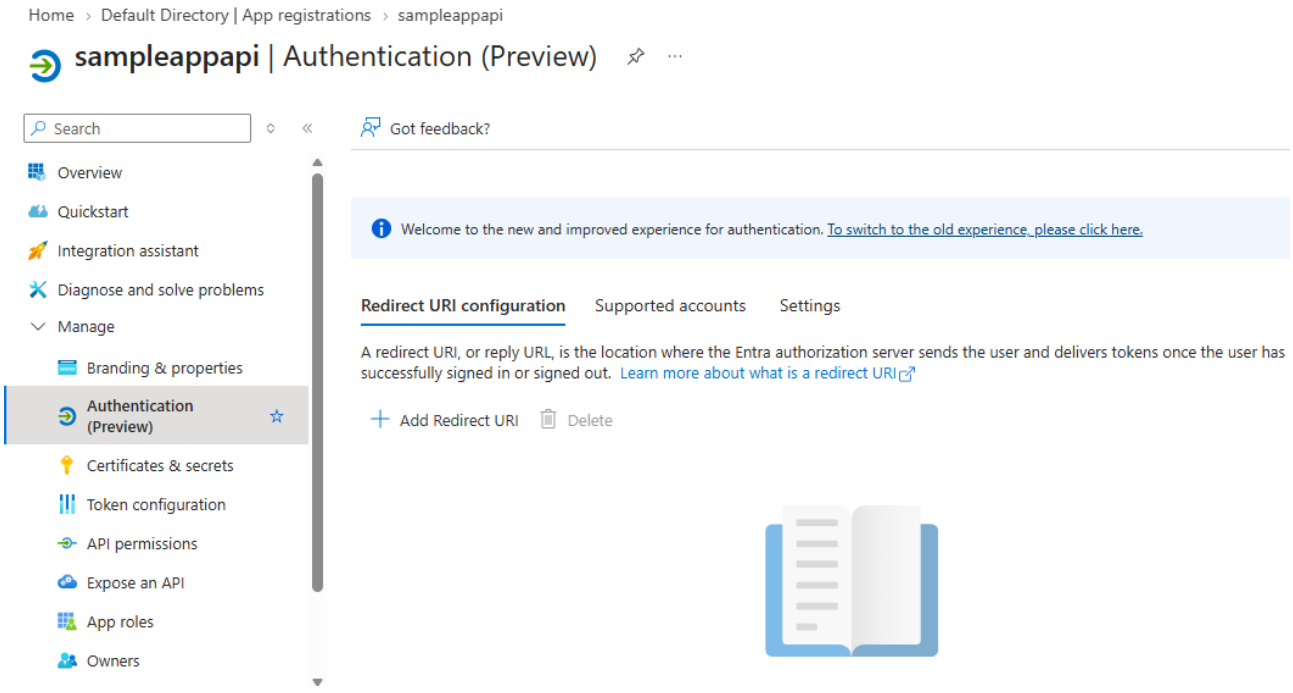
The user has to be register and logs into microsoft

Microsoft will send back a JWT

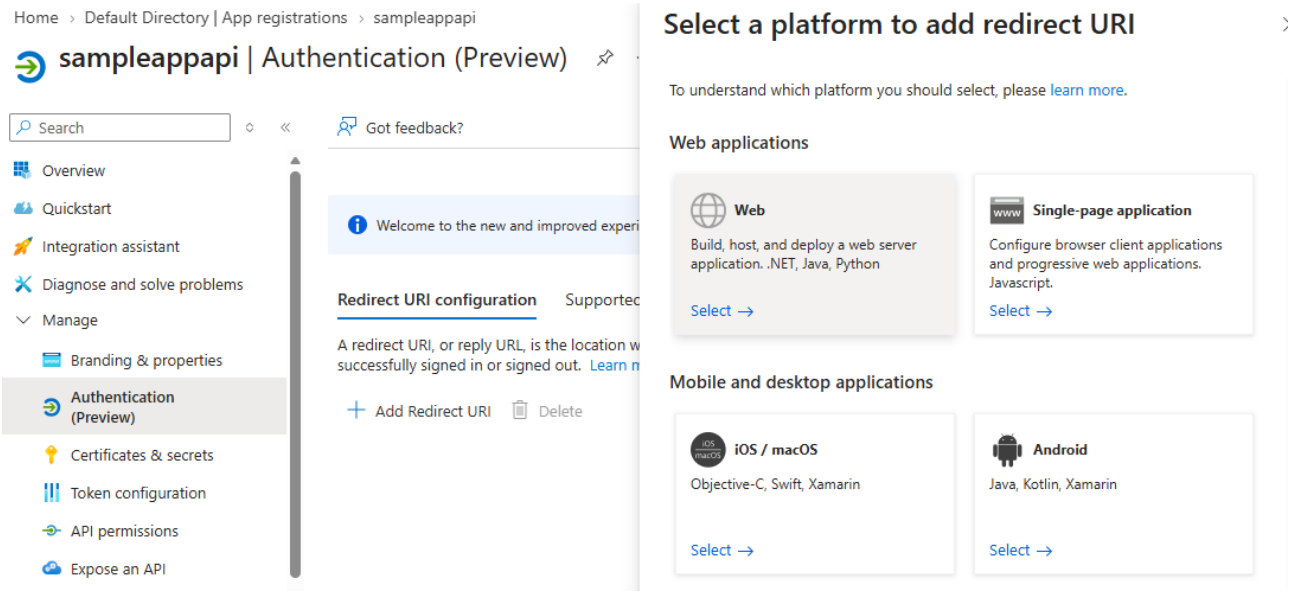
We will check the JWT: signature, valid, rol

When all is valid then the user is authenticated and has authorization to access the database.

Go to EntraId > sampleappapi > Authentication and Add and click on Add Redirect Uri

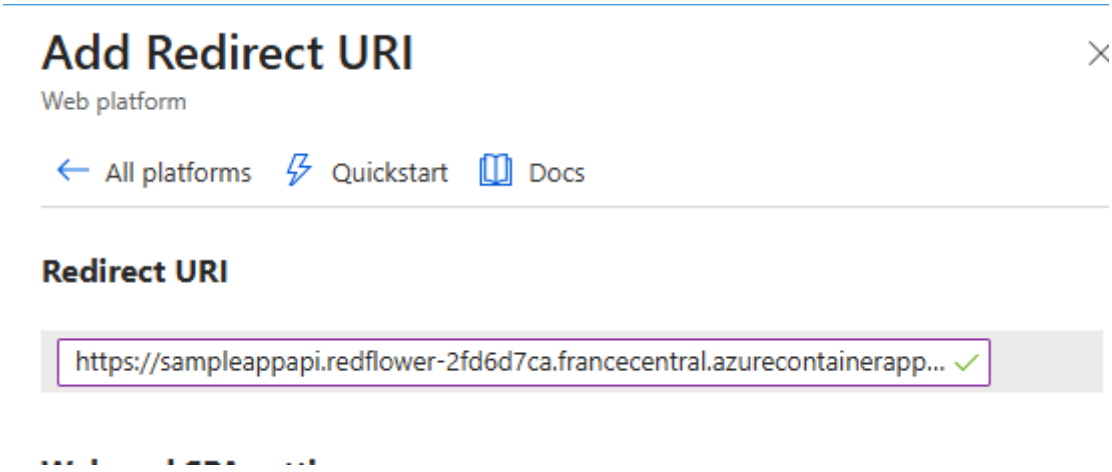


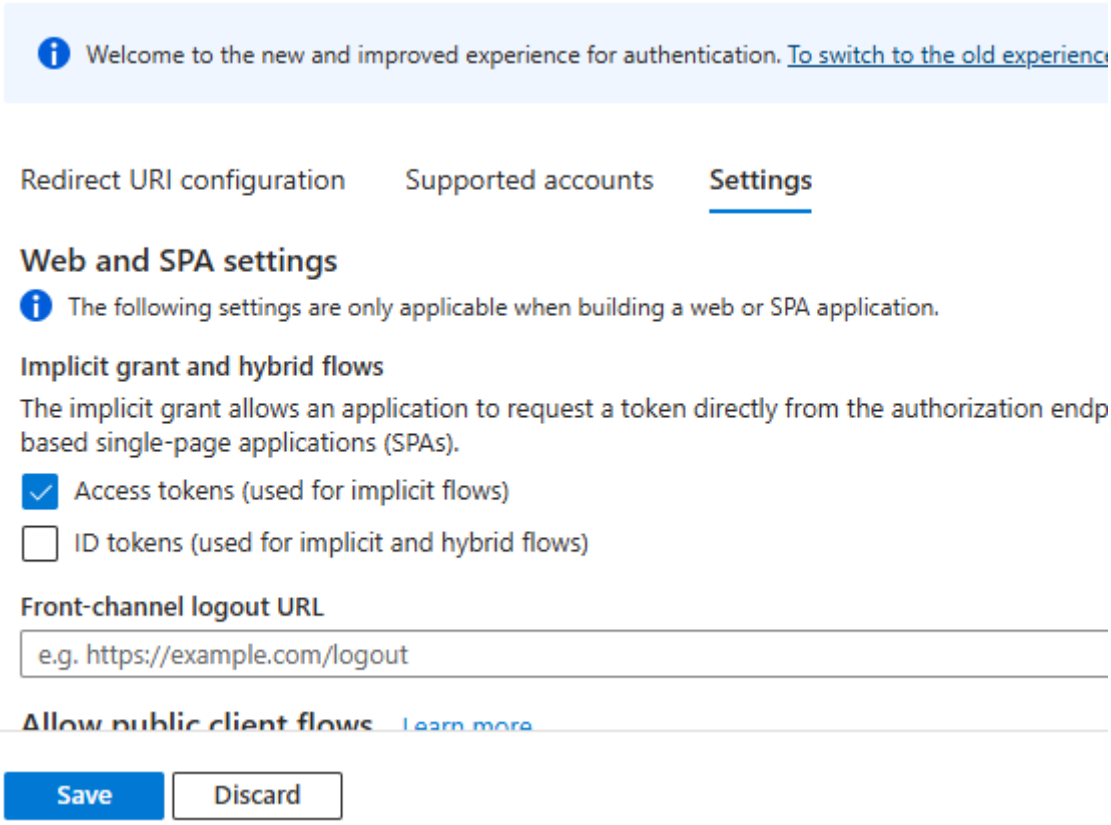
Choose Web



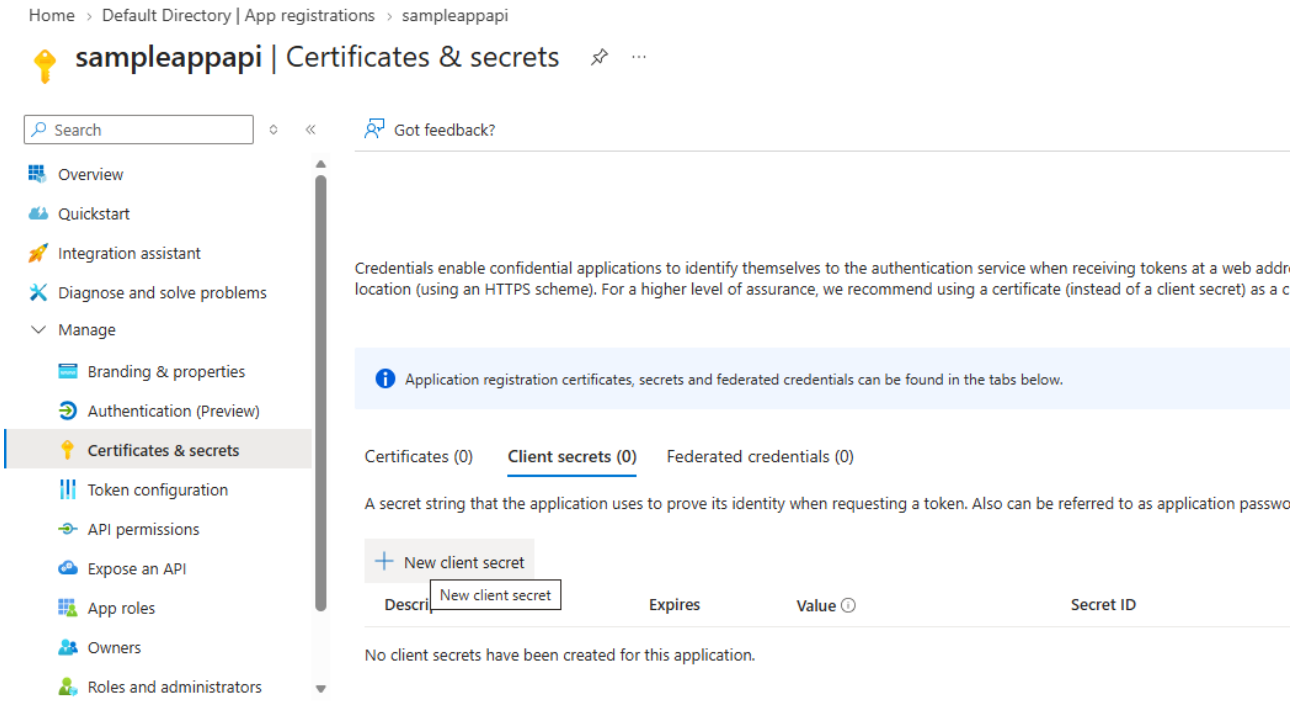
Redirect URI = https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/signin-oidc

Implicit grant and hybrid flows > check Access Tokens





Clcik on Configure. Go to certificates & secrets > click on New Client Secret



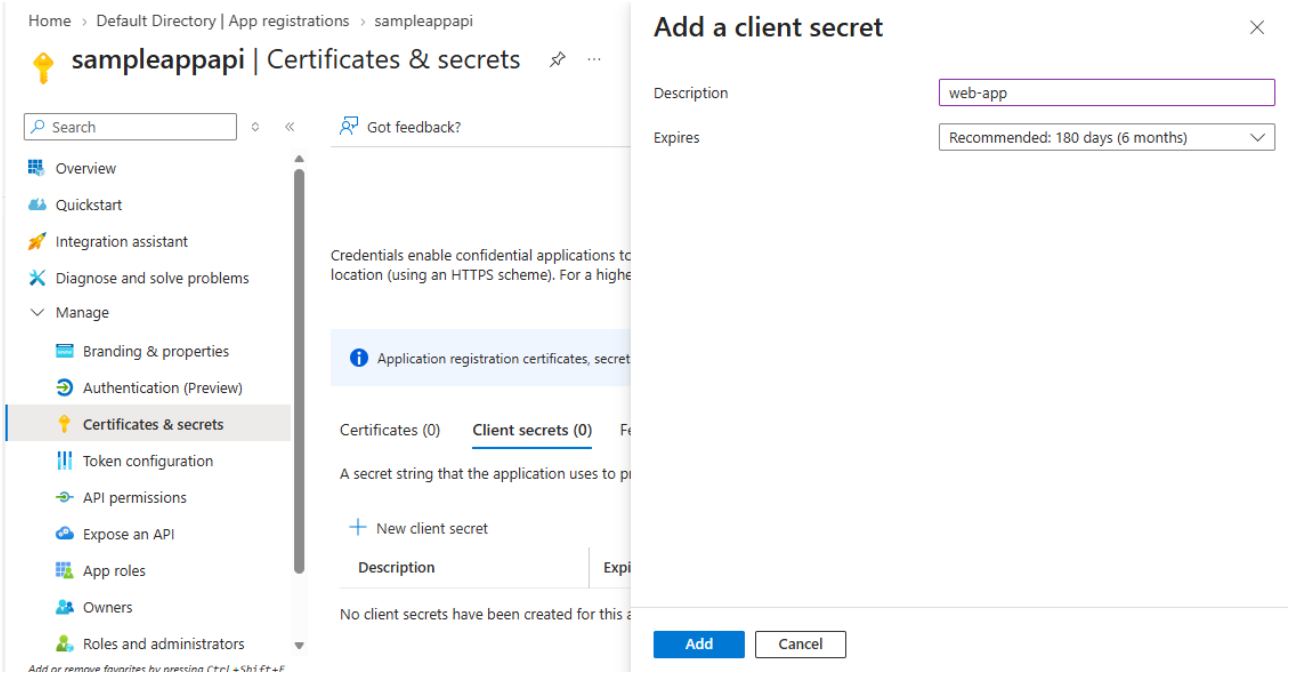
Enter the following values:

Description = web-app

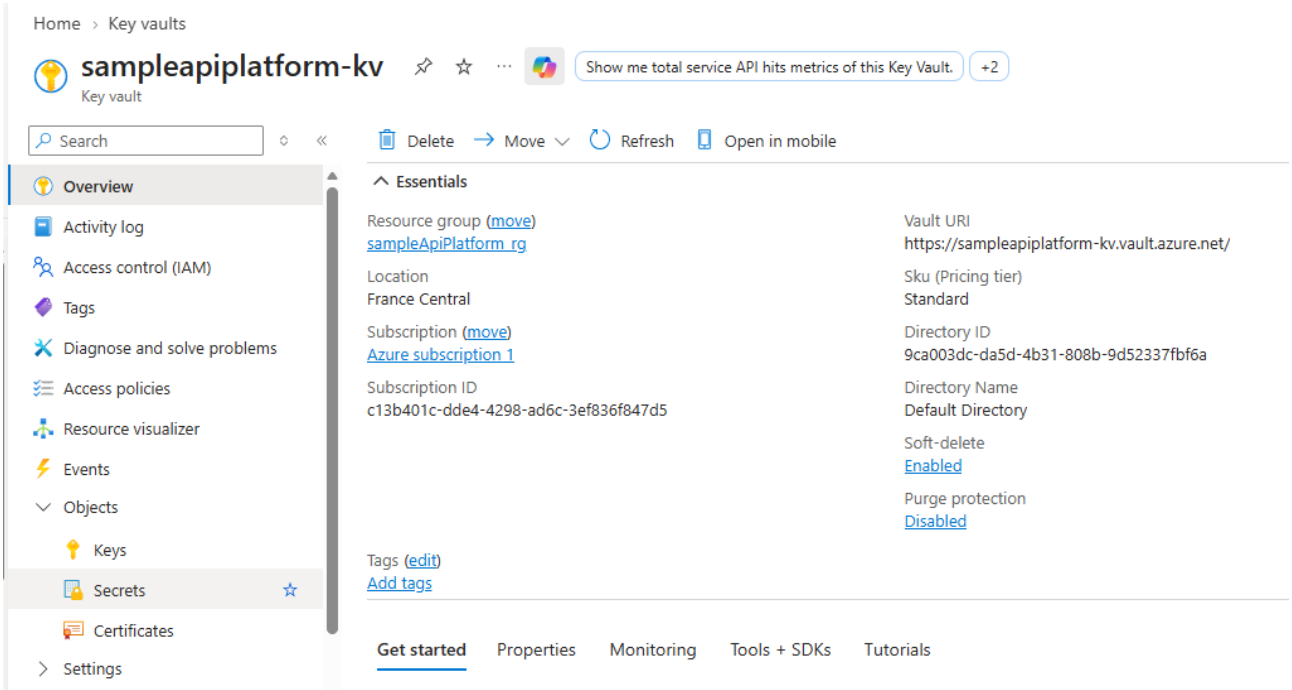
Expires = 6 months

Click on Add

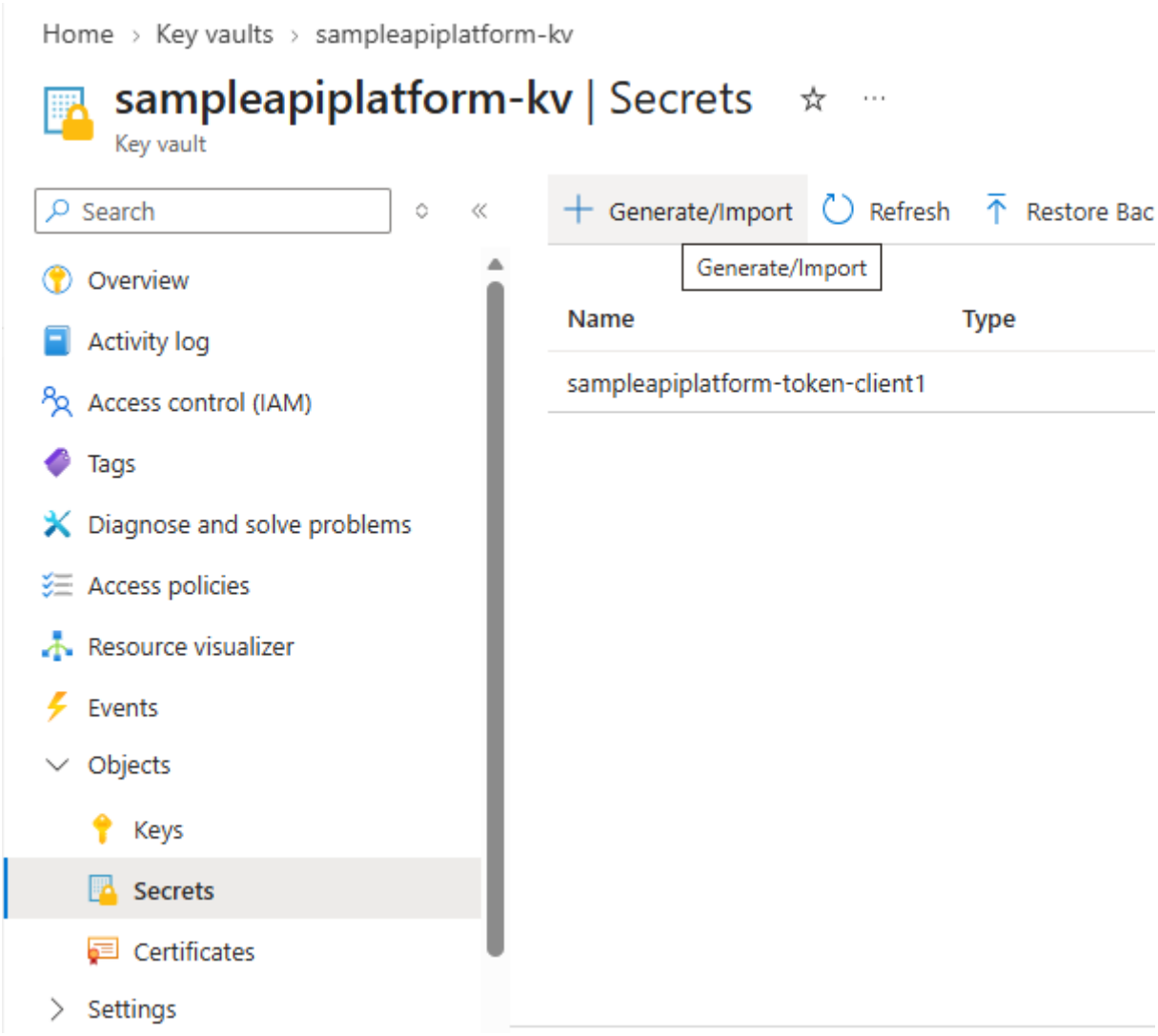
Copy the value = f0W8Q~BTFvq8Mjet_PzFnQ-CYAYuXqevVc.5ddoO



Now go to the Kay Vault > click on sampleapiplatform-kv > Secrets >



Click on Generate/Import



Upload options = Manual

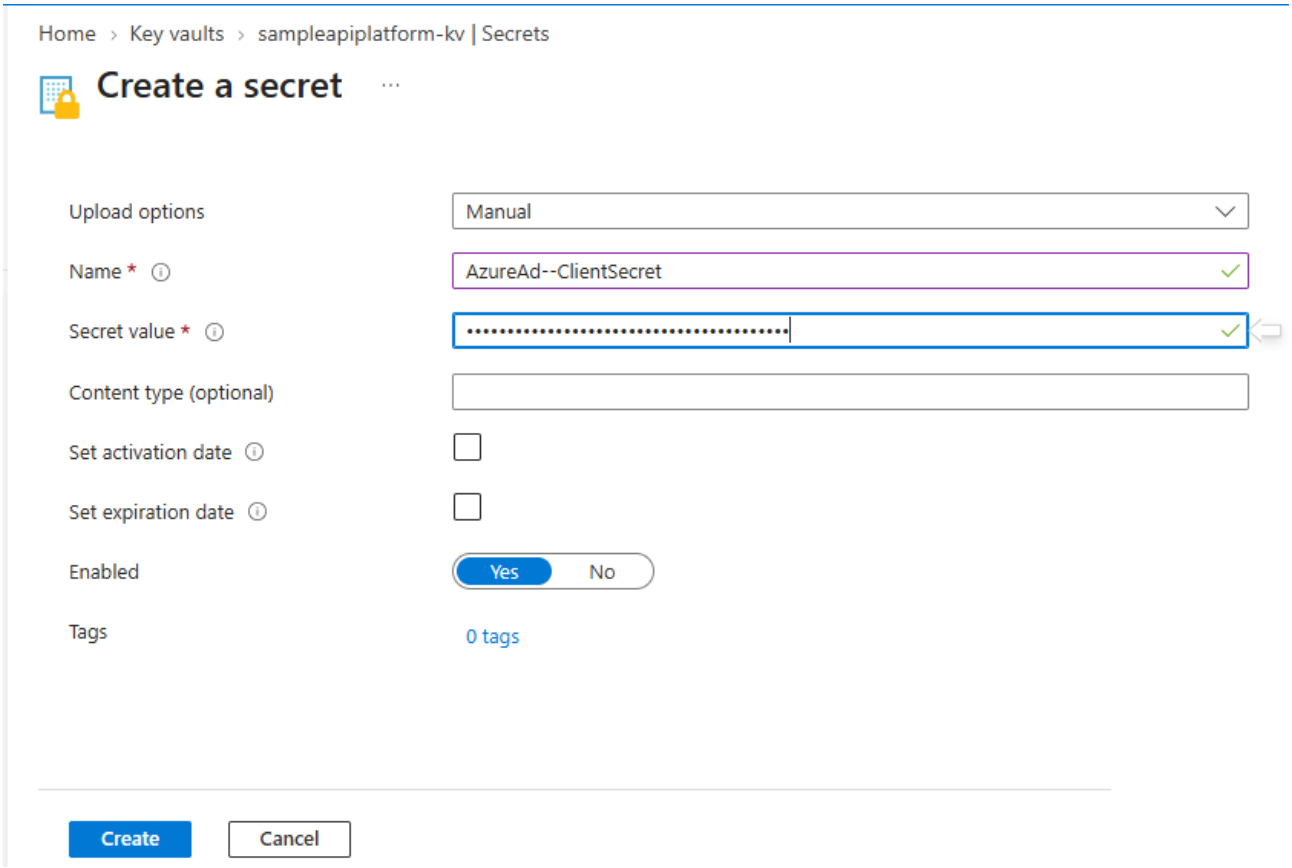
Name = `AzureAdClientSecret1`

Secret Value = f0W8Q~BTFvq8Mjet_PzFnQ-CYAYuXqevVc.5ddoO

Enabled = Yes

~~Then modify the Program.cs: see code~~

~~Then paste in the browser the following url: sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io~~



~~Click on Create~~

For the redirect to work we will need to create the following environment variable in the sampleappapi container:

```
ASPNETCORE_FORWARDEDHEADERS_ENABLED = true
```

Go to Entra Id > app registration > sampleappapi > Manifest > make sure that this field is set to this value:

"requestedAccessTokenVersion": 2,

and also add clientId here (what is in bold)

```
"requiredResourceAccess": [
    {
        "resourceAppId": "00000003-0000-0000-c000-000000000000",
        "resourceAccess": [
            {
                "id": "e1fe6dd8-ba31-4d61-89e7-88639da4683d",
                "type": "Scope"
            }
        ]
    },
    {
      "resourceAppId": "f9a3d163-e6e5-481b-9ea9-869076084bc7",
      "resourceAccess": [
```

```
        { "id": "1f0b1c78-9ed0-4ad8-a4d1-af9cfb207d1d", "type": "Scope" }
      ]
    }
  ]
```

and remove this that is to get Graph tokens:

```
{
                "resourceAppId": "00000003-0000-0000-c000-000000000000",
                "resourceAccess": [
                      {
                            "id": "e1fe6dd8-ba31-4d61-89e7-88639da4683d",
                            "type": "Scope"
                      }
                ]
          }
```

where this f9a3d163-e6e5-481b-9ea9-869076084bc7 comes goes is after the api://f9a3d163-e6e5-481b-9ea9-869076084bc7 when we exposed the api.If we do not do that then Otherwise we get a token for Graph but not for the api.

And also here:

"implicitGrantSettings": {
"enableAccessTokenIssuance": false,
"enableIdTokenIssuance": false
},
fior security reasons to avoid leakage of the token

In the MoviesController :

[ApiController]
[Route("api/movies")]
[Authorize(AuthenticationSchemes = JwtBearerDefaults.AuthenticationScheme, Policy = "AccessAsUser")]
public class MoviesController : ControllerBase

AuthController

using Microsoft.AspNetCore.Mvc; using Microsoft.Identity.Web;

[ApiController] [Route("auth")] public class AuthController : ControllerBase { private readonly ITokenAcquisition _tokenAcquisition; private readonly ILogger _logger;

```
public AuthController(ITokenAcquisition tokenAcquisition,
ILogger<AuthController> logger)
{
    _tokenAcquisition = tokenAcquisition;
    _logger = logger;
}
```

```
// GET /auth/login
// This triggers the Microsoft login redirect. After login, the token is stored
// in the user's session. The client then calls /auth/token to retrieve it.
[HttpGet("login")]
public IActionResult Login()
{
    // Challenge the user to authenticate via Microsoft
    return Challenge(new AuthenticationProperties
    {
        RedirectUri = "/auth/callback"  // after login, redirect here
    }, OpenIdConnectDefaults.AuthenticationScheme);
}

// GET /auth/callback
// Microsoft redirects here after successful login. The cookie is now set.
[HttpGet("callback")]
public IActionResult Callback()
{
    return Ok(new { message = "Login successful. Now call GET /auth/token to get
your JWT." });
}

// GET /auth/token
// Returns the access token as JSON for the console client to copy.
[HttpGet("token")]
[Authorize]  // User must be logged in first
public async Task<IActionResult> GetToken()
{
    try
    {
        // Request an access token for your API (same audience)
        var accessToken = await _tokenAcquisition.GetAccessTokenForUserAsync(
            scopes: new[] {
"api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user" });

        return Ok(new
        {
            access_token = accessToken,
            token_type = "Bearer",
            expires_in = 3600
        });
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Failed to acquire token");
        return StatusCode(500, new { error = ex.Message });
    }
}
}
```

Test:

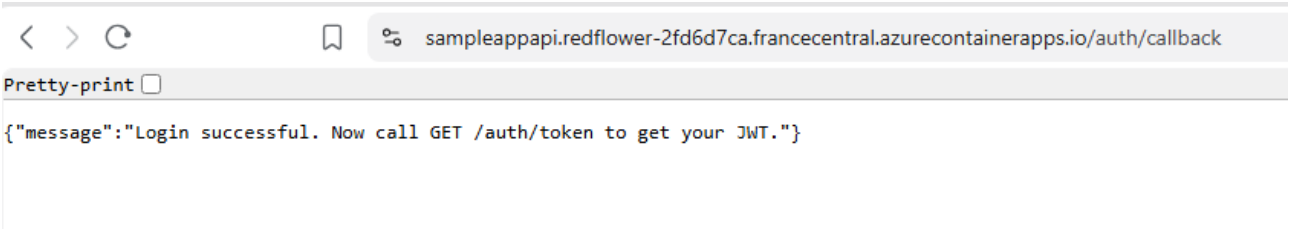
~~https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/whoami~~

~~https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/login~~

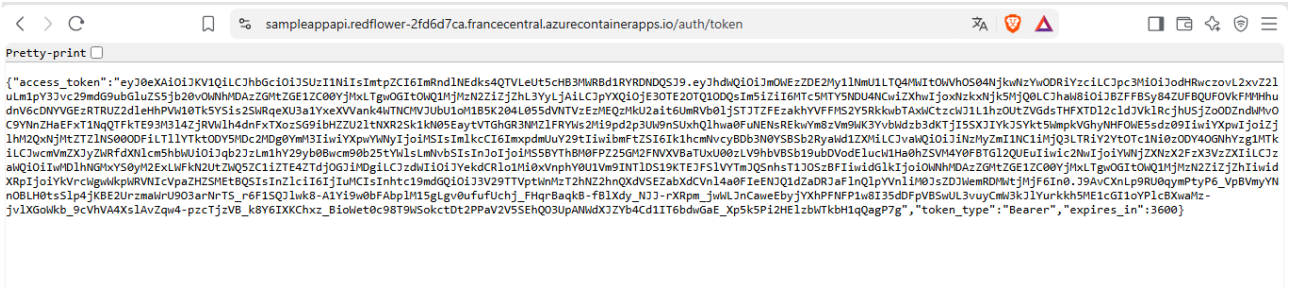
~~curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/login -H "X-Service-Token: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a"~~

Notice that the TokenVaidationMiddleware in the sampleappapi only require the X-Service-token for the paths starting with api, the others do not require the X-Service-Token.
We are expecting that the user logs into microsoft, and this only works if we call our application from a browser. The curl will not work, so we need to type this url in the browser window:

https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/login



Then we call in a separate tab:
https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/token




```
{"access_token":"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt
5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJmOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRiYz
ciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMtZGE1ZC00Yj
MxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTE2OTQ1ODQsIm5iZiI6MTc5MTY5NDU4NC
wiZXhwIjoxNzkxNjk5MjQ0LCJhaW8iOiJBZFFBSy84ZUFBQUFOVkFMMHhudnV6cDNYVGEzRTRUZ2dleH
hPVW10Tk5YSis2SWRqeXU3a1YxeXVVank4WTNCMVJUbU1oM1B5K204L055dVNTVzEzMEQzMkU2ait6Um
RVb0ljSTJTZFEzakhYVFFMS2Y5RkkwbTAxWCtzcWJ1L1hzOUtZVGdsTHFXTDl2cldJVklRcjhUSjZoOD
ZndWMvOC9YNnZHaEFxT1NqQTFkTE93M3l4ZjRVWlh4dnFxTXozSG9ibHZZU2ltNXR2Sk1kN05EaytvTG
```

```
hGR3NMZlFRYWs2Mi9pd2p3UW9nSUxhQlhwa0FuNENsREkwYm8zVm9WK3YvbWdzb3dKTjI5SXJIYkJSYk
t5WmpkVGhyNHFOWE5sdz09IiwiYXpwIjoiZjlhM2QxNjMtZTZlNS00ODFiLTllYTktODY5MDc2MDg0Ym
M3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvcyBDb3N0YSBSb2RyaW
d1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY4OGNhYzg1MTkiLCJwcmVmZXJyZW
RfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2
FNVXVBaTUxU00zLV9hbVBSb19ubDVodElucW1Ha0hZSVM4Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2
FzX3VzZXIiLCJzaWQiOiIwMDlhNGMxYS0yM2ExLWFkN2UtZWQ5ZC1iZTE4ZTdjOGJiMDgiLCJzdWIiOi
JYekdCRlo1Mi0xVnphY0U1Vm9INTlDS19KTEJFSlVYTmJQSnhsT1JOSzBFIiwidGlkIjoiOWNhMDAzZG
MtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoiYkVrcWgwWkpWRVNIcVpaZHZSMEtBQS
IsInZlciI6IjIuMCIsInhtc19mdGQiOiJ3V29TTVptWnMzT2hNZ2hnQXdVSEZabXdCVnl4a0FIeENJQ1
dZaDRJaFlnQlpYVnliM0JsZDJWemRDMWtjMjF6In0.J9AvCXnLp9RU0qymPtyP6_VpBVmyYNnOBLH0ts
Slp4jKBE2UrzmaWrU9O3arNrTS_r6F1SQJlwk8-A1Yi9w0bFAbplM15gLgv0ufufUchj_FHqrBaqkB-
fBlXdy_NJJ-
rXRpm_jwWLJnCaweEbyjYXhPFNFP1w8I35dDFpVBSwUL3vuyCmW3kJlYurkkh5ME1cGI1oYPlcBXwaMz
-jvlXGoWkb_9cVhVA4XslAvZqw4-
pzcTjzVB_k8Y6IXKChxz_BioWet0c98T9WSokctDt2PPaV2V5SEhQO3UpANWdXJZYb4Cd1IT6bdwGaE_
Xp5k5Pi2HElzbWTkbH1qQagP7g","token_type":"Bearer","expires_in":3600}
```


and we have the token that now we can use to query the API

curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies -H "X-Service-Token: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer

eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1R
YRDNDQSJ9.eyJhdWQiOiJmOWEzZDE2My1lNmU1LTQ4MWItOWVhOS04NjkwNzYwODRi
YzciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMt
ZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTE2OTQ1ODQsI
m5iZiI6MTc5MTY5NDU4NCwiZXhwIjoxNzkxNjk5MjQ0LCJhaW8iOiJBZFFBSy84ZUFBQUF
OVkFMMHhudnV6cDNYVGEzRTRUZ2dleHhPVW10Tk5YSis2SWRqeXU3a1YxeXVVank4WT
NCMVJUbU1oM1B5K204L055dVNTVzEzMEQzMkU2ait6UmRVb0ljSTJTZFEzakhYVFFMS2
Y5RkkwbTAxWCtzcWJ1L1hzOUtZVGdsTHFXTDl2cldJVklRcjhUSjZoODZndWMvOC9YNnZH
aEFxT1NqQTFkTE93M3l4ZjRVWlh4dnFxTXozSG9ibHZZU2ltNXR2Sk1kN05EaytVTGhGR3N
MZlFRYWs2Mi9pd2p3UW9nSUxhQlhwa0FuNENsREkwYm8zVm9WK3YvbWdzb3dKTjI5SXJI
YkJSYkt5WmpkVGhyNHFOWE5sdz09IiwiYXpwIjoiZjlhM2QxNjMtZTZlNS00ODFiLTllYTktO
DY5MDc2MDg0YmM3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcm
NvcyBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zO
DY4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b2
5tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9hbVBSb19ubDVod
ElucW1Ha0hZSVM4Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIw
MDlhNGMxYS0yM2ExLWFkN2UtZWQ5ZC1iZTE4ZTdjOGJiMDgiLCJzdWIiOiJYekdCRlo1Mi
0xVnphY0U1Vm9INTlDS19KTEJFSlVYTmJQSnhsT1JOSzBFIiwidGlkIjoiOWNhMDAzZGMtZ
GE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoiYkVrcWgwWkpWRVNIcVpaZHZ
SMEtBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJ3V29TTVptWnMzT2hNZ2hnQXdVSEZabXdCVn
l4a0FIeENJQ1dZaDRJaFlnQlpYVnliM0JsZDJWemRDMWtjMjF6In0.J9AvCXnLp9RU0qymPtyP
6_VpBVmyYNnOBLH0tsSlp4jKBE2UrzmaWrU9O3arNrTS_r6F1SQJlwk8-
A1Yi9w0bFAbplM15gLgv0ufufUchj_FHqrBaqkB-fBlXdy_NJJ-
rXRpm_jwWLJnCaweEbyjYXhPFNFP1w8I35dDFpVBSwUL3vuyCmW3kJlYurkkh5ME1cGI1oY
PlcBXwaMz-jvlXGoWkb_9cVhVA4XslAvZqw4-
pzcTjzVB_k8Y6IXKChxz_BioWet0c98T9WSokctDt2PPaV2V5SEhQO3UpANWdXJZYb4Cd1IT

6bdwGaE_Xp5k5Pi2HElzbWTkbH1qQagP7g" [{"id":"f3e36699-dbe5-4a6e-9ab2-28adeb559490","title":"The Matrix Reloaded","year":2003}]

Understanding what we have so far

Location: France Central

Subscription: Azure subscription 1 (c13b401c-dde4-4298-ad6c-3ef836f847d5)

We have a monolith application running on an app service

A distributed application running on containers accessing the same database as the monolith application.

Container app: sampleappapi
Resource Group : SampleAppApi_rg

Container Apps Environment: sampleapi-env

Application Url: https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io

Log Analytics: workspace-samplepizureevpscrrgFWPG

Secrets: **keyvaulturi** / Container Apps Secret / Value = https://sampleapiplatform-kv.vault.azure.net

Ingress: Ingress: enabled, Ingress traffic: Accepting traffic from anywhere, Ingress Type: HTTP, Client Certificate Mode: Ignore, Transport=Auto, Insecure connections = unchecked, Target Port=8080, Endpoint(s) = https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io , Session Affinity = unchecked, IP Restrictions > IP Security Restrictions Mode > Allow All Traffic (default)

System Assigned Identity:Object ID = a36e4b77-d461-4de8-8b56-93d91857369a / Role Assignment = Role: Key Vault Secrets User , Resource Name : sampleapiplatform-kv; Resource Type: Key Vault; assign to: sampleappapi

User Assigned Identity = SampleDataAccessApi_UserKey

Environment Variables = Name= FORCE_REVISION, Source=Manual Entry , Value= 202610110617 / Name= KeyVaultUri : Source=Reference a secret , Value= **keyvaulturi** / Name= ASPNETCORE_FORWARDEDHEADERS_ENABLED, Source=Manual Entry, Value=true

Dapr: enabled , App Id=sampleappapi, App port=8080, Protocol=HTTP, HTTP Head read buffer size= 65 KB, HTTP max request size=4 MB, Log Level=Debug, API Logging= true, Components = empty

Container app: sampledataaccessapi
Resource Group : sampleDataAccessApi_rg

Container Apps Environment: sampleapi-env

Application Url: None (Ingress disabled)

Log Analytics: workspace-samplepizureevpscrrgFWPG

Secrets: moviesdbsecret / Container Apps Secret / Value (connection string) = Server=tcp:sampleapiserver.database.windows.net,1433;Initial Catalog=free-sql-db-5542591;Persist Security Info=False;User ID=sampleapiserveradmin;Password=L@rossadelbar69;MultipleActiveResultSets=False;Encrypt=True;TrustServerCertificate=False;Connection Timeout=30;

System Assigned Identity: None

User Assigned Identity = SampleDataAccessApi_UserKey

Environment Variables = Name=ConnectionStrings__MoviesDb : Source=Reference a secret , Value= moviesdbsecret / Name= FORCE_REVISION, Source=Manual Entry , Value= 202610110617 / Name= AZURE_CLIENT_ID, Source=Manual Entry , Value= 562e5afa-70d7-4dc8-8fa7-13082ba5d3c6

Dapr: enabled , App Id=sampledataccessapi, App port=8080, Protocol=HTTP, HTTP Head read buffer size= 65 KB, HTTP max request size=4 MB, Log Level=Debug, API Logging= true, Components = empty

Container app environmenr: sampleapi-env
Resource Group: sampleApi_AzureDevOps_Acr_rg

Environment type: workload profiles

Static IP: 172.189.88.176

Applications: 3

KEDA Version: 2.20.2

Dapr Version: 1.16.4-msft.13

User Assigned Identity:

SampleDataAccessApi_UserKey

Resource Group: sampleapiplatform_rg

Client ID: dd0c5b2a-7b64-487b-9280-32fbd4718a55

Object (principal) ID = f94144d4-e6eb-4c84-9d3e-e86d39497cbb

Key Vault
Name = sampleapiplatform-kv
Reource Group = sampleApiPlatform_rg
Vault URI = https://sampleapiplatform-kv.vault.azure.net/
Sku (Pricing Tier) = Standard

Directory ID = 9ca003dc-da5d-4b31-808b-9d52337fbf6a
Directory name = Default Directory
Soft Delete = Enabled
Purge Protection = Dosabled

Objects > Secrets
AzureAdClientSecret1
sampleapiplatform-token-client1

Entra ID
Tenant ID = 9ca003dc-da5d-4b31-808b-9d52337fbf6a
Primary Domain (url) = jobsmacoroprotonmail.onmicrosoft.com
License = Microsoft Entra ID Free
Users 1
Groups: 0
Applications : 5
Devicces: 0

App Registrations:
sampleappapi : Display Name= sampleappapi , Application (client) ID = f9a3d163-e6e5-481b-9ea9-869076084bc7, Object ID= 02519300-3705-4d65-a1f9-bd08ae109d73, Directory (tenant) ID = 9ca003dc-da5d-4b31-808b-9d52337fbf6a, Supported Account Types = My organization Only, Client credentials = 0 certificate, 1 secret, Redirects URI = 1 web 0 spa 0 public client, Application ID Uri = api://f9a3d163-e6e5-481b-9ea9-869076084bc7, Managed application in local directoy = sampleappapi, State = Activated

Authentication > redirect URI configuration > Redirect URI = https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/signin-oidc,
Authentication > Supported Account Types = Single Tenant Only – Default Directory (My organization),
Authentication > Settings > Web And Spa > implicit grant and hybrid flow > Access tokes (used for implicit flows) = false, ID Tokens (used for implicit and hybrid flows) = false, Front channel logour url = empty, Allow public client flows = disabled

Expose an API> Scopes > Scope Name = access_as_user (api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user), Who can consent = Admins and Users, Admin Consent Display Type = Access the sampleappapi application, Admin Consent Description = Allow the application to access sampleappapi on behalf of the signed-in user, State = Enabled

Expose an API> Authorized Client Applications > Client ID = 04b07795-8ddb-461a-bbee-02f9e1bf7b46, Authorized Scopes = api://f9a3d163-e6e5-481b-9ea9-869076084bc7/access_as_user

API Permissions > Configured Permissions > Grant Admin Consent for Default Directory = enabled

API Permissions > Configured Permissions > sampleappapi (1) > access_as_user , Type = delegated, Description=Access the sampleappapi application, Admin consent required = No, Status = empty

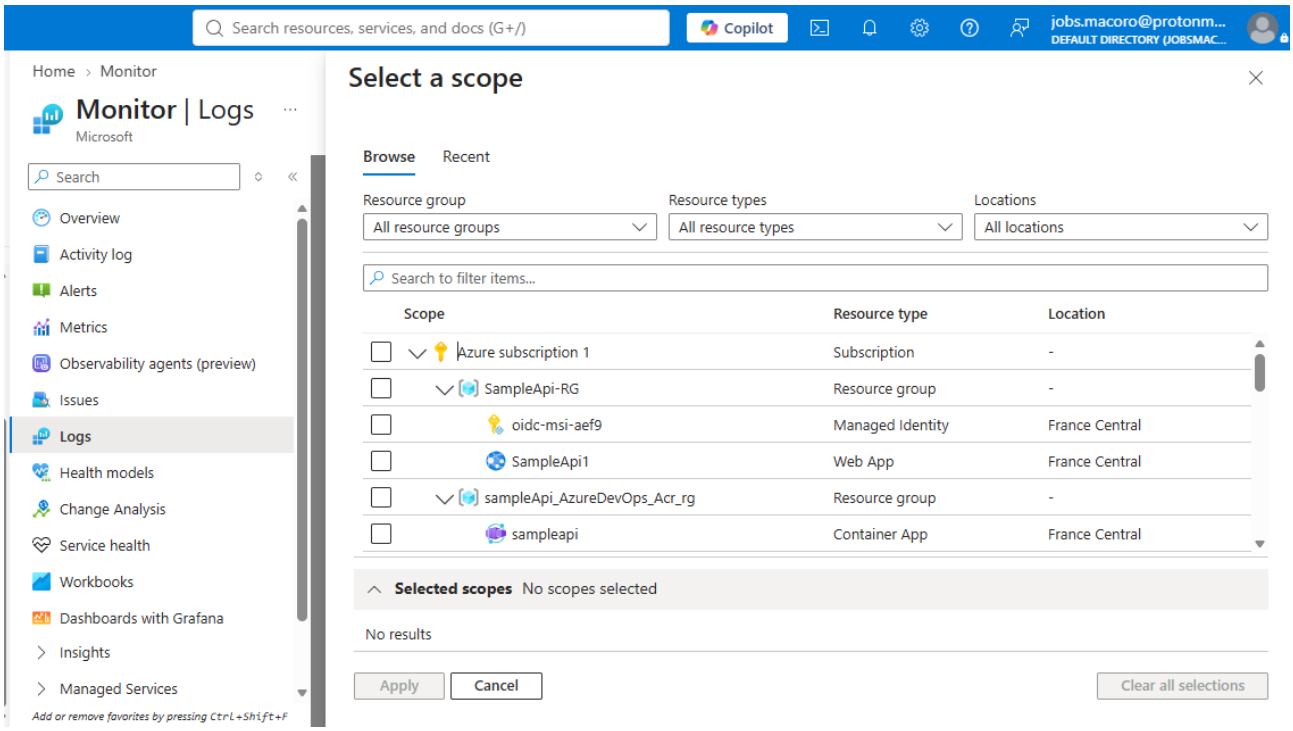
API Permissions > Other Permissions Granted for Default Directory > Microsoft Graph (1) > User.Read, Type=Delegated, Description = Sign in and read user profile, Admin Consent Required = No, Status = Granted for Default Directory

Certificates And Secrets > client secrets > Descriptiion = web-app, expires= 4/8/2027, Value= *****, Secret-ID = d84fa024-8448-4a76-8a82-84b8816659ee

Azure SQL
SQL Databases > free-sql-db-5542591 (sampleapiserve/free-sql-db- 5542591)>
Resource group : DefaultResourceGroup-CCAN, Server name : sampleapiserver.database.windows.net, Pricing tier : Free

What we have by resource group

If we go to Azure Portal > Monitor > prtal > Logs > we can see a list of resource groups with all the instances that they have inside.



Azure subscription 1
SampleApi-RG> oidc-msi-aef9 (Managed Identity , France Central), SampleApi1 (Web App, France Central)

sampleApi_AzureDevOps_Acr_rg > sampleapi(Container App,France Central),sampleapi-env(Container Apps Environment, France Central),workspace-samplepizureevpscrrgFWPG(Log Analytics workspace,France Central), workspacesampleapiazuredevopsacrrg9687(Log Analytics workspace,France Central)

SampleApi_AzureDevOps_Linux_RG > SampleApiAzureDevOpsLinux (Web App, France Central)

sampleApiPlatform_rg > sampleapiplatform-kv (Key vault,France Central), SampleDataAccessApi_UserKey (Managed Identity,France Central)

SampleAppApi_rg > sampleappapi (Container App,France Central), workspacesampleappapirg810e(Log Analytics workspace, France Central), workspacesampleappapirg9a54 (Log Analytics workspace, France Central), workspacesampleappapirgaa98 (Log Analytics workspace, France Central), workspacesampleappapirgac3a (Log Analytics

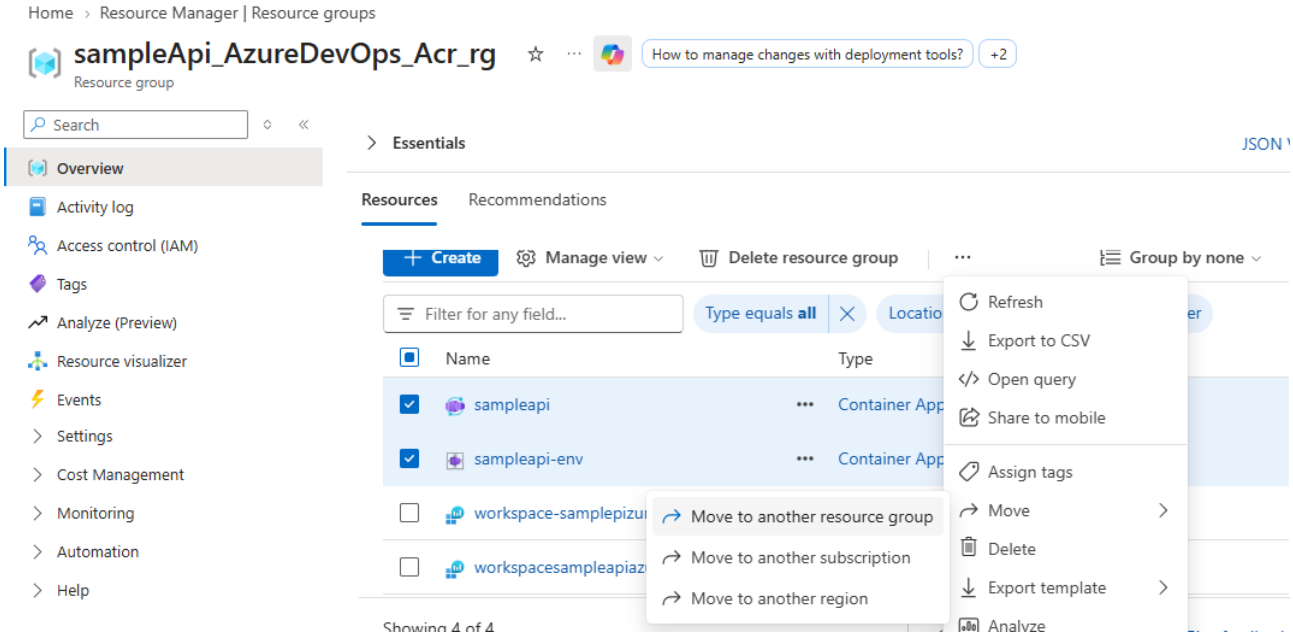
workspace, France Central), workspacesampleappapirgb264 (Log Analytics workspace, France Central), workspacesampleappapirgbe45 (Log Analytics workspace, France Central),

sampleDataAccessApi_rg > sampledataaccessapi (Container App, France Central), sampledataaccessapi-env2 (Container Apps Environment, France Central), workspacesampledataaccessapirg8896 (Log Analytics workspace, France Central), workspacesampledataaccessapirg8f23 (Log Analytics workspace, France Central), workspacesampledataaccessapirg9935 (Log Analytics workspace, France Central), workspacesampledataaccessapirgba6b (Log Analytics workspace, France Central)

We will move all our dsitributed contenaired application to one resource group: sampleApiPlatform_rg

1) Delete the sampledataaccessapi-env2 as is has 0 applications.

2) Resource Groups > click on sampleApi_AzureDevOps_Acr_rg > select the sampleapi and the sampleapi-env under the … select move > to another resource group > sampleApiPlatform_rg



3) Resource Groups > click on `SampleAppApi_rg` > select sampleappapi and under the … select move > to another resource group > sampleApiPlatform_rg

4) Resource Groups > click on `sampleDataAccessApi_rg` > select `sampledataaccessapi` and under the … select move > to another resource group > sampleApiPlatform_rg

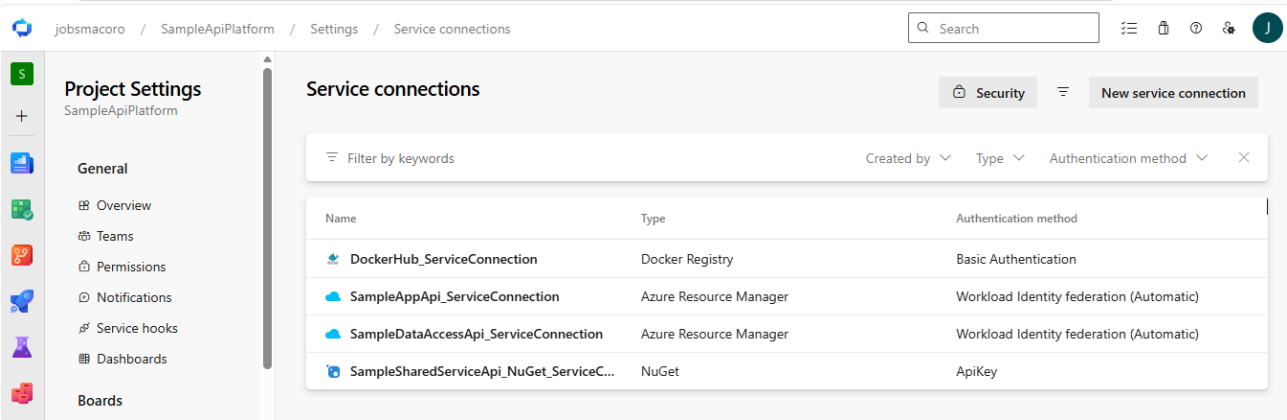
5) Go to each of the two RGs (`SampleAppApi_rg` and `sampleDataAccessApi_rg`):

1.Select **all** the Log Analytics workspaces (you can select multiple by checking boxes)

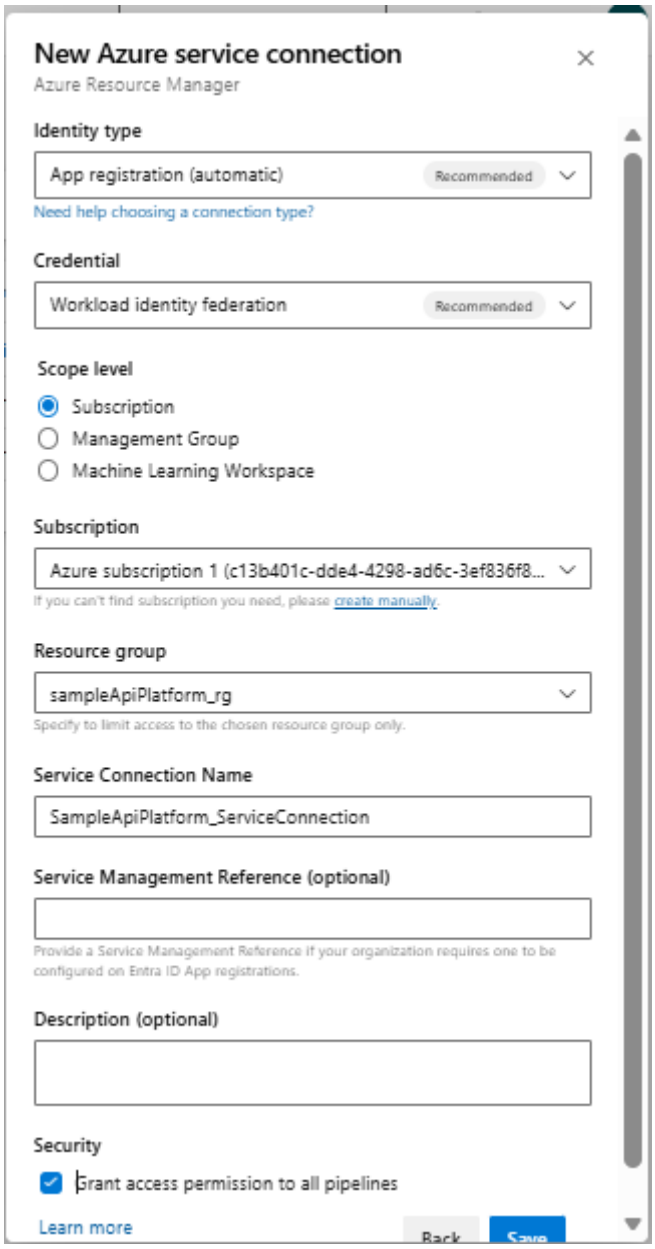
2.Click **Move → Move to another resource group**
3.Destination: `sampleApiPlatform_rg`

6) `SampleAppApi_rg` and `sampleDataAccessApi_rg`) and Click **"Delete resource group"**

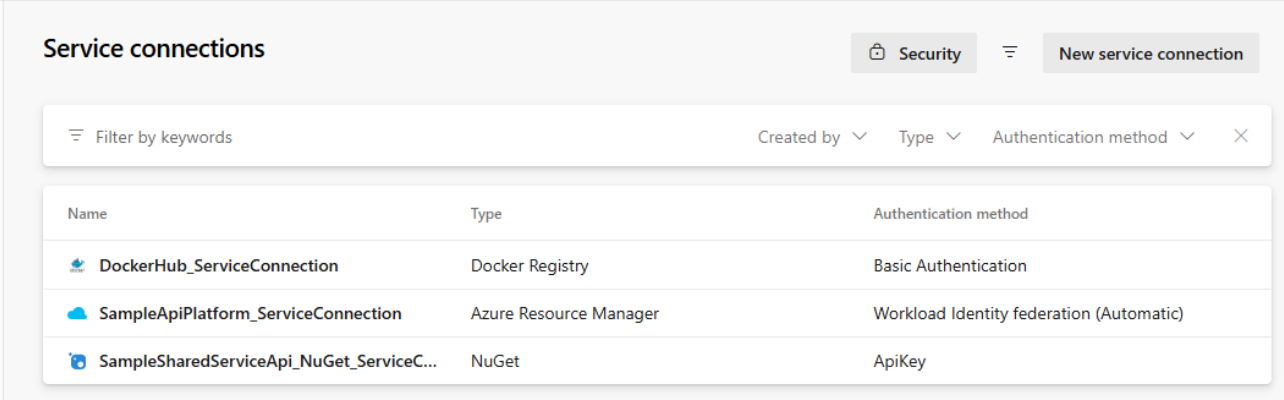
7) In Azure devops delete the service connections that were used before for the resource groups that have been deleted: SampleAppApi_ServiceConnection and SampleDataAccessApi_ServiceConnection.



9) Create a new ARM service connection: SampleApiPlatform_ServiceConnection:



Leaving these service connections:



10) Modify the pipelines that were point to the resources deleted and point them to the new resource group: sampleApiPlatform_rg

```
task: AzureCLI@2
  displayName: Update ACA image
  inputs:
    #we reuse the same service connection as in SampleDataAccessApi because it is only to login into Azure
    azureSubscription: 'SampleApiPlatform_ServiceConnection'
    scriptType: bash
    scriptLocation: inlineScript
    inlineScript: |
      az containerapp update \
        --name sampleappapi \
        --resource-group sampleApiPlatform_rg \
        --image docker.io/$(DOCKERHUB_USERNAME)/sampleappapi:latest
```

Then I deleted the workspaces because they seemed to be related by their name to the resource groups that we have just deleted.

After all these changes this is what we have:

Azure subscription 1
Location: France Central

SampleApi-RG (monolitic application deployed in web app)	oidc-msi-aef9 (Managed Identity), SampleApi1 (Web App)
sampleApi_AzureDevOps_Acr_rg (monolitic application deployed as a container)	sampleapi(Container App), workspace-samplepizureevpscrrgFWPG(Log Analytics workspace), workspacesampleapiazuredevopsacrrg9687(Log Analytics workspace)
SampleApi_AzureDevOps_Linux_RG > sampleApiPlatform_rg	SampleApiAzureDevOpsLinux (Web App) sampleapiplatform-kv (Key vault),

(distributed platform)	SampleDataAccessApi_UserKey (Managed Identity), sampleapi-env (Container Apps Environment), sampleappapi (Container App), sampledataaccessapi (Container App, France Central),
DefaultResourceGroup-CCAN (this DB is accessed by monolthic and distributed apps and therefore for the time being it make sense that has its own resource group)	free-sql-db-5542591 (SQL Database) sampleapiserver (Sql Server)

New container: SampleAuthApi

The current sampleappapi container does two things : authenticate users and queries for movies. These are two different concerns that we would like to split. Besides authenticating users can be also interesting for other projects as well in the same organization and therefore can be resued.

For this purpose we have copied the code from sampleappapi into sampleauthapi.

Reengineering Tasks SampleAuthApi

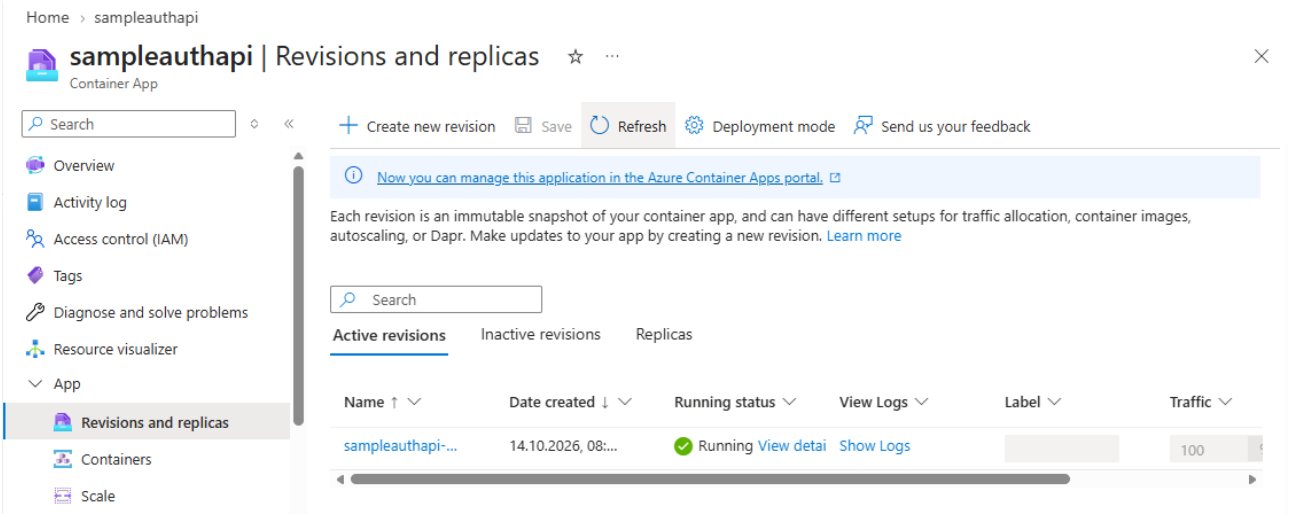
We have removed all concerning the moviescontroller and connecting to the sampledataccessapi.

We have updated the pipeline file.

The tasks for the sampleauthapi:

- register the container in entraid in the same way that we did for sampleappapi. This is the first thing because the Program.cs contains code that can throw an exception when the entraid values are not found. This can be done before the container exist.
- Create an image in dockerhub with the dedicated pipeline, as we did before with the other containers.
- create a container for sampleauthapi similar to sampleappapi, configure Ingress, add a system key, I think for this container we do not need the X-Token-Service as it is public,

We correct the errors until the container is up and running:



We need to modify the manifest of the sampleauthapi like we did initially for the sampleappiapi:

"requestedAccessTokenVersion": 2,

and also add clientId here (what is in bold)

```
"requiredResourceAccess": [
    {
~~~~~~~~"resourceAppId": "00000003-0000-0000-c000-000000000000",~~~~
~~~~~~~~"resourceAccess": [~~~~
~~~~~~~~~~{~~~~
~~~~~~~~~~"id": "e1fe6dd8-ba31-4d61-89e7-88639da4683d",~~~~
~~~~~~~~~~"type": "Scope"~~~~
~~~~~~~~~~}~~~~
~~~~~~~~]~~~~
~~~~~~},~~~~
    {
      "resourceAppId": "f9a3d163-e6e5-481b-9ea9-869076084bc7",
      "resourceAccess": [
        { "id": "1f0b1c78-9ed0-4ad8-a4d1-af9cfb207d1d", "type": "Scope" }
      ]
    }
  ]
```

and remove this that is to get Graph tokens:

```
{
                "resourceAppId": "00000003-0000-0000-c000-000000000000",
                "resourceAccess": [
                        {
                                "id": "e1fe6dd8-ba31-4d61-89e7-88639da4683d",
                                "type": "Scope"
                        }
                ]
```

```
            }
```

`where this f9a3d163-e6e5-481b-9ea9-869076084bc7 comes goes is after the api://f9a3d163-e6e5-481b-9ea9-869076084bc7 when we exposed the api.If we do not do that then Otherwise we get a token for Graph but not for the api.`

And also here:

"implicitGrantSettings": {
"enableAccessTokenIssuance": false,
"enableIdTokenIssuance": false

The tasks for the sampleappapi:

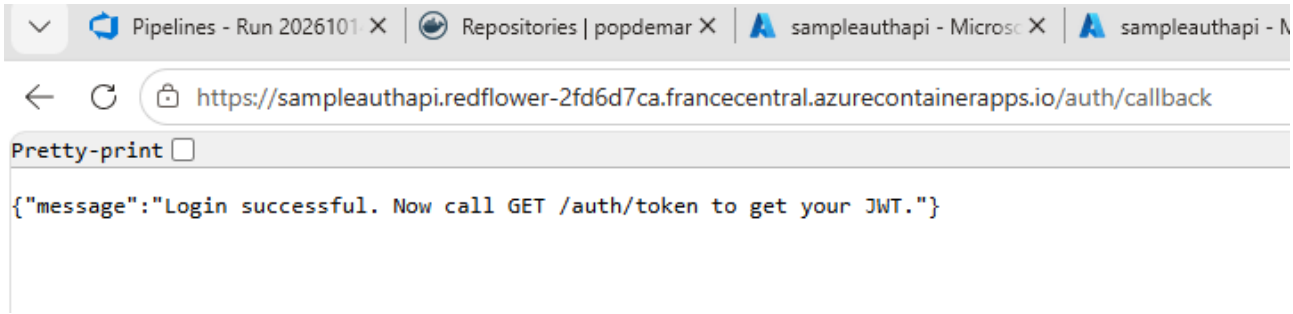
- it has to keep ingress as still will have queries coming from outside, it needs to verify the token,
- we can delete sampleappapi from the app registrations in entraid in the same way that the sampledataaccess is neither registered
- the configuration has to be the same as sampleauthapi

The tasks for the sampledataaccessapi

- the configuration has to be the same as sampleauthapi

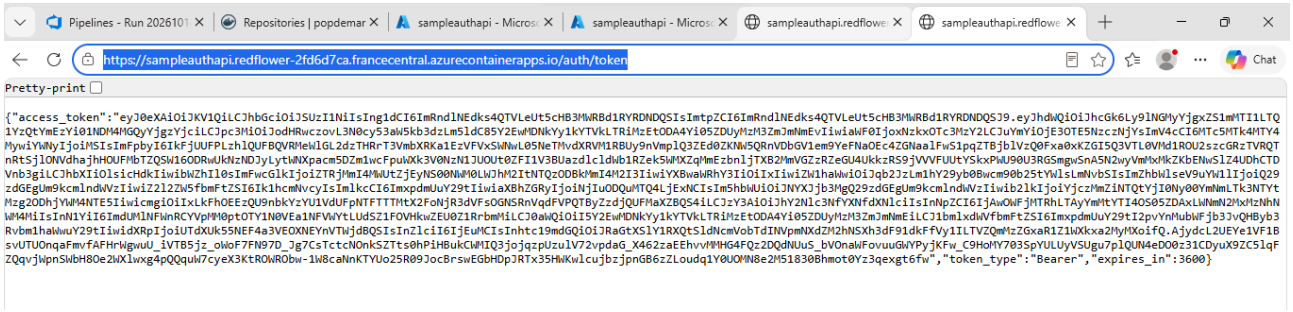
Now we can test the sampleauthapi. We expect to get a JWT token:

https://sampleauthapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/login



Once we get a successfult message then we can get the token by typing in the browser the following url:

sampleauthapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/auth/token




```
{"access_token":"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6ImRndlNEdks4QTVLeUt
5cHB3MWRBd1RYRDNDQSIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOi
JhcGk6Ly9lNGMyYjgxZS1mMTI1LTQ1YzQtYmEzYi01NDM4MGQyYjgzYjciLCJpc3MiOiJodHRwczovL3
N0cy53aW5kb3dzLm5ldC85Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi05ZDUyMzM3ZmJmNmEvIiwiaWF0Ij
oxNzkxOTc3MzY2LCJuYmYiOjE3OTE5NzczNjYsImV4cCI6MTc5MTk4MTY4MywiYWNyIjoiMSIsImFpby
I6IkFjUUFPLzhlQUFBQVRMeWlGL2dzTHRrT3VmbXRKa1EzVFVxSWNwL05NeTMvdXRVM1RBUy9nVmplQ3
ZEd0ZKNW5QRnVDbGV1em9YeFNaOEc4ZGNaalFwS1pqZTBjblVzQ0Fxa0xKZGI5Q3VTL0VMd1ROU2szcG
RzTVRQTnRtSjlONVdhajhHOUFMbTZQSW16ODRwUkNzNDJyLytWNXpacm5DZm1wcFpuWXk3V0NzN1JUOU
t0ZFI1V3BUazdlcldWb1RZek5WMXZqMmEzbnljTXB2MmVGZzRZeGU4UkkzRS9jVVVFUUtYSkxPWU90U3
RGSmgwSnA5N2wyVmMxMkZKbENwSlZ4UDhCTDVnb3giLCJhbXIiOlsicHdkIiwibWZhIl0sImFwcGlkIj
oiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNTQzODBkMmI4M2I3IiwiYXBwaWRhY3IiOiIxIiwiZW1haW
wiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsImZhbWlseV9uYW1lIjoiQ29zdGEgUm9kcmlndW
VzIiwiZ2l2ZW5fbmFtZSI6Ik1hcmNvcyIsImlkcCI6ImxpdmUuY29tIiwiaXBhZGRyIjoiNjIuODQuMT
Q4LjExNCIsIm5hbWUiOiJNYXJjb3MgQ29zdGEgUm9kcmlndWVzIiwib2lkIjoiYjczMmZiNTQtYjI0Ny
00YmNmLTk3NTYtMzg2ODhjYWM4NTE5IiwicmgiOiIxLkFhOEEzQU9nbkYzYU1VdUFpNTFTTTMtX2FoNj
R3dVFsOGNSRnVqdFVPQTByZzdjQUFMaXZBQS4iLCJzY3AiOiJhY2Nlc3NfYXNfdXNlciIsInNpZCI6Ij
AwOWFjMTRhLTAyYmMtYTI4OS05ZDAxLWNmN2MxMzNhNWM4MiIsInN1YiI6ImdUMlNFWnRCYVpMM0ptOT
Y1N0VEa1NFVWYtLUdSZ1FOVHkwZEU0Z1RrbmMiLCJ0aWQiOiI5Y2EwMDNkYy1kYTVkLTRiMzEtODA4Yi
05ZDUyMzM3ZmJmNmEiLCJ1bmlxdWVfbmFtZSI6ImxpdmUuY29tI2pvYnMubWFjb3JvQHByb3Rvbm1haW
wuY29tIiwidXRpIjoiUTdXUk55NEF4a3VEOXNEYnVTWjdBQSIsInZlciI6IjEuMCIsInhtc19mdGQiOi
JRaGtXSlY1RXQtSldNcmVobTdINVpmNXdZM2hNSXh3dF91dkFfVy1ILTVZQmMzZGxaR1Z1WXkxa2MyMX
oifQ.AjydcL2UEYe1VF1BsvUTUOnqaFmvfAFHrWgwuU_iVTB5jz_oWoF7FN97D_Jg7CsTctcNOnkSZTt
s0hPiHBukCWMIQ3jojqzpUzulV72vpdaG_X462zaEEhvvMMHG4FQz2DQdNUuS_bVOnaWFovuuGWYPyjK
Fw_C9HoMY703SpYULUyVSUgu7plQUN4eDO0z31CDyuX9ZC5lqFZQqvjWpnSWbH8Oe2WXlwxg4pQQquW7
cyeX3KtROWRObw-1W8caNnKTYUo25R09JocBrswEGbHDpJRTx35HWKwlcujbzjpnGB6zZLoudq1Y0UOM
N8e2M51830Bhmot0Yz3qexgt6fw","token_type":"Bearer","expires_in":3600}
```


We can verify the token in: jwt.io

Then we call the same methods as we did before:

1) GetAll

curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies -H "X-Service-Token: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer "eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJlNGMyYjgxZS1mMTI1LTQ1YzQtYmEzYi01NDM4MGQyYjgzYjciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTIwMzIwMDAsIm5iZiI6MTc5MjAzMjAwMCwiZXhwIjoxNzkyMDM3MDkwLCJhaW8iOiJBZFFBSy84ZUFBQUFPS2lkUmtsekpkbmNwQnI5MUJkanRJYkp0NUdCd2loTmp1eGZWK1BicXgzdnJlN05pMWhiNDFjTHM4K01iTUZhdlh4dkR0ckl5NlY5SE9PUmdqeUovdGJvRUh3QkRTbTJ3U3ppQ25xTEVWR0pMU0toT1kzTzJNbVhTeGdWWEM3ZFh6b2x0UVBpZkwrb0NsejVSV1hZV0poZ08ralRpUkZ6eENDc1A1WU44QzdYWTRLaWJNa1JXWURWdzFockFSYktKNE9PQjJtdGpSNlhpcDRwakU1LzQ0WEM0ZDNrdHhyL3ExbDZvcmpGSU1keUR3ckRIeEZVbnJBQm1SY1N1QURvYS81NGpCZUJHSzY4UEVyRDE0amhWZz09IiwiYXpwIjoiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNT

QzODBkMmI4M2I3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvcyBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9haDY0d3VRbDhjUkZ1anRVT0Ewcmc3Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIwMDlhZTg1YS1lNzlkLTc1ODQtODVmZS1jOWJhMmJiMjY3MzQiLCJzdWIiOiJnVDJTRVp0QmFaTDNKbTk2NTdFRGtTRVVmLS1HUmdRTlR5MGRFNGdUa25jIiwidGlkIjoiOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoib2NvWWtOUWtSVWVvYjdHU2k4dWZBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJfWksxNDFFR1pWZGRCYVVJekUwMXcyVlcxMjVHQ2FLbTNleEhIakh5N1BVQlpYVnliM0JsYm05eWRHZ3RaSE50Y3cifQ.I_FUSKWS2zZ3S0fMT0A21YhaSmeXopx29UD_sf33RostjVsgJoMp_ZIVMcLNZNgxI-x6RlsEtRIsKg3O316TfOkyNn-EQr5aHqz7nOaWWUHHLw3fRzuhwba6vJxJcgAhFYF8mZei5CFbzFEMNRSu1RhB3WqtfWPVeRRIe6kqsQ5DfOXQvaKCJ6LdVEwALwVlI71QSQZQCWzCZkYh1nIZpSG7Vb-Kk7_QSL-UTlwHZrUqW3DU1I1jIL1TNl0RsXIV7SHMZz4Wel-OrQtu5UvsmxDSLj05aWXOFOtJSsaRE8xXblLga4FVXX2QC-WAWCC69JxS7UCuLa707eUGwMkmcg" [{"id":"f3e36699-dbe5-4a6e-9ab2-28adeb559490","title":"The Matrix Reloaded","year":2003}

2)GetById

curl https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies/f3e36699-dbe5-4a6e-9ab2-28adeb559490 -H "X-Service-Token: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJlNGMyYjgxZS1mMTI1LTQ1YzQtYmEzYi01NDM4MGQyYjgzYjciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTIwMzIwMDAsIm5iZiI6MTc5MjAzMjAwMCwiZXhwIjoxNzkyMDM3MDkwLCJhaW8iOiJBZFFBSy84ZUFBQUFPS2lkUmtsekpkbmNwQnI5MUJkanRJYkp0NUdCd2loTmp1eGZWK1BicXgzdnJlN05pMWhiNDFjTHM4K01iTUZhdlh4dkR0ckl5NlY5SE9PUmdqeUovdGJvRUh3QkRTbTJ3U3ppQ25xTEVWR0pMU0toT1kzTzJNbVhTeGdWWEM3ZFh6b2x0UVBpZkwrb0NsejVSV1hZV0poZ08ralRpUkZ6eENDc1A1WU44QzdYWTRLaWJNa1JXWURWdzFockFSYktKNE9PQjJtdGpSNlhpcDRwakU1LzQ0WEM0ZDNrdHhyL3ExbDZvcmpGSU1keUR3ckRIeEZVbnJBQm1SY1N1QURvYS81NGpCZUJHSzY4UEVyRDE0amhWZz09IiwiYXpwIjoiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNTQzODBkMmI4M2I3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvcyBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9haDY0d3VRbDhjUkZ1anRVT0Ewcmc3Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIwMDlhZTg1YS1lNzlkLTc1ODQtODVmZS1jOWJhMmJiMjY3MzQiLCJzdWIiOiJnVDJTRVp0QmFaTDNKbTk2NTdFRGtTRVVmLS1HUmdRTlR5MGRFNGdUa25jIiwidGlkIjoiOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoib2NvWWtOUWtSVWVvYjdHU2k4dWZBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJfWksxNDFFR1pWZGRCYVVJekUwMXcyVlcxMjVHQ2FLbTNleEhIakh5N1BVQlpYVnliM0JsYm05eWRHZ3RaSE50Y3cifQ.I_FUSKWS2zZ3S

0fMT0A21YhaSmeXopx29UD_sf33RostjVsgJoMp_ZIVMcLNZNgxI-x6RlsEtRIsKg3O316TfOkyNn-EQr5aHqz7nOaWWUHHLw3fRzuhwba6vJxJcgAhFYF8mZei5CFbzFEMNRSu1RhB3WqtfWPVeRRIe6kqsQ5DfOXQvaKCJ6LdVEwALwVlI71QSQZQCWzCZkYh1nIZpSG7Vb-Kk7_QSL-UTlwHZrUqW3DU1I1jIL1TNl0RsXIV7SHMZz4Wel-OrQtu5UvsmxDSLj05aWXOFOtJSsaRE8xXblLga4FVXX2QC-WAWCC69JxS7UCuLa707eUGwMkmcg" {"id":"f3e36699-dbe5-4a6e-9ab2-28adeb559490","title":"The Matrix Reloaded","year":2003}

3)Add

curl -X POST https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies -H "Content-Type: application/json" -H "X-Service-Token:4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJlNGMyYjgxZS1mMTI1LTQ1YzQtYmEzYi01NDM4MGQyYjgzYjciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTIwMzIwMDAsIm5iZiI6MTc5MjAzMjAwMCwiZXhwIjoxNzkyMDM3MDkwLCJhaW8iOiJBZFFBSy84ZUFBQUFPS2lkUmtsekpkbmNwQnI5MUJkanRJYkp0NUdCd2loTmp1eGZWK1BicXgzdnJlN05pMWhiNDFjTHM4K01iTUZhdlh4dkR0ckl5NlY5SE9PUmdqeUovdGJvRUh3QkRTbTJ3U3ppQ25xTEVWR0pMU0toT1kzTzJNbVhTeGdWWEM3ZFh6b2x0UVBpZkwrb0NsejVSV1hZV0poZ08ralRpUkZ6eENDc1A1WU44QzdYWTRLaWJNa1JXWURWdzFockFSYktKNE9PQjJtdGpSNlhpcDRwakU1LzQ0WEM0ZDNrdHhyL3ExbDZvcmpGSU1keUR3ckRIeEZVbnJBQm1SY1N1QURvYS81NGpCZUJHSzY4UEVyRDE0amhWZz09IiwiYXpwIjoiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNTQzODBkMmI4M2I3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvcyBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9haDY0d3VRbDhjUkZ1anRVT0Ewcmc3Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIwMDlhZTg1YS1lNzlkLTc1ODQtODVmZS1jOWJhMmJiMjY3MzQiLCJzdWIiOiJnVDJTRVp0QmFaTDNKbTk2NTdFRGtTRVVmLS1HUmdRTlR5MGRFNGdUa25jIiwidGlkIjoiOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoib2NvWWtOUWtSVWVvYjdHU2k4dWZBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJfWksxNDFFR1pWZGRCYVVJekUwMXcyVlcxMjVHQ2FLbTNleEhIakh5N1BVQlpYVnliM0JsYm05eWRHZ3RaSE50Y3cifQ.I_FUSKWS2zZ3S0fMT0A21YhaSmeXopx29UD_sf33RostjVsgJoMp_ZIVMcLNZNgxI-x6RlsEtRIsKg3O316TfOkyNn-EQr5aHqz7nOaWWUHHLw3fRzuhwba6vJxJcgAhFYF8mZei5CFbzFEMNRSu1RhB3WqtfWPVeRRIe6kqsQ5DfOXQvaKCJ6LdVEwALwVlI71QSQZQCWzCZkYh1nIZpSG7Vb-Kk7_QSL-UTlwHZrUqW3DU1I1jIL1TNl0RsXIV7SHMZz4Wel-OrQtu5UvsmxDSLj05aWXOFOtJSsaRE8xXblLga4FVXX2QC-WAWCC69JxS7UCuLa707eUGwMkmcg" -d "{"Title": "The Matrix151020260248", "Year": 1999}" {"id":"fd7464a9-ba3d-470a-8787-8041ba6f4c4f","title":"The Matrix151020260248","year":1999}

4)Uddate

curl -X PUT https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies/fd7464a9-ba3d-470a-8787-8041ba6f4c4f -H "Content-Type: application/json" -H "X-Service-Token: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJlNGMyYjgxZS1mMTI1LTQ1YzQtYmEzYi01NDM4MGQyYjgzYjciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTIwMzIwMDAsIm5iZiI6MTc5MjAzMjAwMCwiZXhwIjoxNzkyMDM3MDkwLCJhaW8iOiJBZFFBSy84ZUFBQUFPS2lkUmtsekpkbmNwQnI5MUJkanRJYkp0NUdCd2loTmp1eGZWK1BicXgzdnJlN05pMWhiNDFjTHM4K01iTUZhdlh4dkR0ckl5NlY5SE9PUmdqeUovdGJvRUh3QkRTbTJ3U3ppQ25xTEVWR0pMU0toT1kzTzJNbVhTeGdWWEM3ZFh6b2x0UVBpZkwrb0NsejVSV1hZV0poZ08ralRpUkZ6eENDc1A1WU44QzdYWTRLaWJNa1JXWURWdzFockFSYktKNE9PQjJtdGpSNlhpcDRwakU1LzQ0WEM0ZDNrdHhyL3ExbDZvcmpGSU1keUR3ckRIeEZVbnJBQm1SY1N1QURvYS81NGpCZUJHSzY4UEVyRDE0amhWZz09IiwiYXpwIjoiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNTQzODBkMmI4M2I3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvcyBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9haDY0d3VRbDhjUkZ1anRVT0Ewcmc3Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIwMDlhZTg1YS1lNzlkLTc1ODQtODVmZS1jOWJhMmJiMjY3MzQiLCJzdWIiOiJnVDJTRVp0QmFaTDNKbTk2NTdFRGtTRVVmLS1HUmdRTlR5MGRFNGdUa25jIiwidGlkIjoiOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoib2NvWWtOUWtSVWVvYjdHU2k4dWZBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJfWksxNDFFR1pWZGRCYVVJekUwMXcyVlcxMjVHQ2FLbTNleEhIakh5N1BVQlpYVnliM0JsYm05eWRHZ3RaSE50Y3cifQ.I_FUSKWS2zZ3S0fMT0A21YhaSmeXopx29UD_sf33RostjVsgJoMp_ZIVMcLNZNgxI-x6RlsEtRIsKg3O316TfOkyNn-EQr5aHqz7nOaWWUHHLw3fRzuhwba6vJxJcgAhFYF8mZei5CFbzFEMNRSu1RhB3WqtfWPVeRRIe6kqsQ5DfOXQvaKCJ6LdVEwALwVlI71QSQZQCWzCZkYh1nIZpSG7Vb-Kk7_QSL-UTlwHZrUqW3DU1I1jIL1TNl0RsXIV7SHMZz4Wel-OrQtu5UvsmxDSLj05aWXOFOtJSsaRE8xXblLga4FVXX2QC-WAWCC69JxS7UCuLa707eUGwMkmcg" -d "{"Title": "The Matrix Reloaded 151120260252", "Year": 2003}" {"id":"fd7464a9-ba3d-470a-8787-8041ba6f4c4f","title":"The Matrix Reloaded 151120260252","year":2003}

5)Delete

curl -X DELETE https://sampleappapi.redflower-2fd6d7ca.francecentral.azurecontainerapps.io/api/movies/fd7464a9-ba3d-470a-8787-8041ba6f4c4f -H "Content-Type: application/json" -H "X-Service-Token: 4f8b2e1c-9a2d-4c3f-8a1e-2c9d7f1e3b4a" -H "Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6ImRndlNEdks4QTVLeUt5cHB3MWRBd1RYRDNDQSJ9.eyJhdWQiOiJlNGMyYjgxZS1mMTI1LTQ1YzQtYmEzYi01NDM4MGQyYjgzYjciLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhL3YyLjAiLCJpYXQiOjE3OTIwMzIwMDAsIm5iZiI6MTc5MjAzMjAwMCwiZXhwIjoxNzkyMDM3MDkwLCJhaW8iOiJBZFFBSy84ZUFBQUFPS2lkUmtsekpkbmNwQnI5MUJkanRJYkp0NUdCd2loTmp1eGZWK1BicXgzdnJlN05pMWhiNDFjTHM4K01iTUZhdlh4dkR0ckl5NlY5SE9PUmdqeUovdGJvRUh3QkRTbTJ3U3ppQ25xTEVWR0pMU0toT1kzTzJNbVhTeGdWWEM3ZFh6b2x0UVBpZkwrb0NsejVSV1hZV0poZ08ralRpUkZ6eENDc1A1WU44QzdYWTRLaWJNa1JXWURWdzFockFSYktKNE9PQjJtdGpSNlhpcDRwakU1LzQ0WEM0ZDNrdHhyL3ExbDZvcmpGSU1keUR3ckRIeEZVbnJBQm1SY1N1QURvYS81NGpCZUJHSzY4UEVyRDE0amhWZz09IiwiYXpwIjoiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNTQzODBkMmI4M2I3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvcyBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tYWlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9haDY0d3VRbDhjUkZ1anRVT0Ewcmc3Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIwMDlhZTg1YS1lNzlkLTc1ODQtODVmZS1jOWJhMmJiMjY3MzQiLCJzdWIiOiJnVDJTRVp0QmFaTDNKbTk2NTdFRGtTRVVmLS1HUmdRTlR5MGRFNGdUa25jIiwidGlkIjoiOWNhMDAzZGMtZGE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoib2NvWWtOUWtSVWVvYjdHU2k4dWZBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJfWksxNDFFR1pWZGRCYVVJekUwMXcyVlcxMjVHQ2FLbTNleEhIakh5N1BVQlpYVnliM0JsYm05eWRHZ3RaSE50Y3cifQ.I_FUSKWS2zZ3S0fMT0A21YhaSmeXopx29UD_sf33RostjVsgJoMp_ZIVMcLNZNgxI-x6RlsEtRIsKg3O316TfOkyNn-EQr5aHqz7nOaWWUHHLw3fRzuhwba6vJxJcgAhFYF8mZei5CFbzFEMNRSu1RhB3WqtfWPVeRRIe6kqsQ5DfOXQvaKCJ6LdVEwALwVlI71QSQZQCWzCZkYh1nIZpSG7Vb-Kk7_QSL-UTlwHZrUqW3DU1I1jIL1TNl0RsXIV7SHMZz4Wel-OrQtu5UvsmxDSLj05aWXOFOtJSsaRE8xXblLga4FVXX2QC-WAWCC69JxS7UCuLa707eUGwMkmcg

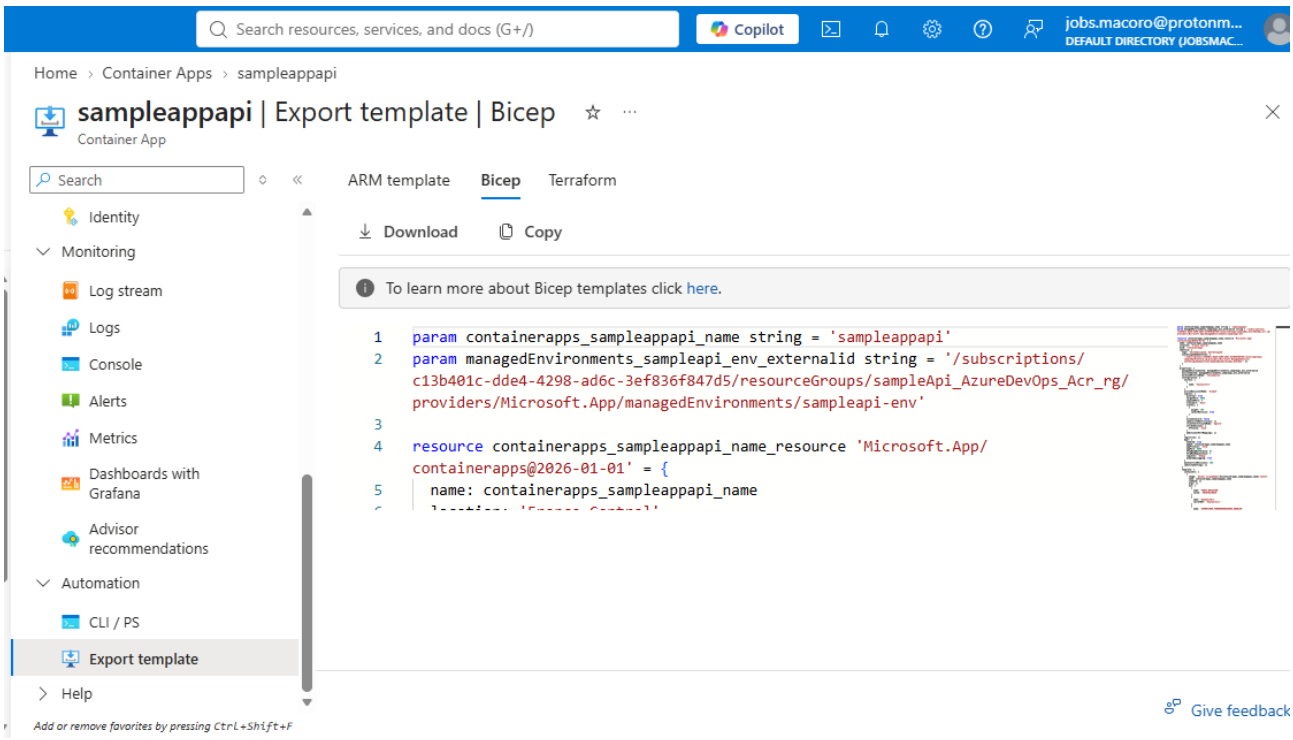
HM4K01iTUZhdlh4dkR0ckl5NlY5SE9PUmdqeUovdGJvRUh3QkRTbTJ3U3ppQ25xTEVWR0p
MU0toT1kzTzJNbVhTeGdWWEM3ZFh6b2x0UVBpZkwrb0NsejVSV1hZV0poZ08ralRpUkZ6eE
NDc1A1WU44QzdYWTRLaWJNa1JXWURWdzFockFSYktKNE9PQjJtdGpSNlhpcDRwakU1Lz
Q0WEM0ZDNrdHhyL3ExbDZvcmpGSU1keUR3ckRIeEZVbnJBQm1SY1N1QURvYS81NGpCZ
UJHSzY4UEVyRDE0amhWZz09IiwiYXpwIjoiZTRjMmI4MWUtZjEyNS00NWM0LWJhM2ItNT
QzODBkMmI4M2I3IiwiYXpwYWNyIjoiMSIsImlkcCI6ImxpdmUuY29tIiwibmFtZSI6Ik1hcmNvc
yBDb3N0YSBSb2RyaWd1ZXMiLCJvaWQiOiJiNzMyZmI1NC1iMjQ3LTRiY2YtOTc1Ni0zODY
4OGNhYzg1MTkiLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJqb2JzLm1hY29yb0Bwcm90b25tY
WlsLmNvbSIsInJoIjoiMS5BYThBM0FPZ25GM2FNVXVBaTUxU00zLV9haDY0d3VRbDhjUkZ
1anRVT0Ewcmc3Y0FBTGl2QUEuIiwic2NwIjoiYWNjZXNzX2FzX3VzZXIiLCJzaWQiOiIwMDl
hZTg1YS1lNzlkLTc1ODQtODVmZS1jOWJhMmJiMjY3MzQiLCJzdWIiOiJnVDJTRVp0QmFaT
DNKbTk2NTdFRGtTRVVmLS1HUmdRTlR5MGRFNGdUa25jIiwidGlkIjoiOWNhMDAzZGMtZ
GE1ZC00YjMxLTgwOGItOWQ1MjMzN2ZiZjZhIiwidXRpIjoib2NvWWtOUWtSVWVvYjdHU2
k4dWZBQSIsInZlciI6IjIuMCIsInhtc19mdGQiOiJfWksxNDFFR1pWZGRCYVVJekUwMXcyVlcx
MjVHQ2FLbTNleEhIakh5N1BVQlpYVnliM0JsYm05eWRHZ3RaSE50Y3cifQ.I_FUSKWS2zZ3S
0fMT0A21YhaSmeXopx29UD_sf33RostjVsgJoMp_ZIVMcLNZNgxI-
x6RlsEtRIsKg3O316TfOkyNn-
EQr5aHqz7nOaWWUHHLw3fRzuhwba6vJxJcgAhFYF8mZei5CFbzFEMNRSu1RhB3WqtfWPVe
RRIe6kqsQ5DfOXQvaKCJ6LdVEwALwVlI71QSQZQCWzCZkYh1nIZpSG7Vb-Kk7_QSL-
UTlwHZrUqW3DU1I1jIL1TNl0RsXIV7SHMZz4Wel-
OrQtu5UvsmxDSLj05aWXOFOtJSsaRE8xXblLga4FVXX2QC-
WAWCC69JxS7UCuLa707eUGwMkmcg"

C:\Users\

~~So as an exercise, we will try to create this container using Bicep:~~

~~But before that we need to upload a first image to Docker Hub and to do that like in the previous containers, we need to build a Dockerfile and to create a first pipeline that will just create the image.~~

~~Go to Container Apps > sampleappapi > Automation > Export Template > Bicep Tab~~



We will get a file like this:

```
param containerapps_sampleappapi_name string = 'sampleappapi'
param managedEnvironments_sampleapi_env_externalid string =
'/subscriptions/c13b401c-dde4-4298-ad6c-3ef836f847d5/resourceGroups/
sampleApiPlatform_rg/providers/Microsoft.App/managedEnvironments/sampleapi-env'

resource containerapps_sampleappapi_name_resource
'Microsoft.App/containerapps@2026-01-01' = {
  name: containerapps_sampleappapi_name
  location: 'France Central'
  kind: 'containerapps'
  identity: {
    type: 'SystemAssigned, UserAssigned'
    userAssignedIdentities: {
      '/subscriptions/c13b401c-dde4-4298-ad6c-3ef836f847d5/resourcegroups/
sampleApiPlatform_rg/providers/Microsoft.ManagedIdentity/userAssignedIdentities/
SampleDataAccessApi_UserKey': {}
    }
  }
  properties: {
    managedEnvironmentId: managedEnvironments_sampleapi_env_externalid
    environmentId: managedEnvironments_sampleapi_env_externalid
    workloadProfileName: 'Consumption'
    configuration: {
      secrets: [
        {
          name: 'keyvaulturi'
        }
      ]
```

```
    activeRevisionsMode: 'Single'
    ingress: {
      external: true
      targetPort: 8080
      exposedPort: 0
      transport: 'Auto'
      traffic: [
        {
          weight: 100
          latestRevision: true
        }
      ]
      allowInsecure: false
      ipSecurityRestrictions: []
      clientCertificateMode: 'Ignore'
      stickySessions: {
        affinity: 'none'
      }
      additionalPortMappings: []
    }
    registries: []
    dapr: {
      enabled: true
      appId: containerapps_sampleappapi_name
      appProtocol: 'http'
      appPort: 8080
      httpReadBufferSize: 65
      httpMaxRequestSize: 4
      logLevel: 'debug'
      enableApiLogging: true
    }
    maxInactiveRevisions: 100
    identitySettings: []
  }
  template: {
    containers: [
      {
        image: 'docker.io/popdemar/${containerapps_sampleappapi_name}:latest'
        name: containerapps_sampleappapi_name
        command: []
        args: []
        env: [
          {
            name: 'FORCE_REVISION'
            value: '202610130850'
          }
          {
            name: 'KeyVaultUri'
            secretRef: 'keyvaulturi'
          }
          {
```

```
          name: 'ASPNETCORE_FORWARDEDHEADERS_ENABLED'
          value: 'true'
        }
      ]
      resources: {
        cpu: json('0.5')
        memory: '1Gi'
      }
      probes: [
        {
          type: 'Liveness'
          failureThreshold: 3
          initialDelaySeconds: 0
          periodSeconds: 10
          successThreshold: 1
          tcpSocket: {
            port: 8080
          }
          timeoutSeconds: 5
        }
        {
          type: 'Readiness'
          failureThreshold: 48
          initialDelaySeconds: 0
          periodSeconds: 5
          successThreshold: 1
          tcpSocket: {
            port: 8080
          }
          timeoutSeconds: 5
        }
        {
          type: 'Startup'
          failureThreshold: 240
          initialDelaySeconds: 1
          periodSeconds: 1
          successThreshold: 1
          tcpSocket: {
            port: 8080
          }
          timeoutSeconds: 3
        }
      ]
    }
  ]
  scale: {
    minReplicas: 1
    maxReplicas: 1
    cooldownPeriod: 300
    pollingInterval: 30
  }
```

```
      volumes: []
    }
  }
}
```

From here we create a template by extracting the values and putting in properties.

`sampleauthapi_container.bicep`

```
// ============================================
// Parameters
// ============================================
@description('Name of the container app')
param containerAppName string = 'sampleauthapi'

param location string = resourceGroup().location

@description('Resource ID of the Container Apps managed environment')
param managedEnvironmentId string

param containerImage string = 'docker.io/popdemar/${containerAppName}:latest'
param managedIdentityId string =
'/subscriptions/c13b401c-dde4-4298-ad6c-3ef836f847d5/resourcegroups/sampleApiPlatform_rg/
providers/Microsoft.ManagedIdentity/userAssignedIdentities/SampleDataAccessApi_UserKey'
param minReplicas int = 1
param maxReplicas int = 5
param targetPort int = 8080
param cpu string = '0.5'
param memory string = '1Gi'

@description('Name of the Key Vault holding the app secrets')
param keyVaultName string = 'sampleapiplatform-kv'

@secure()
@description('URI of the Key Vault passed to the container as a secret (leave empty to derive it
from keyVaultName)')
param keyVaultUri string = ''

// ============================================
// Variables
// ============================================
var resolvedKeyVaultUri = keyVaultUri != '' ? keyVaultUri : 'https://${keyVaultName}.$
{environment().suffixes.keyvaultDns}'

// ============================================
// Resources
// ============================================
resource containerApp 'Microsoft.App/containerApps@2023-05-01' = {
  name: containerAppName
```

```
  location: location

  identity: {
    type: 'SystemAssigned, UserAssigned'
    userAssignedIdentities: {
      '${managedIdentityId}': {}
    }
  }

  properties: {
    managedEnvironmentId: managedEnvironmentId
    workloadProfileName: 'Consumption'

    configuration: {
      secrets: [
        {
          name: 'keyvaulturi'
          value: resolvedKeyVaultUri
        }
      ]

      activeRevisionsMode: 'Single'

      ingress: {
        external: true
        targetPort: targetPort
        exposedPort: 0
        transport: 'Auto'

        traffic: [
          {
            weight: 100
            latestRevision: true
          }
        ]

        allowInsecure: false
        ipSecurityRestrictions: []
        clientCertificateMode: 'Ignore'

        stickySessions: {
          affinity: 'none'
        }
      }

      registries: []

      dapr: {
        enabled: true
        appId: containerAppName
        appProtocol: 'http'
        appPort: targetPort
```

```
    httpReadBufferSize: 65
    httpMaxRequestSize: 4
    logLevel: 'info'
    enableApiLogging: true
  }

  maxInactiveRevisions: 100
 }

 template: {
  containers: [
   {
    name: containerAppName
    image: containerImage

    env: [
     {
      name: 'ASPNETCORE_FORWARDEDHEADERS_ENABLED'
      value: 'true'
     }
     {
      name: 'KeyVaultUri'
      secretRef: 'keyvaulturi'
     }
    ]

    resources: {
     cpu: json(cpu)
     memory: memory
    }

    probes: [
     {
      type: 'Liveness'
      failureThreshold: 3
      initialDelaySeconds: 10
      periodSeconds: 10
      successThreshold: 1
      tcpSocket: {
       port: targetPort
      }
      timeoutSeconds: 5
     }
     {
      type: 'Readiness'
      failureThreshold: 3
      initialDelaySeconds: 5
      periodSeconds: 5
      successThreshold: 1
      tcpSocket: {
       port: targetPort
      }
```

```
        timeoutSeconds: 5
      }
    ]
  }
]

  scale: {
    minReplicas: minReplicas
    maxReplicas: maxReplicas
  }
 }
}

// ================================================
// Outputs
// ================================================
output fqdn string = containerApp.properties.configuration.ingress.fqdn
output containerAppId string = containerApp.id
```

Then we validate the file

```
az login
az bicep build --file sampleauthapi_container.bicep
```

Then we create the container with the following command:

```
az deployment group create --resource-group sampleApiPlatform_rg --template-file sampleauthapi_container.bicep --parameters containerAppName=sampleauthapi managedEnvironmentId='/subscriptions/c13b401c-dde4-4298-ad6c-3ef836f847d5/resourceGroups/sampleApiPlatform_rg/providers/Microsoft.App/managedEnvironments/sampleapi-env'
```

Centralizing configuration: Azure App Configuration

We have 3 containers running which share some configuration parameters from EntraId.
We can centralize them and have them all in one place.
The optimal location for that is the Azure App Configuration, because we do not need to build any infrastructure to get these values from a central location and all is implemented internally by azure.

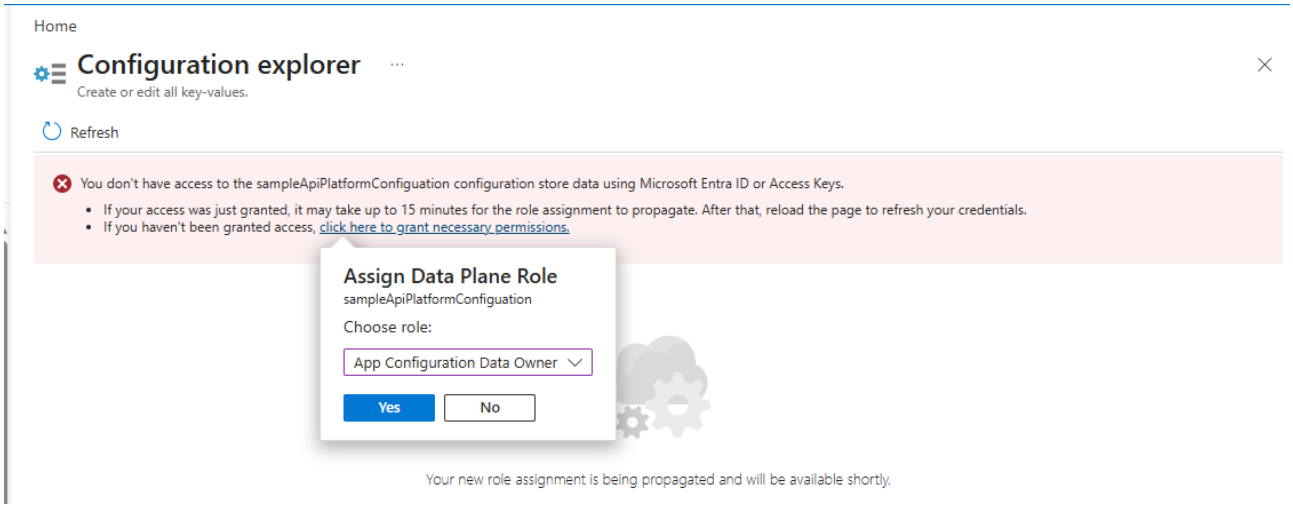
[Create App Configuration - Microsoft Azure]

We create a new one:

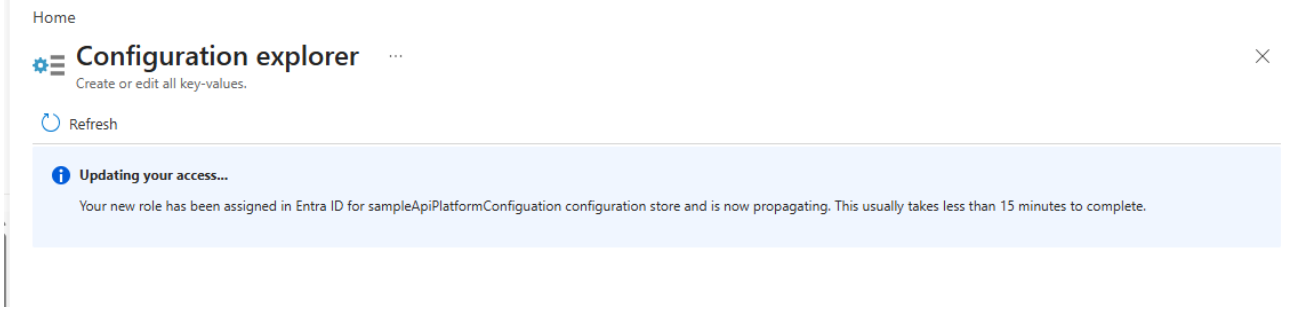
Subscription = Azure subscription 1
Resource Group = sampleApiPlatform_rg
Location = Central France
resource Name = sampleApiPlatformConfiguration
Pricing Tier = Free

Next is to go to App Configuration > Configuration Explorer >

It is possible that we get a message that we are not granted to add Key-Value pairs. There is a link in the same error message that if we click on it will give us rights to enter the values. Choose App Configuration Data Owner (not data reader).



It can take up to 15 minutes to Update this setting



Enter the Key value pairs that we share in these 3 containers.

sampleauthapi

```
  "AzureAd": {
    "Instance":
"https://login.microsoftonline.com/",
    "Domain":
"jobsmacoroprotonmail.onmicrosoft.com",
    "TenantId": "9ca003dc-
da5d-4b31-808b-9d52337fbf6a",
    "ClientId": "e4c2b81e-f125-
45c4-ba3b-54380d2b83b7",
    "Audience": "api://e4c2b81e-
f125-45c4-ba3b-54380d2b83b7",
    "Scope": "access_as_user"
  },
  "AuthClient": {
    "TenantId": "9ca003dc-
da5d-4b31-808b-9d52337fbf6a",
    "ClientId": "aa8df142-9e42-
4ec3-bd1c-238313856a84",
    "Scope": "api://e4c2b81e-
f125-45c4-
ba3b-54380d2b83b7/access_as_
user"
  },
```

sampleappapi

```
  "AzureAd": {
    "Instance":
"https://login.microsoftonline.com/",
    "Domain":
"jobsmacoroprotonmail.onmicrosoft.com",
    "TenantId": "9ca003dc-
da5d-4b31-808b-9d52337fbf6a",
    "ClientId": "e4c2b81e-f125-
45c4-ba3b-54380d2b83b7",
    "Audience": "api://e4c2b81e-
f125-45c4-ba3b-54380d2b83b7",
    "Scope": "access_as_user"
  },
```

sampledataaccessapi

```
"AzureAd": {
    "Instance":
"https://login.microsoftonline.com/",
    "Domain":
"jobsmacoroprotonmail.onmicrosoft.com",
    "TenantId": "9ca003dc-
da5d-4b31-808b-9d52337fbf6a",
    "ClientId": "e4c2b81e-f125-
45c4-ba3b-54380d2b83b7",
    "Audience": "api://e4c2b81e-
f125-45c4-ba3b-54380d2b83b7",
    "Scope": "access_as_user"
  },
```

Issues with Azure

Complex environement: until things wotk it takes long time to set up

Environment difficult to reproduce in Dev: secrets, dapr, tokens

Long process to debug: correct code, push into repo, run pipeline, update the container (cached)

Azure can get very messi: we start adding things and adding and not deleting what has been abandoned, leaving the whole status in a situation of do not touch this as it can have an impact somewhere that we don't know.

HTTP Codes:

401= Un Authorized